

Solutions to Exercises in *Ordinary Differential Equations*

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Preface

鉴于之前流传的《常微分方程》答案错误太多,有些题目的解答步骤过于繁琐,且有些题目并没有给出解答.为了帮助同学们和学弟学妹们复习这门课程,故撰写《常微分方程》第一至四章的习题解答(绝大部分,个人时间有限),这是我们期末考试的考试范围.由于个人时间精力有限,难免出现错误,但错误肯定是比原解答要少的.鉴于大家的英文阅读水平,我们只在题目和解答的地方使用英文,其他地方使用中文(或者中英双语),方便大家理解.

In view of the fact that the previously circulated answers to 'Ordinary Differential Equations' contained too many errors, some solution steps were overly complicated, and some problems had no solutions provided, this work compiles solutions to problems from Chapters 1 to 4 of 'Ordinary Differential Equations' (most of them, limited by personal time) to help students review this course. These chapters are within the scope of our final exam. Due to limited personal time and energy, errors are inevitable, but they are certainly fewer than in the original solutions. Considering everyone's English reading ability, we only use English for the problems and solutions, while the rest is in Chinese (or bilingual Chinese-English) to facilitate understanding.

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Chapter 1

Elementary Integration Method

Exercise 1.1

Trivially.

Exercise 1.2

Find general solutions of the differential equations in Problems 1 through 10.

Problem 1.2.1. $\frac{dy}{dx} + 4xy = 0$.

Solution. $y = Ce^{-2x^2} (C > 0)$. □

Problem 1.2.2. $\frac{dy}{dx} + 3x^2y^2 = 0$.

Solution. $y = \frac{1}{x^3 - C}$. □

Problem 1.2.3. $\frac{dy}{dx} = e^y \sin x$.

Solution. $y = -\ln(\cos x - c)$. □

Problem 1.2.4. $y^3 \frac{dy}{dx} = (y^4 + 1) \sin x$.

Solution. Separate variables to get $\frac{y^3}{y^4 + 1} dy = \sin x dx$, which gives $\frac{1}{4} \ln(1 + y^4) = -\cos x + C$. □

Problem 1.2.5. $x^2y' = 1 - 2x^2 + y^2 - 2x^2y^2$.

Solution. Separate variables to get $\frac{dy}{y^2 + 1} = \frac{1 - 2x^2}{x^2} dx$, which gives $\arctan y = -\frac{1}{x} - 2x + C$. □

Problem 1.2.6. $x \sin y + (x^2 + 1) \cos y \cdot y' = 0$.

Solution. Separate variables to get $\cot y dy = -\frac{x}{x^2 + 1} dx$, which gives $\ln |\sin y| = -\frac{1}{2} \ln(x^2 + 1) + C$. □

Problem 1.2.7. $(x^2 + 1) \cdot (\tan y)y' = 2x$.

Solution. $\ln |\sec y| = \ln(x^2 + 1) + C$. □

Problem 1.2.8. $y' = 1 + x + y + xy$.

Solution. $\ln |1 + y| = x + \frac{1}{2}x^2 + C$. □

Problem 1.2.9. $\frac{dy}{dx} = y^2 \cos x$.

Solution. $y = -\frac{1}{\sin x + C}$. □

Problem 1.2.10. $\frac{dy}{dx} = \frac{x + x^{\frac{3}{2}}}{1 + y^{\frac{5}{2}}}$.

Solution. $y + \frac{2}{7}y^{7/2} = \frac{1}{2}x + \frac{2}{5}x^{5/2} + C$. □

Find the particular solutions of the initial value problems in Problems 11 through 15.

Problem 1.2.11. $\frac{dy}{dx} = 2ye^x$, $y(0) = e$.

Solution. $y = e^{2e^x - 1}$. □

Problem 1.2.12. $\frac{dy}{dx} = 4x^2y - y$, $y(1) = 3$.

Solution. $y = 3e^{\frac{4x^3}{3} - x - \frac{1}{3}}$. □

Problem 1.2.13. $\frac{dy}{dx} = e^{2x-y}$, $y(0) = 0$.

Solution. $y = \ln \left(\frac{e^{2x} + 1}{2} \right)$. □

Problem 1.2.14. $\frac{dy}{dx} = xy^2 + 2x^2y^2$, $y(1) = 1$.

Solution. $y = -\frac{6}{4x^3 + 3x^2 - 13}$. □

Problem 1.2.15. $(y^2 + xy^2) dx - (x^2y^2) dy = 0$, $y(1) = 2$.

Solution. $y = 2e^{-\frac{1}{3x^3} - \frac{1}{2x^2} + \frac{5}{6}}$. □

Problem 1.2.16. Show that the transformation $y = xu$ changes the differential equation

$$y^n f(x) + g\left(\frac{y}{x}\right)(y - xy') = 0$$

into a separable equation.

Proof. You can directly verify it. □

Problem 1.2.17. When sugar is dissolved in water, the amount A that remains undissolved after t minutes satisfies the differential equation $\frac{dA}{dt} = -kA$ ($k > 0$). If 25% of the sugar dissolves after 1 minute, how long does it take for half of the sugar to dissolve?

Solution. Please solve it yourself (answer not provided). □

Problem 1.2.18. Solve the differential equation

$$(y - xy^2) dx - (x + x^2y) dy = 0$$

by making the substitution $y = ux^n$ for an appropriate n .

Solution. Let $y = ux^{-1}$, which gives $\left(1 + \frac{1}{u}\right) du = \frac{2}{x} dx$. Then the general solution is $xy + \ln|xy| + C = 2 \ln|x|$. □

Exercise 1.3

Find general solutions of the differential equations.

Problem 1.3.1. $(x + y)y' = x - y$.

Solution. Let $y/x := u$. $y = -x \pm \sqrt{2x^2 + e^{2c}}$. □

Problem 1.3.2. $x^2y' = xy + x^2e^{y/x}$;

Solution. Let $y/x := u$. $y = -x \ln(-c - \ln x)$ □

Problem 1.3.3. $x^2y' = xy + y^2$.

Solution. Let $y/x := u$. $y = \frac{x}{-\ln|x| + c_1}$. □

Problem 1.3.4. $y' = \frac{2xy}{x^2 - y^2}$.

Solution. Let $y/x := u$. $y = \frac{1}{2} \left(e^c \pm \sqrt{-4x^2 + e^{2c}} \right)$. □

Problem 1.3.5. $xyy' = y^2 + x\sqrt{4x^2 + y^2}$.

Solution. Let $y/x := u$. $y = \pm x \sqrt{\ln^2(x) + 2c \ln(x) - 4 + c^2}$. □

Problem 1.3.6. $2xyy' = x^2 + 3y^2$.

Solution. Let $y/x := u$. $y = \pm x \sqrt{-1 + cx}$. □

Problem 1.3.7. $y' = (x + y + 1)^3$.

Solution. Let $u := x + y + 1$, then $\frac{du}{dx} = 1 + \frac{dy}{dx}$. The original equation can be written as

$$\frac{du}{dx} = 1 + u^3.$$

Hence, when $u \neq -1$, we get

$$\frac{du}{u^3 + 1} = dx \implies x = -\frac{1}{6} \ln(u^2 - u + 1) + \frac{1}{3} \ln|u + 1| + \frac{1}{\sqrt{3}} \arctan\left(\frac{2u - 1}{\sqrt{3}}\right) + C.$$

Substituting $u = x + y + 1$ into it gives the implicit general solution of the original differential equation. When $u = -1$, note that $y = -x - 2$ is a particular solution. □

Problem 1.3.8. $y' = x^2 + 2xy + y^2$.

Solution. Let $u := x + y$, then $\frac{du}{dx} = 1 + \frac{dy}{dx}$. The original equation can be written as

$$\frac{du}{dx} = 1 + u^2.$$

Hence, we get $u = \tan(x + C)$, which implies $y = -x + \tan(x + C)$. □

Problem 1.3.9. $xy' - y = (x + y) \ln \frac{x + y}{x}$.

Solution. Let $u := y/x$. $y = x(-1 + e^{-u})$. □

Problem 1.3.10. $(2x - 4y + 6) dx + (x + y - 3) dy = 0$.

Solution. Let

$$\begin{cases} u := x - 1; \\ v := y - 2, \end{cases} \implies \begin{cases} x = u + 1; \\ y = v + 2. \end{cases}$$

Hence, $\frac{dv}{du} = \frac{dy}{dx}$. Though simplification, we can get

$$2(u - 2v) du + (u + v) dv = 0.$$

Therefore, when $u \neq 2v$, we get

$$\frac{dv}{du} = \frac{4v - 2u}{u + v}.$$

Let $k := \frac{v}{u}$, then

$$-\frac{1 + k}{k^2 - 3k + 2} dk = \frac{du}{u}.$$

Integrating both sides,

$$2 \ln |1 - k| - 3 \ln |2 - k| = \ln |u| + C.$$

Substituting $k = \frac{v}{u} = \frac{y - 2}{x - 1}$ into it gives the implicit general solution of the original differential equation. When $u = 2v$, note that $y \equiv 2$ is a particular solution. □

Problem 1.3.11. $(2x + y + 1) dx + (4x + 2y + 3) dy = 0$.

Solution. Let $u := 2x + y + 1$, then $2 + \frac{dy}{dx} = \frac{du}{dx}$. Therefore, we have $\frac{2u + 1}{3u + 2} du = dx$ when $3u + 2 \neq 0$, then $6u - \ln |2 + 3u| = 9(x + c) - 4 \implies 12x + 6y - \ln |6x + 3y + 5| = 9(x + c) - 10$. □

Problem 1.3.12. $(x + 3y) y' = x + y + 5$.

Solution. Let

$$\begin{cases} u := x + 15/2; \\ v := y - 5/2, \end{cases} \iff \begin{cases} x = u - 15/2; \\ y = v + 5/2. \end{cases}$$

Hence, $\frac{dv}{du} = \frac{dy}{dx}$. Let $k := \frac{v}{u}$, then

$$\frac{1 + 3k}{1 - 3k^2} dk = \frac{du}{u}, \quad 3k^2 \neq 1.$$

Integrating both sides,

$$\frac{1}{6} \left[(\sqrt{3} - 3) \ln |3k + \sqrt{3}| - (3 + \sqrt{3}) \ln |\sqrt{3} - 3k| \right] = \ln |u| + C.$$

Substituting $k = \frac{v}{u} = \frac{y-2}{x-1}$ into the above equation gives

$$\frac{1}{6} \left((\sqrt{3} - 3) \ln \left| \frac{3(2y-5)}{2x+15} + \sqrt{3} \right| - (3 + \sqrt{3}) \ln \left| \frac{15-6y}{2x+15} + \sqrt{3} \right| \right) = \ln \left| x + \frac{15}{2} \right| + C.$$

When $3k^2 = 1 \iff u = \pm\sqrt{3}v$, note that $x + \frac{15}{2} = \pm\sqrt{3} \left(y - \frac{5}{2} \right)$ are two particular solutions. □

Problem 1.3.13. $(x^2 + 2y^2 - 1)x dx - (x^2 + y^2 + 2)y dy = 0$.

Solution. Let us introduce the substitutions: $u = x^2$, $v = y^2$, so that

$$du = 2x dx \implies x dx = \frac{1}{2} du, \quad dv = 2y dy \implies y dy = \frac{1}{2} dv.$$

Substituting into the original equation gives $(u + 2v - 1) du - (u + v + 2) dv = 0$, or equivalently, $\frac{du}{dv} = \frac{u + v + 2}{u + 2v - 1}$. Setting $u = w - 5$, $v = z + 3$ gives $\frac{dw}{dz} = \frac{w + z}{w + 2z}$. Let $w = tz$, so that $dw = t dz + z dt$. After arranging the equation, we obtain

$$-\frac{t+2}{t^2+t-1} dt = \frac{dz}{z}.$$

Therefore:

$$-\frac{1}{10} (5 + 3\sqrt{5}) \ln |-2t + \sqrt{5} - 1| - \frac{1}{10} |5 - 3\sqrt{5}| \ln |2t + \sqrt{5} + 1| = \ln |z| + C,$$

where $C > 0$ is an arbitrary constant.

Recall that $t = \frac{w}{z} = \frac{u+5}{v-3} = \frac{x^2+5}{y^2-3}$, $z = v-3 = y^2-3$. Thus:

$$-\frac{1}{10} \left[(5 + 3\sqrt{5}) \ln \left| -\frac{2(x^2+5)}{y^2-3} + \sqrt{5} - 1 \right| + (5 - 3\sqrt{5}) \ln \left| \frac{2(x^2+5)}{y^2-3} + \sqrt{5} + 1 \right| \right] = \ln |y^2-3| + C,$$

or equivalently,

$$-\frac{1}{20} \left[(5 + 3\sqrt{5}) \ln |-2x^2 + (\sqrt{5} - 1)y^2 - 3\sqrt{5} - 7| + (5 - 3\sqrt{5}) \ln |2x^2 + (1 + \sqrt{5})y^2 - 3\sqrt{5} + 7| \right] = C.$$

□

Problem 1.3.14. $y' = \frac{x+y-1}{x-y+1}$.

Solution. $\ln \left(\frac{x^2 + y^2 - 2y + 1}{2x^2} \right) + 2 \arctan \left(1 - \frac{2x}{-y + x + 1} \right) + 2 \ln(x) + C = 0$. □

Problem 1.3.15. $y' = (x + y + 1)^2 - 4$.

Solution. $y = -x + \frac{1}{\frac{1}{2\sqrt{3}} + Ce^{2\sqrt{3}x}} - \sqrt{3} - 1.$ □

Problem 1.3.16. Show that the substitution $u = \ln y$ transforms the differential equation

$$\frac{dy}{dx} + p(x)y = q(x)(y \ln y)$$

into the linear equation

$$\frac{du}{dx} + p(x) = q(x)u(x).$$

Proof. It suffices to use the following facts:

$$\frac{du}{dx} = \frac{du}{dy} \frac{dy}{dx} = \frac{1}{y} \frac{dy}{dx} = e^{-u} \frac{dy}{dx}. \quad \square$$

Problem 1.3.17. Show that if the differential equation

$$P(x, y) dx + Q(x, y) dy = 0$$

is homogeneous and P and Q possess continuous partial derivatives in some domain D , then

$$\frac{x \frac{\partial P}{\partial x} + y \frac{\partial P}{\partial y}}{x \frac{\partial Q}{\partial x} + y \frac{\partial Q}{\partial y}} = \frac{P}{Q}.$$

Proof. Since $P(x, y)$ and $Q(x, y)$ are homogeneous functions of degree n , by Euler's theorem for homogeneous functions, we have

$$x \frac{\partial P}{\partial x} + y \frac{\partial P}{\partial y} = nP(x, y) \quad \text{and} \quad x \frac{\partial Q}{\partial x} + y \frac{\partial Q}{\partial y} = nQ(x, y).$$

Therefore,

$$\frac{x \frac{\partial P}{\partial x} + y \frac{\partial P}{\partial y}}{x \frac{\partial Q}{\partial x} + y \frac{\partial Q}{\partial y}} = \frac{nP(x, y)}{nQ(x, y)} = \frac{P}{Q}. \quad \square$$

Remark 1.1. 齐次函数的欧拉定理是指:

Theorem 1.3.1. 设 f 具有连续偏导数, 则 f 是 n 次齐次函数的充分必要条件是

$$x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y} = nf. \quad (1.1)$$

Proof. 设 f 是 n 次齐次函数, 则 $nf = \frac{df(tx, ty)}{dt} = \frac{\partial f}{\partial u} \frac{du}{dt} + \frac{\partial f}{\partial v} \frac{dv}{dt} = x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y} = nf.$

反之, 令 $\varphi(t) := \frac{f(tx, ty)}{t^n}$. 则

$$\begin{aligned}\varphi'(t) &= \frac{t^n \left[x \frac{\partial f}{\partial x}(tx, ty) + y \frac{\partial f}{\partial y}(tx, ty) \right] - nt^{n-1} f(tx, ty)}{t^{2n}} \\ &= \frac{t^{n-1} \left[x \frac{\partial f}{\partial x}(tx, ty) + y \frac{\partial f}{\partial y}(tx, ty) - nf(tx, ty) \right]}{t^{2n}} = 0.\end{aligned}$$

故

$$\varphi(t) \equiv \varphi(1) = f(x, y) \implies f(tx, ty) = t^n f(x, y) \quad (\forall t > 0). \quad \square$$

Remark 1.2. 可以看出在充分性的证明中, 本质上是解了微分方程.

Problem 1.3.18. Show that the substitution $y = x^n u$ in the differential equation

$$y' = x^{n-1} f\left(\frac{y}{x^n}\right)$$

leads to a separable equation. Hence, solve the differential equation

$$x^3 y' = 2y(x^2 - y).$$

Proof. The substitution $y = x^n u$ gives the following equation:

$$\frac{dy}{dx} = nx^{n-1}u + x^n \frac{du}{dx}.$$

Therefore, we have that

$$nx^{n-1}u + x^n \frac{du}{dx} = x^{n-1} f(u) \implies x \frac{du}{dx} = f(u) - nu,$$

which is a separable equation. Hence, the equation $x^3 y' = 2y(x^2 - y)$ can be easily solved as $y = \frac{x^2}{2 \ln(x) + c}$ by setting $u := y/x$. $\square$

Exercise 1.4

Find general solutions of the differential equations in Problems 1 through 15. If an initial condition is given, find the corresponding particular solution.

Remark 1.3. 对于下面的绝大部分题目, 总是可以套用通解公式

$$y = ce^{-\int p(x) dx} + e^{-\int p(x) dx} \int q(x) e^{\int p(x) dx} dx \quad (1.2)$$

或

$$y(x) = y_0 e^{-\int_{x_0}^x p(t) dt} + \int_{x_0}^x q(s) e^{-\int_s^x p(t) dt} ds \quad (1.3)$$

算出答案. 因此我们省略大部分步骤.

Problem 1.4.1. $y' + 2y = 1, y(0) = 0$.

Solution. $y = \frac{1}{2}e^{-2x}(e^{2x} - 1)$. □

Problem 1.4.2. $y' + 2y = xe^{-2x}$.

Solution. $y = \frac{1}{2}e^{-2x}(x^2 + 2C)$. □

Problem 1.4.3. $y' - 2xy = e^{x^2}$.

Solution. $y = e^{x^2}(x + C)$. □

Problem 1.4.4. $y' + \frac{3}{x}y = 2x^4, y(2) = 1$.

Solution. The general solution is $y = \frac{x^5}{4} + \frac{c}{x^3}$. With $y(2) = 1$, we can get $c = -56$. □

Problem 1.4.5. $y' - (\cot x)y = 2x \sin x$.

Solution. The general solution is $y = (x^2 + c) \sin(x)$. □

Problem 1.4.6. $y' + y + 2x + x^2 + x^3 = 0$.

Solution. The general solution is $y = -x^3 + 2x^2 - 6x + c_1e^{-x} + 6$. □

Problem 1.4.7. $y' + y \tan x = \sec x$.

Solution. $y = \sin(x) + C \cos(x)$. □

Problem 1.4.8. $2x(y' - y) = e^x$.

Solution. $y = \frac{1}{2}e^x(\ln(x) + 2C)$. □

Problem 1.4.9. $y' + xy = 2x, y(0) = 1$.

Solution. The general solution is $y = 2 + ce^{-\frac{x^2}{2}}$. With $y(0) = 1$, we can get $c = -1$. □

Problem 1.4.10. $xy' + 5y = 6x^2, y(1) = 4$.

Solution. The general solution is $y = \frac{6x^2}{7} + \frac{C}{x^5}$. With $y(1) = 4$, we can get $C = \frac{6x^7 + 22}{7x^5}$. □

Remark 1.4. 下面 5 题都是 Bernoulli 方程, 可以通过作换元 $y \mapsto y^{1-n}$ 将其变成一阶线性微分方程, 从而可以套用公式(1.2)解决. 我们只给出 Problem 1.4.11 的步骤.

Problem 1.4.11. $\frac{dy}{dx} + y = xy^5$.

Solution. This is a **Bernoulli's equation**. We perform the substitution $z := y^{-4}$ to transform it into a first-order linear differential equation

$$\frac{dz}{dx} - 4z = -4x.$$

Solving it gives $z = x + ce^{4x} + \frac{1}{4}$. Therefore, the general solution is $y^4 = \frac{4}{4x + 4ce^{4x} + 1}$. □

Problem 1.4.12. $y' + xy + xy^3 = 0$.

Solution. This a Bernoulli's equation. Its general solution is $y = \pm \frac{e^C}{\sqrt{e^{x^2} - e^{2C}}}$. □

Problem 1.4.13. $y' + 2xy = x^3y^3$.

Solution. This a Bernoulli's equation. The general solution is $y = \pm \frac{2}{\sqrt{2x^2 + 4ce^{2x^2} + 1}}$. □

Problem 1.4.14. $y' + y = y^2e^{-x}$.

Solution. This a Bernoulli's equation. The general solution is $y = \frac{2e^x}{1 + 2c_1e^{2x}}$. □

Problem 1.4.15. $xy' + 6y = 3xy^{4/3}$.

Solution. This a Bernoulli's equation. The general solution is $y = \frac{1}{x^3(1 + Cx)^3}$. □

Solve the differential equations in Problems 16 through 20 by regarding y as the independent variable rather than x .

Problem 1.4.16. $(2x + ye^y) \frac{dy}{dx} = 1$.

Solution. This is the first-order linear differential equation of x with respect to y , which can be directly solved by applying the formula (1.2) that $x = e^y(-y - 1) + Ce^{2y}$. □

Problem 1.4.17. $(1 - 2xy^2) \frac{dy}{dx} = y^3$.

Solution. This is the first-order linear differential equation of x with respect to y , which can be directly solved by applying the formula (1.2) that $x = \frac{\ln(y)}{y^2} + \frac{C}{y^2}$. □

Problem 1.4.18. $(y + 2xy) \frac{dy}{dx} = 1 + y^2$.

Solution. The general solution is $x = -\frac{1}{2} + C(y^2 + 1)$. □

Problem 1.4.19. $y'(x^3 - xy) = 1$.

Solution. If it is regarded as the equation of x with respect to y , then this is the Bernoulli equation. We can simply substitute $z := x^{-2}$, which gives that

$$\frac{dz}{dy} - 2yz = -2.$$

Then the general solution is

$$z = Ce^{-\int 2y dy} + e^{-\int 2y dy} \int (-2) e^{-\int 2y dy} dy = Ce^{-y^2} - 2e^{-y^2} \int e^{-y^2} dy.$$

Let $\text{Erf}(x) := \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt$, then the general solution can be written as

$$x = \pm \frac{1}{\sqrt{e^{y^2} (-\sqrt{\pi} \text{Erf}(y) + c_1)}}. \quad \square$$

Problem 1.4.20. If $y_1(x)$ and $y_2(x)$ are solutions of $y' + p(x)y = q_i(x)$, $i = 1, 2$, respectively, then $C_1y_1(x) + C_2y_2(x)$ is a solution of the differential equation

$$y' + p(x)y = C_1q_1(x) + C_2q_2(x),$$

where C_1 and C_2 are constants.

Proof. Obviously. □

Problem 1.4.21. The differential operator

$$\mathcal{L}y = y' + p(x)y \quad (1.4)$$

can be used to write the differential equation $y' + p(x)y = q(x)$ in the form

$$\mathcal{L}y = y' + p(x)y = 0 \quad (\text{homogeneous}) \quad (1.5)$$

and

$$\mathcal{L}y = y' + p(x)y = q(x) \quad (\text{nonhomogeneous}). \quad (1.6)$$

Show that the following properties hold:

- (1) If $\phi_1(x)$ and $\phi_2(x)$ are solutions of the homogeneous linear equation (1.5), then a linear combination $C_1\phi_1(x) + C_2\phi_2(x)$ is also a solution of equation (1.5), where C_1 and C_2 are constants.
- (2) Suppose that $\phi_h(x)$ and $\phi_p(x)$ are solutions of the differential equations (1.5) and (1.6) respectively. Then $\phi_h(x) + \phi_p(x)$ is a solution of equation (1.6).
- (3) Let $\phi_1(x)$ and $\phi_2(x)$ be solutions of the differential equation (1.6). Then $\phi_1(x) - \phi_2(x)$ is a solution of equation (1.5).
- (4) Let $\phi(x)$ be any one solution of the differential equation (1.5). Then either $\phi(x) = 0$ or $\phi(x) \neq 0$ at each point of I .
- (5) The initial value problem

$$\begin{cases} \frac{dy}{dx} + p(x)y = q(x), \\ y(t_0) = y_0, \end{cases} \quad (1.7)$$

has a unique solution.

Proof. (1)-(3) are obvious. For (4), we can get the general solution to (1.5) is $\phi(x) = Ce^{-\int p(x)dx}$. If $C = 0$, then $\phi \equiv 0$; otherwise, $\phi(x) \neq 0$ ($\forall x \in I$).

For (5). Firstly, we know that the solution of (1.7) exists, which can be written in the form of (1.2). Let ϕ_1, ϕ_2 be a solution of the differential equation (1.7), then by (3), $\phi_1 - \phi_2$ is a solution of equation (1.5). Therefore, by (4), we know either $\phi_1 - \phi_2 \equiv 0$ or $\phi_1 \neq \phi_2$ at each point of I . If it is the latter, this will contradict the initial value condition, so there can only be $\phi_1 \equiv \phi_2$, which indicates that there is a unique solution to the initial value problem (1.7). □

Problem 1.4.22. Let $g \in C[0, +\infty)$, $\lim_{x \rightarrow +\infty} g(x) = \beta$. For the differential equation $y' + by = g(x)$, show the following:

(1) If $b > 0$, every solution tends to $\frac{\beta}{b}$ as $x \rightarrow \infty$.

(2) If $b < 0$, there exists a unique solution which approaches $\frac{\beta}{b}$ as $x \rightarrow \infty$.

Proof. (1) If $b > 0$, note that

$$\lim_{x \rightarrow +\infty} y(x) = \lim_{x \rightarrow +\infty} \frac{e^{bx}(y(x))}{e^{bx}} = \lim_{x \rightarrow +\infty} \frac{e^{bx}(y'(x) + by(x))}{be^{bx}} = \lim_{x \rightarrow +\infty} \frac{g(x)}{b} = \frac{\beta}{b}.$$

(2) If $b < 0$, note that

$$\lim_{x \rightarrow +\infty} e^{-bx} \int_0^x g(t)e^{bt} dt = \lim_{x \rightarrow +\infty} \frac{g(x)e^{bx}}{be^{bx}} = \frac{\beta}{b}.$$

Therefore, the solution of a differential equation tends to $\frac{\beta}{b}$ if and only if $C = 0$, which proves the existence and uniqueness. □

Problem 1.4.23. Let $y \in C^1[0, +\infty)$ and $\lim_{x \rightarrow +\infty} [y'(x) + 2y(x)] = 0$. Show that $\lim_{x \rightarrow +\infty} y(x) = 0$.

Proof. Note that

$$\lim_{x \rightarrow +\infty} y(x) = \lim_{x \rightarrow +\infty} \frac{e^{2x}y(x)}{e^{2x}} = \lim_{x \rightarrow +\infty} \frac{e^{2x}(y'(x) + 2y(x))}{2e^{2x}} = 0. \quad \square$$

Remark 1.5. 这个题目也可以通过求解微分方程 $y' + 2y = 0$ 得到 y 的表达式解决, 细节留给读者解决.

Problem 1.4.24. Consider the differential equation

$$\frac{dy}{dx} + p(x)y = q(x) \quad (1.8)$$

where $p(x)$ and $q(x)$ are continuous and ω -periodic. Show the following:

(1) If $q(x) \equiv 0$, then any nonzero solution of this differential equation is ω -periodic if and only if $\bar{p} = \frac{1}{\omega} \int_0^\omega p(x) dx = 0$.

(2) If $q(x) \neq 0$, then this differential equation exists a unique ω -periodic solution if and only if $\bar{p} \neq 0$. Find this solution.

Proof. (1) Any nonzero solution of this differential equation can be written as

$$\varphi(x) = y(0)e^{-\int_0^x p(t) dt}, \quad C \neq 0.$$

Then φ is ω -periodic iff $\varphi(x + \omega) = \varphi(x), \forall x \iff \int_0^{x+\omega} p(t) dt = \int_0^x p(t) dt \iff \int_x^{x+\omega} p(t) dt = 0$. Note that p is ω -periodic, then $\left(\int_x^{x+\omega} p(t) dt\right)' = p(x + \omega) - p(x) = 0 \iff \int_x^{x+\omega} p(t) dt \equiv \int_0^\omega p(t) dt = 0 \iff \bar{p} = 0$.

(2) If $q(x) \neq 0$, then the general solution of equation (1.8) is

$$y(x) = e^{-\int_0^x p(x) dx} \left(C + \int_0^x q(s) e^{\int_0^s p(t) dt} ds \right).$$

Choose a constant C such that $y(x)$ becomes an ω -periodic function, i.e.,

$$y(x + \omega) \equiv y(x), \quad \forall x \in \mathbb{R}.$$

We claim: if $y(x)$ is a solution of equation (1.8), and satisfies

$$y(\omega) = y(0), \tag{1.9}$$

then $y(x)$ is an ω -periodic solution of equation (1.8). In fact, since $y(x)$ is a solution of equation (1.8), then $y(x + \omega)$ is also a solution of equation (1.8). Let

$$u(x) := y(x + \omega) - y(x),$$

then $y = u(x)$ is a solution of the corresponding homogeneous linear equation. If (1.9) holds, then $u(0) = 0$. By the uniqueness of the solution, we get $u(x) \equiv 0$. Therefore, $y(x)$ is an ω -periodic solution of equation (1.8). Substitute the general solution formula into (1.9), we get

$$C = \frac{1}{e^{\int_0^\omega p(x) dx} - 1} \int_0^\omega q(s) e^{\int_0^s p(t) dt} ds.$$

Therefore, when

$$C = \frac{1}{e^{\int_0^\omega p(x) dx} - 1} \int_0^\omega q(s) e^{\int_0^s p(t) dt} ds,$$

equation (1.8) has an ω -periodic solution $y(x)$. Hence, (1.8) has a unique ω -periodic solution if and only if $\bar{p} \neq 0$. Below we find the expression for $y(x)$.

Since $y(x)$ is a solution of equation (1.8), we have

$$\frac{dy(x)}{dx} + p(x)y(x) = q(x).$$

Multiply both sides by $e^{\int_0^x p(t) dt}$, we get

$$\frac{d}{dx} \left(e^{\int_0^x p(t) dt} y(x) \right) = e^{\int_0^x p(t) dt} q(x).$$

Integrate both sides from x to $x + \omega$, noting that $y(x + \omega) = y(x)$, we get

$$e^{\int_0^{x+\omega} p(t) dt} y(x + \omega) - e^{\int_0^x p(t) dt} y(x) = \int_x^{x+\omega} e^{\int_0^t p(s) ds} q(t) dt.$$

Since

$$e^{\int_0^{x+\omega} p(t) dt} y(x + \omega) = e^{\int_0^x p(t) dt} e^{\int_0^\omega p(t) dt} y(x) = e^{\int_0^x p(t) dt} e^{\int_0^\omega p(t) dt} y(x),$$

we have

$$y(x) = \frac{1}{e^{\int_0^\omega p(t) dt} - 1} \int_x^{x+\omega} e^{\int_x^t p(s) ds} q(t) dt,$$

which is the expression for the ω -periodic solution of equation (1.8). $\square$

Remark 1.6. 若记

$$G(x, t) := \frac{e^{\int_x^t p(s) ds}}{e^{\int_0^\omega p(t) dt} - 1}, \quad x \leq t \leq x + \omega,$$

则(1.8)的 ω -周期解可写作

$$y(x) = \int_x^{x+\omega} G(x, t) q(t) dt.$$

此时我们称 $G(x, t)$ 为 **Green 函数**.

Exercise 1.5

In Problems 1 through 10, verify that the given differential equation is exact, then solve it.

Remark 1.7. 验证是否恰当的步骤就是根据教材 21 页 Theorem 1.5.1, 我们省略不写.

Problem 1.5.1. $(x^2 + 3y) dx + (3x + e^y) dy = 0$.

Proof. $x^3/3 + 3xy + e^y = C$. $\square$

Problem 1.5.2. $(x - y) dx + (5y - x) dy = 0$.

Solution. $\frac{x^2}{2} - xy + \frac{5y^2}{2} = C$. $\square$

Problem 1.5.3. $(x^3 + 2y^3) dx + (6xy^2 + \cos y) dy = 0$.

Solution. $\frac{x^4}{4} + 2xy^3 + \sin(y) = C$. $\square$

Problem 1.5.4. $\left(x^2 + \frac{y}{x}\right) dx + (2y + \ln x) dy = 0$.

Solution. $\frac{x^3}{3} + y \ln(x) + y^2 = C$. $\square$

Problem 1.5.5. $(3x^2y^3 + y^4) dx + (3x^3y^2 + y^4 + 4xy^3) dy = 0$.

Solution. $x^3y^3 + xy^4 + \frac{y^5}{5} = C$. $\square$

Problem 1.5.6. $(e^x + 2xy) dx + (x^2 - 3y^2) dy = 0$.

Solution. $x^2y + e^x - y^3 = C$. $\square$

Problem 1.5.7. $(1 + y^2 \sin 2x) dx + (e^y - y \cos 2x) dy = 0$.

Solution. $-\frac{1}{2}y^2 \cos(2x) + x + e^y = C$. □

Problem 1.5.8. $(x^2 + \arctan y) dx + \left(\frac{x+y}{1+y^2}\right) dy = 0$.

Solution. $\frac{x^3}{3} + x \arctan(y) + \frac{1}{2} \ln(y^2 + 1) = C$. □

Problem 1.5.9. $(2e^x \sin y + \tan y) dx + (2e^x \cos y + x \sec^2 y) dy = 0$.

Solution. $2e^x \sin(y) + x \tan(y) = C$. □

Problem 1.5.10. $(\sin x + 2 \ln y) dx + \left(\frac{2x}{y} + 2 \cos y\right) dy = 0$.

Solution. $2x \ln(y) - \cos(x) + 2 \sin(y) = C$. □

Solve the following initial value problems.

Problem 1.5.11. $(4x^3y + 8xy^2) dx + (x^4 + 8x^2y + y^2) dy = 0, y(2) = 1$.

Solution. This is an exact equation. Its general solution is $x^4y + 4x^2y^2 + \frac{y^3}{3} = C$. Combining the initial conditions, we calculate that $C = 97/3$. □

Problem 1.5.12. $(2x - y \cos x) dx + (y - \sin x) dy = 0, y\left(\frac{\pi}{2}\right) = 1$.

Solution. This is an exact equation. Its general solution is $x^2 - y \sin(x) + \frac{y^2}{2} = C$. Combining the initial conditions, we calculate that $C = \frac{\pi^2}{4} - \frac{1}{2}$. □

Problem 1.5.13. $(2xy + x^2) dx + (x^2 + 2y) dy = 0, y(0) = 1$.

Solution. This is an exact equation. Its general solution is $\frac{x^3}{3} + x^2y + y^2 = C$. Combining the initial conditions, we calculate that $C = 1$. □

Problem 1.5.14. $(ye^{xy} + 4y^3) dx + (xe^{xy} + 12xy^2 - 2y) dy = 0, y(0) = 2$.

Solution. This is an exact equation. Its general solution is $4xy^3 + e^{xy} - y^2 = C$. Combining the initial conditions, we calculate that $C = -3$. □

In Problems 15 through 18, determine the constant λ so that each equation is exact and solve the resulting differential equations.

Problem 1.5.15. $(\lambda^3 + 3\lambda x) dt + (\lambda t^2 + 4x) dx = 0$.

Solution. We calculate that

$$\frac{\partial (t^3 + 3tx)}{\partial x} = 3t, \quad \frac{\partial (\lambda t^2 + 4x)}{\partial t} = 2\lambda t.$$

Then the equation is exact iff $\lambda = 3/2$. Therefore, the general solution of the equation is $\frac{1}{4}t^4 + \frac{3}{2}t^2x + 2x^2 = 0$. □

Problem 1.5.16. $(x^{-2} + 2y^{-2}) dx + (\lambda x + 2)y^{-3} dy = 0$.

Solution. $\lambda = -4$. $\frac{2x}{y^2} - \frac{1}{x} - \frac{1}{y^2} = C$. □

Problem 1.5.17. $(x^2 + \lambda xy^3) dx + (2x^2y^2 + 2y) dy = 0$.

Solution. $\lambda = \frac{4}{3}$. $\frac{x^3}{3} + \frac{2x^2y^3}{3} + y^2 = C$. □

Problem 1.5.18. $(e^x + \lambda y^2 \sin x) dx + (y^3 - y \cos x) dy = 0$.

Solution. $\lambda = \frac{1}{2}$. $-\frac{1}{2}y^2 \cos(x) + e^x + \frac{y^4}{4} = C$. □

Exercise 1.6

Problem 1.6.1. Solve the following differential equations.

(1) $(3x^2y + 2xy + y^3) dx + (x^2 + y^2) dy = 0$.

(2) $y dx + (4xy - e^{-y}) dy = 0$.

(3) $y dx - (2x^2 + 2y^2 + x) dy = 0$.

(4) $xy^3 dx + (x^2y^2 - 4) dy = 0$.

(5) $y(1 + e^x y) dx - x dy = 0$.

(6) $(x^2 + 2y^2 + 4x) dx + 2xy dy = 0$.

(7) $(x^4 + y^4) dx - 2xy^3 dy = 0$.

(8) $\frac{dy}{dx} = \frac{xy - y + y^2}{x + 2y}$.

(9) $x \frac{dy}{dx} + y \ln y - x^2 y = 0$.

(10) $\frac{dy}{dx} = \frac{e^{2x} - e^y - y}{1 + e^y}$.

Solution. (1) Firstly, we have $\frac{\partial P}{\partial y} \neq \frac{\partial Q}{\partial x}$, indicating that it's not an exact equation. Note that

$$\frac{1}{Q} \left(\frac{\partial P}{\partial y} - \frac{\partial Q}{\partial x} \right) = 3.$$

Therefore, one of the integration factors of the equation is $\mu = e^{\int 3 dx}$. At this time, the equation is an exact equation, and its general solution is $\frac{1}{3}e^{3x}y(3x^2 + y^2) = C$.

(2) Firstly, we have $\frac{\partial P}{\partial y} \neq \frac{\partial Q}{\partial x}$, indicating that it's not an exact equation. Note that

$$\frac{1}{Q} \left(\frac{\partial P}{\partial y} - \frac{\partial Q}{\partial x} \right) = 4 - \frac{1}{y}.$$

Therefore, one of the integration factors of the equation is $\mu = e^{\int (4-1/y) dy} = e^{4y}/y$. At this time, the equation is an exact equation, and its general solution is $xe^{4y} - \ln|y| = C$.

(3) Rewrite the original equation as

$$ydx - xdy = (2x^2 + 2y^2)dy.$$

Note that

$$d\left(\frac{x}{y}\right) = \frac{ydx - xdy}{y^2} = 2\left(\frac{x^2}{y^2} + 1\right)dy,$$

Let $u := \frac{x}{y}$, then the equation becomes:

$$du = 2(u^2 + 1)dy,$$

Separating variables and integrating both sides yields

$$\arctan(u) = 2y + C,$$

where C is the constant of integration. Substitute back $u = \frac{x}{y}$:

$$\arctan\left(\frac{x}{y}\right) = 2y + C \implies x = y \tan(2y + C).$$

$$(4) \quad \mu = \frac{1}{y}, \quad \frac{x^2 y^2}{2} - 4 \ln|y| = C.$$

$$(5) \quad \mu = 2/y^2, \quad e^x + x/y = C.$$

$$(6) \quad \mu = x, \quad \frac{x^4}{4} + \frac{4x^3}{3} + x^2 y^2 = C.$$

$$(7) \quad \mu = \frac{1}{x^3}, \quad \frac{x^4 - y^4}{2x^2} = C.$$

$$(8) \quad \mu = e^{-x}, \quad e^{-x}(-xy - y^2) = C.$$

$$(9) \quad \mu = \frac{1}{y}, \quad -\frac{x^3}{3} + x \ln|y| = C.$$

$$(10) \quad \mu = e^x, \quad \frac{1}{3}e^x(e^{2x} - 3y - 3e^y) = C. \quad \square$$

Problem 1.6.2. Solve the following differential equations for which the type of integrating factor has been indicated.

$$(1) \quad y' = \frac{y^2 - x}{xy} [\mu(x)].$$

$$(2) \quad 2y dx + (y^2 - 2x) dy = 0 [\mu(y)].$$

$$(3) \quad y + x(1 - 3x^2 y^2) \frac{dy}{dx} = 0 [\mu(xy)].$$

$$(4) \quad y' = -\frac{3xy + y^2}{3xy + x^2} [\mu(x + y)].$$

$$(5) \quad (x + x^4 + 2x^2y^2 + y^4) dx + y dy = 0 [\mu(x^2 + y^2)].$$

$$(6) \quad (y^2 + 2x^2y) dx + (xy + x^3) dy = 0 [\mu(x^m y^n)].$$

Solution. (1) $\mu = \frac{1}{x^3}, \quad -\frac{y^2}{2x^2} + \frac{1}{x} = C.$

(2) $\mu = \frac{1}{y^2}, \quad \frac{2x}{y} + y = C.$

(3) Rewrite the original equation as

$$y dx + x(1 - 3x^2 y^2) dy := P(x, y) dx + Q(x, y) dy = 0.$$

Note that

$$\frac{\frac{\partial P}{\partial y} - \frac{\partial Q}{\partial x}}{yQ - xP} = -\frac{3}{xy},$$

Therefore, the integrating factor is $\mu = e^{-3 \int \frac{1}{xy} d(xy)} = \frac{1}{(xy)^3}$. Therefore, the general solution is $-\frac{1}{2x^2 y^2} - 3 \ln |y| + c_1 = C.$

(4) $\mu = x + y, \quad xy(x + y)^2 = C.$

(5) $\mu = \frac{1}{(x^2 + y^2)^2}, \quad x - \frac{1}{2(x^2 + y^2)} = C.$

(6) Let $P(x, y) := y^2 + 2x^2 y, \quad Q(x, y) := xy + x^3$. After calculation, we have

$$\frac{\frac{\partial P}{\partial y} - \frac{\partial Q}{\partial x}}{\alpha \cdot Q/x - \beta \cdot P/y} \cdot \frac{1}{x^\alpha y^\beta} = \frac{x^{-\alpha} (y - x^2) y^{-\beta}}{x^2(\alpha - 2\beta) + y(\alpha - \beta)}.$$

Therefore, we choose

$$\begin{cases} \alpha - 2\beta = -1, \\ \alpha - \beta = 1. \end{cases} \iff \begin{cases} \alpha = 3, \\ \beta = 2. \end{cases}$$

Therefore, an integrating factor is $x^3 y^2$. Then the general solution is $x^4 y^3 (4x^2 + 3y) = C.$ □

Problem 1.6.3. Find an integrating factor of the following equations.

(1) First-order linear differential equation $y' + p(x)y = q(x)$.

(2) Bernoulli's equation $y' + p(x)y = q(x)y^n \quad (n \neq 0, 1).$

Solution. (1) Rewrite the original equation as

$$(q(x) - p(x)y) dx - dy = 0.$$

Then the integrating factor is $\mu(x) = e^{\int p(x) dx}.$

(2) Substituting $z := y^{1-n}$ gives

$$z' + (1-n)p(x)z = (1-n)q(x) \iff ((1-n)p(x)z - (1-n)q(x))dx + dz = 0.$$

Then the integrating factor is $\mu(x) = e^{\int (1-n)p(x)dx}$. □

Problem 1.6.4. The differential equation $(e^x \sec y - \tan y)dx + dy = 0$ has an integrating factor of the form $e^{-ax} \cos y$ for some constant a . Find a , and solve this equation.

Solution. Let $P(x, y) := e^{-ax} \cos y (e^x \sec y - \tan y)$, $Q(x, y) := e^{-ax} \cos y$, then

$$\frac{\partial P}{\partial y} = -e^{-ax} \cos(y), \quad -ae^{-ax} \cos(y).$$

Therefore, $a = 1$, and the general solution is $e^{-x} \sin(y) + x = C$. □

Problem 1.6.5. Suppose that $f(u), g(u)$ are continuously differentiable and $\varphi(x, y) = [f(xy) - g(xy)]xy \neq 0$. Prove that

$$\mu(x, y) = \frac{1}{\varphi(x, y)}$$

is an integrating factor of the equation

$$f(xy)y dx + g(xy)x dy = 0.$$

Proof. • (Method 1) Let $u := xy$, then $y = \frac{u}{x}$, $du = y dx + x dy$. Thus,

$$x dy = du - y dx = du - \frac{u}{x} dx.$$

Therefore, the original equation becomes

$$\frac{u}{x} f(u) dx + g(u) \left(du - \frac{u}{x} dx \right) = 0,$$

i.e.,

$$\frac{u}{x} (f(u) - g(u)) dx + g(u) du = 0.$$

This is a separable differential equation, and it has an integrating factor

$$\mu = \frac{1}{u(f(u) - g(u))} = \frac{1}{xy[f(xy) - g(xy)]}.$$

• (Method 2) It is only necessary to prove:

$$\frac{\partial}{\partial y} \left[\frac{yf(xy)}{xy[f(xy) - g(xy)]} \right] = \frac{\partial}{\partial x} \left[\frac{xg(xy)}{xy[f(xy) - g(xy)]} \right]$$

By calculation,

$$\begin{aligned}\text{RHS} &= \frac{1}{x} \frac{\partial}{\partial y} \frac{f(xy)}{[f(xy) - g(xy)]} = \frac{1}{x} \frac{xf'(xy)[f(xy) - g(xy)] - f(xy)[xf'(xy) - xg'(xy)]}{[f(xy) - g(xy)]^2} \\ &= \frac{-f'(xy)g(xy) + f(xy)g'(xy)}{[f(xy) - g(xy)]^2}, \\ \text{LHS} &= \frac{1}{y} \frac{\partial}{\partial x} \frac{g(xy)}{f(xy) - g(xy)} = \frac{1}{y} \frac{yg'(xy)[f(xy) - g(xy)] - g(xy)[yf'(xy) - yg'(xy)]}{[f(xy) - g(xy)]^2} \\ &= \frac{g'(xy)f(xy) - g(xy)f'(xy)}{[f(xy) - g(xy)]^2}.\end{aligned}$$

Therefore, $\text{RHS}=\text{LHS}$. □

Problem 1.6.6. Verify that $\mu(x, y) = (x^2 + y^2)^{-1}$ is an integrating factor of a differential equation of the form

$$[y + xf(x^2 + y^2)]dx + [yf(x^2 + y^2) - x]dy = 0.$$

Hence, solve the differential equation

$$[y + x(x^2 + y^2)^2]dx + [y(x^2 + y^2)^2 - x]dy = 0.$$

Solution. • Let $M := y + xf(x^2 + y^2)$ and $N := yf(x^2 + y^2) - x$, then we have

$$\mu M = \frac{y + xf(x^2 + y^2)}{x^2 + y^2}, \quad \mu N = \frac{yf(x^2 + y^2) - x}{x^2 + y^2}.$$

Let $u = x^2 + y^2$, so $\frac{\partial u}{\partial x} = 2x$ and $\frac{\partial u}{\partial y} = 2y$. Hence, we have

$$\begin{aligned}\frac{\partial}{\partial y}(\mu M) &= \frac{\partial}{\partial y} \left(\frac{y + xf(u)}{u} \right) = \frac{u - 2y^2 + 2xy(f'(u)u - f(u))}{u^2} = \frac{x^2 - y^2 + 2xy(f'(u)u - f(u))}{u^2}, \\ \frac{\partial}{\partial x}(\mu N) &= \frac{\partial}{\partial x} \left(\frac{yf(u) - x}{u} \right) = \frac{2xy(f'(u)u - f(u)) - u + 2x^2}{u^2} = \frac{x^2 - y^2 + 2xy(f'(u)u - f(u))}{u^2}.\end{aligned}$$

Therefore, $\mu(x, y) = (x^2 + y^2)^{-1}$ is indeed an integrating factor.

• This is a special case where $f(u) = u^2$. The solution to the differential equation is

$$\arctan\left(\frac{x}{y}\right) + \frac{1}{4}(x^2 + y^2)^2 = C. \quad \square$$

Problem 1.6.7. Suppose that $\mu = \mu(x, y)$ is an integrating factor of the differential equation of

$$M(x, y)dx + N(x, y)dy = 0$$

such that

$$d\Phi(x, y) = \mu(x, y)M(x, y)dx + \mu(x, y)N(x, y)dy.$$

Show that $\mu(x, y)g[\Phi(x, y)]$ is also an integrating factor of this equation, where $g(t)$ is any one differentiable non-zero function.

Proof. From

$$\mu g(\Phi(x, y))P(x, y) dx + \mu g(\Phi(x, y))Q(x, y) dy = g(\Phi(x, y))d\Phi(x, y) = dG(\Phi(x, y))$$

we know $G(s) = \int g(s) ds$. Proof is complete. $\square$

Problem 1.6.8. Suppose that $M(x, y) dx + N(x, y) dy = 0$ is exact and has an integrating factor $\mu(x, y)$, where $\mu(x, y)$ is not a constant. Show that $\mu(x, y) = c$ is a solution of this equation.

Proof. Note that

$$\frac{\partial(\mu M)}{\partial y} = \frac{\partial(\mu N)}{\partial x}, \quad \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x},$$

we have

$$M \frac{\partial \mu}{\partial y} = N \frac{\partial \mu}{\partial x}.$$

Therefore,

$$d\mu = \frac{\partial \mu}{\partial x} dx + \frac{\partial \mu}{\partial y} dy = 0. \quad \square$$

Problem 1.6.9. If $\mu_1(x)$ and $\mu_2(x)$ are two integrating factors of $M(x, y) dx + N(x, y) dy = 0$ such that their ratio is not a constant, then $\mu_1(x, y) = c\mu_2(x, y)$ is a solution of this equation.

Proof. Note that the equation

$$\mu_2 M dx + \mu_2 N dy$$

is exact, and has an integrating factor $\mu_1 \mu_2^{-1}$, so $\mu_1 \mu_2^{-1} = c$ is a solution of $\mu_1 M dx + \mu_1 N dy = 0$ by Problem 1.6.8, i.e., the solution of $M(x, y) dx + N(x, y) dy = 0$. $\square$

Problem 1.6.10. Show the following:

(1) $u(x, y) = c$ is the general solution of the differential equation

$$M(x, y) dx + N(x, y) dy = 0$$

if and only if

$$M(x, y) \frac{\partial u}{\partial y} = N(x, y) \frac{\partial u}{\partial x}.$$

(2) If $\frac{\partial M}{\partial x} = \frac{\partial N}{\partial y}$, $\frac{\partial M}{\partial y} = -\frac{\partial N}{\partial x}$, then the equation $M(x, y) dx + N(x, y) dy = 0$ has an integrating factor

$$\mu(x, y) = (M^2 + N^2)^{-1}.$$

Proof. (1) If $u(x, y) = c$ is the general solution of the differential equation $M(x, y) dx + N(x, y) dy = 0$, then along the solution curve, we have $du = \frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy = 0$, which implies $\frac{dy}{dx} = -\frac{\partial u / \partial x}{\partial u / \partial y}$. And from the original equation, we get $\frac{dy}{dx} = -\frac{M}{N}$. Combining

these, we obtain $M \frac{\partial u}{\partial y} = N \frac{\partial u}{\partial x}$. Conversely, if this equation holds, then $u(x, y) = c$ is the general solution.

- (2) When $\frac{\partial M}{\partial x} = \frac{\partial N}{\partial y}$ and $\frac{\partial M}{\partial y} = -\frac{\partial N}{\partial x}$, it is verified that $\mu = (M^2 + N^2)^{-1}$ satisfies $\frac{\partial}{\partial y}(\mu M) = \frac{\partial}{\partial x}(\mu N)$. The calculation yields:

$$\begin{aligned}\frac{\partial}{\partial y}(\mu M) &= \frac{1}{(M^2 + N^2)^2} (M^2 \frac{\partial N}{\partial x} - 2MN \frac{\partial M}{\partial x} - N^2 \frac{\partial N}{\partial x}) \\ \frac{\partial}{\partial x}(\mu N) &= \frac{1}{(M^2 + N^2)^2} (M^2 \frac{\partial N}{\partial x} - 2MN \frac{\partial M}{\partial x} - N^2 \frac{\partial N}{\partial x}).\end{aligned}$$

The two are equal, so μ is an integrating factor. □

Exercise 1.7

Solve the following differential equations.

Problem 1.7.1. $\frac{dy}{dx} = x^2 + 4xy + 4y^2$.

Solution. Rewrite the original equation as $\frac{dy}{dx} = (x + 2y)^2$. Let $t := x + 2y \Rightarrow \frac{dy}{dx} = \frac{1}{2} \frac{dt}{dx} - \frac{1}{2} \Rightarrow \frac{dt}{dx} = 2t^2 + 1 \Rightarrow \frac{1}{\sqrt{2}} \arctan(\sqrt{2}t) = x + c \Rightarrow y = \frac{1/\sqrt{2} \tan(\sqrt{2}(x+c)) - x}{2}$. □

Problem 1.7.2. $\frac{dy}{dx} = \cos^2(x - y)$.

Solution. Let $u := x - y \Rightarrow \frac{dy}{dx} = 1 - \frac{du}{dx} \Rightarrow \frac{du}{dx} = \sin^2 u$. When $\sin^2 u \neq 0$, we have

$$\tan u = -\frac{1}{x+c} \iff y = x + \arctan \frac{1}{x+c}.$$

Note that $y = k\pi + x, k \in \mathbb{Z}$ are the particular solutions as well. □

Problem 1.7.3. $\frac{dy}{dx} = 2xe^{x+y} - 1$.

Solution. Let $u := x + y \Rightarrow \frac{dy}{dx} = \frac{du}{dx} - 1 \Rightarrow \frac{du}{dx} = 4x \tan u$. Therefore, we have

$$\frac{du}{e^u} = 2x dx \iff -e^{-u} = x^2 + C \iff e^{-(x+y)} = -x^2 - C. \quad \square$$

Problem 1.7.4. $\frac{dy}{dx} = 4x \tan(x + y) - 1$.

Solution. Let $u := x + y \Rightarrow \frac{dy}{dx} = \frac{du}{dx} - 1 \Rightarrow \frac{du}{dx} = 4x \tan u$. When $\tan u \neq 0$, we have

$$\frac{du}{\tan u} = 4x dx \iff \ln |\sin u| = 2x^2 + C \iff \ln |\sin(x+y)| = 2x^2 + C.$$

Note that $y + x = k\pi, k \in \mathbb{Z}$ are the particular solutions as well. □

Problem 1.7.5. $\frac{dy}{dx} = -\frac{y}{x} - x^2y^2 - 4$.

Solution. Setting $u := xy$ gives

$$\frac{du}{dx} = -(u^2 + 4)x \iff u = -2 \tan(x^2 - 2C) \iff y = -\frac{2 \tan(x^2 - 2C)}{x}. \quad \square$$

Problem 1.7.6. $xy' + y = \sin^2(xy)$.

Solution. Let $xy = t \implies \frac{dy}{dx} = \frac{1}{x} \frac{dt}{dx} - \frac{t}{x^2} \implies \frac{dt}{dx} = \sin^2 t$. When $\sin^2 t \neq 0$, $y = -\frac{1}{x} \arctan \frac{1}{x+c}$. Note that $xy = k\pi, k \in \mathbb{Z}$ are the particular solutions as well. $\square$

Problem 1.7.7. $y' = x \left(x + \frac{y}{x}\right)^2 + \frac{y}{x}$.

Solution. By substituting $v := x + \frac{y}{x}$, the original differential equation $y' = x \left(x + \frac{y}{x}\right)^2 + \frac{y}{x}$ can be converted to $v' = v^2 + 1$, which is an equation for separable variables. The solution is $\arctan(v) = x + C$, i.e. $v = \tan(x + C)$. Substituting v back to the original variable gives the general solution $y = x \tan(x + C) - x^2$, where C is an arbitrary constant. $\square$

Problem 1.7.8. $xy' - x^2e^{y/x} - y = 0$.

Solution. Let $y = tx \implies \frac{dy}{dx} = t + x \frac{dt}{dx} \implies \frac{dt}{dx} - e^t = 0$. Therefore, the solution is $\frac{dt}{e^t} = dx \implies y = -x \ln(-x - c)$. $\square$

Problem 1.7.9. $y' = \frac{y}{2x} + \frac{1}{2y} \tan \frac{y^2}{x}$.

Solution. Let $u := y^2/x$, then $2y \frac{dy}{dx} = u + x \frac{du}{dx} \implies xdu = \tan u dx$. If $\tan u \neq 0$, then $\ln |\sin u| = \ln |x| + C \iff \ln \left| \sin \left(\frac{y^2}{x} \right) \right| = \ln |x| + C$. When $\tan u = 0$, note that $y^2 = x \cdot k\pi (k \in \mathbb{Z})$ is a particular solution. $\square$

Problem 1.7.10. $y' = 6x\sqrt{x^4 + y}$.

Solution. By replacing $v := \frac{\sqrt{x^4 + y}}{x^2}$ with a variable, Transform the original differential equation $y' = 6x\sqrt{x^4 + y}$ into the separable variable form $\frac{v}{3v + 2 - 2v^2} dv = \frac{1}{x} dx$. Integrating both sides resulting from the integral $(2v + 1)(2 - v)x^4 = Cx^{-10}$, where C is the integration constants. Substitute v back into the original variable and organize it to obtain the general solution $\left(2\sqrt{x^4 + y} + x^2\right) \left(2x^2 - \sqrt{x^4 + y}\right)^4 = C$, where C is any constant. $\square$

Problem 1.7.11. $y' = \frac{xy^2 + y}{x^2y - x}$.

Solution. By making the substitution $u := xy$, the original differential equation can be transformed into the separable form $\frac{u-1}{u^2} du = \frac{2}{x} dx$. Integrating both sides yields $\ln |u| + \frac{1}{u} = 2 \ln |x| + C$, where C is the constant of integration. Substituting back $u = xy$ and simplifying, the general solution is obtained as $\ln \left| \frac{y}{x} \right| + \frac{1}{xy} = C$ or equivalently $\frac{y}{x} \cdot e^{\frac{1}{xy}} = K$, where K is an arbitrary constant. $\square$

Problem 1.7.12. $y' = 3xe^{xy} - \frac{y}{x}$.

Solution. By making the substitution $u := xy$, the original differential equation can be transformed into the separable form $e^{-u} du = 3x^2 dx$. Integrating both sides yields $-e^{-u} = x^3 + C$, where C is the constant of integration. Substituting $u = xy$ back in, we obtain the general solution $e^{-xy} = -x^3 - C$ or equivalently $xy = -\ln(-x^3 + C)$, where C is an arbitrary constant. $\square$

Problem 1.7.13. $\frac{dy}{dx} = 2\sqrt{y} - 2x$.

Solution. Let $\sqrt{y} = t$, then we have $\frac{dt}{dx} = 1 - 2\frac{x}{t}$ if $t \neq 0$. Therefore, let $t/x := k$, we can get that $\frac{dk}{1 - 1/k - k} = \frac{1}{x} dx \implies -\frac{1}{2} \ln(k^2 - k + 1) - \frac{\arctan\left(\frac{2k-1}{\sqrt{3}}\right)}{\sqrt{3}} = \ln|x| + C$ if $k \neq 1$. Note that $y = x^2 (x \geq 0)$ is a particular solution. $\square$

Problem 1.7.14. $\frac{dy}{dx} = e^{-2y} \cos x - 2$.

Solution. By making the substitution $u := e^{2y}$, the original differential equation $\frac{dy}{dx} = e^{-2y} \cos x - 2$ is transformed into the first-order linear differential equation $\frac{du}{dx} + 4u = 2 \cos x$. Solving it using the integrating factor method yields $u = \frac{8}{17} \cos x + \frac{2}{17} \sin x + Ce^{-4x}$, where C is the constant of integration. Substituting $u = e^{2y}$ back in, we obtain the general solution $e^{2y} = \frac{8}{17} \cos x + \frac{2}{17} \sin x + Ce^{-4x}$ or equivalently $y = \frac{1}{2} \ln\left(\frac{8}{17} \cos x + \frac{2}{17} \sin x + Ce^{-4x}\right)$, where C is an arbitrary constant. $\square$

Problem 1.7.15. $\frac{dy}{dx} = \frac{1}{2x + x^2 y}$.

Solution. Viewing the original equation as a differential equation of x with respect to y :

$$\frac{dx}{dy} = 2x + x^2 y,$$

it is a Bernoulli equation. Solving it by the method described in Remark 1.4, the final result is $x = -\frac{4e^{2y}}{2e^{2y}y - e^{2y} - 4C}$. $\square$

Problem 1.7.16. $\frac{dy}{dx} = \frac{y}{x + y^2 \sin y}$.

Solution. It is noted that $y \equiv 0$ is a particular solution. If $y \neq 0$, consider the original equation as a differential equation of x with respect to y and make the substitution $u := x/y$ to transform it into a separable variable equation $du = \sin y dy$. Thus, the solution is obtained as $\frac{x}{y} = -\cos y + C$. $\square$

Problem 1.7.17. $\frac{dy}{dx} = \frac{x^4 y}{x^5 - y^5}$.

Solution. Firstly, we have $\frac{dx}{dy} = \frac{x}{y} - \frac{y^2}{x^4}$. Then let $x = yt$, so we have $\frac{1}{5}t^5 = \frac{1}{2y^2} + C \implies x^5 = \frac{1}{2}y^3 + 5cy^5$. $\square$

Problem 1.7.18. $\frac{dy}{dx} = \frac{y}{e^{2x} - y}$.

Solution. 错题, 没有初等表达. 一个特殊函数解是 $2\text{Ei}(-2y(x)) + e^{-2(y(x)+x)} + 2C = 0$. 其中

$$\text{Ei}(x) := -\int_{-x}^{+\infty} \frac{1}{e^{tt}} dt. \quad (1.10)$$

□

Remark 1.8. 通过变量替换 $u = e^{2x}$, 将原微分方程 $\frac{dy}{dx} = \frac{y}{e^{2x} - y}$ 转化为 Bernoulli 方程 $\frac{du}{dy} + 2u = \frac{2u^2}{y}$. 进一步令 $v = u^{-1}$, 得到一阶线性微分方程 $\frac{dv}{dy} - 2v = -\frac{2}{y}$, 利用积分因子法求解后, 得到隐式解 $C = e^{-2(x+y)} + 2\text{Ei}(-2y)$, 其中 C 为任意常数, Ei 为指数积分函数.

Problem 1.7.19. $\frac{dy}{dx} = \frac{3x^2y^2 + 1}{2x^3y + e^y}$.

Solution. Rewrite the differential equation $\frac{dy}{dx} = -\frac{3x^2y^2 + 1}{2x^3y + e^y}$ in standard form:

$$(3x^2y^2 + 1)dx + (2x^3y + e^y)dy = 0.$$

Let $M(x, y) := 3x^2y^2 + 1$ and $N(x, y) := 2x^3y + e^y$, calculate the partial derivatives:

$$\frac{\partial M}{\partial y} = 6x^2y = \frac{\partial N}{\partial x},$$

then this equation is an exact equation. Therefore, the general solution of the differential equation is: $x^3y^2 + x + e^y = C$ where C is an arbitrary constant. □

Problem 1.7.20. $\frac{dy}{dx} = \frac{2x^2 - y^2}{1 + 2xy}$.

Solution. Rewrite it in standard form:

$$(2x^2 - y^2)dx - (1 + 2xy)dy = 0.$$

Let $M(x, y) := 2x^2 - y^2$ and $N(x, y) := -1 - 2xy$. Verify the exactness.

$$\frac{\partial M}{\partial y} = -2y, \quad \frac{\partial N}{\partial x} = -2y.$$

Therefore, the general solution is:

$$\frac{2}{3}x^3 - xy^2 - y = C \quad \text{or} \quad 2x^3 - 3xy^2 - 3y = C_1. \quad \square$$

Problem 1.7.21. $\frac{dy}{dx} = \frac{x^3 - 2y}{2x + y^2}$.

Solution. Rewrite it as:

$$(x^3 - 2y)dx - (2x + y^2)dy = 0.$$

Let $M(x, y) := x^3 - 2y$ and $N(x, y) := -2x - y^2$. Verify the exactness.

$$\frac{\partial M}{\partial y} = -2, \quad \frac{\partial N}{\partial x} = -2$$

This equation is an exact equation. Hence, the general solution is:

$$\frac{1}{4}x^4 - 2xy - \frac{1}{3}y^3 = C \quad \text{or} \quad 3x^4 - 24xy - 4y^3 = C_1,$$

where C and C_1 are arbitrary constants. □

Problem 1.7.22. $dx - (xy + x^2y^2)dy = 0$.

Solution. The differential equation $dx - (xy + x^2y^2)dy = 0$ is a Bernoulli equation. By substituting $v = \frac{1}{x}$, it is transformed into the linear equation $\frac{dv}{dy} + yv = -y^2$. Solving it with

the integrating factor $e^{\frac{y^2}{2}}$, the general solution is obtained as $x = \frac{e^{\frac{y^2}{2}}}{C - ye^{\frac{y^2}{2}} + \int e^{\frac{y^2}{2}} dy}$, where $\int e^{\frac{y^2}{2}} dy$ cannot be expressed in terms of elementary functions. □

Problem 1.7.23. $(x^2 + y^2)dx + (4xy + y^2 - x^2)dy = 0$.

Solution. Let $y = kx$, then we have $k - 2 \arctan k + 2 \ln(k^2 + 1) = -x + C \implies \frac{y}{x} - 2 \arctan \frac{y}{x} + 2 \ln(y^2/x^2 + 1) = \ln|x| + C$. □

Problem 1.7.24. $\frac{dy}{dx} = \frac{x + 2e^y + 3}{y - 2xe^y}$.

Solution. The differential equation can be rewritten as $(x + 2e^y + 3)dx - (y - 2xe^y)dy := M(x, y)dx + N(x, y)dy = 0$. Verifying that $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} = 2e^y$ shows that it is an exact differential equation. By solving the exact equation, the general solution is obtained as $x^2 + 4xe^y + 6x - y^2 = C$, where C is an arbitrary constant. □

Problem 1.7.25. $x(x^2 + 4y^2)dx + y(3x^2 + y^2)dy = 0$.

Solution. The differential equation is a homogeneous equation. By substituting $y = vx$, it can be transformed into a separable variable equation. After integration, the general solution is obtained as $3 \ln|x^4 + 7x^2y^2 + y^4| - \frac{1}{\sqrt{5}} \ln \left| \frac{(7 - 3\sqrt{5})x^2 + 2y^2}{(7 + 3\sqrt{5})x^2 + 2y^2} \right| = C$, where C is an arbitrary constant. □

Problem 1.7.26. $\frac{dy}{dx} = \frac{2x + 3y}{3x - 2y}$.

Solution. • (Method 1) Using the transformation $y = ux$: Let $y = ux$, then $\frac{dy}{dx} = u + x \frac{du}{dx}$. Substituting this into the original equation gives $u + x \frac{du}{dx} = \frac{2 + 3u}{3 - 2u}$, which simplifies to $\frac{3 - 2u}{u^2 + 1} du = \frac{2}{x} dx$. Integrating yields $3 \arctan(u) - \ln(u^2 + 1) = 2 \ln|x| + C$. Substituting back $u = \frac{y}{x}$ gives the general solution $3 \arctan\left(\frac{y}{x}\right) - \ln(x^2 + y^2) = C$.

- (Method 2) Using polar coordinate substitution: Let $x = r \cos \theta$ and $y = r \sin \theta$. Then, $\frac{dy}{dx} = \frac{\sin \theta dr + r \cos \theta d\theta}{\cos \theta dr - r \sin \theta d\theta}$. The right side of the original equation becomes $\frac{2 \cos \theta + 3 \sin \theta}{3 \cos \theta - 2 \sin \theta}$. Cross-multiplying and simplifying both sides of the equation gives $3r d\theta = 2 dr$. Separating variables and integrating yields $3\theta = 2 \ln |r| + C$. Substituting $\theta = \arctan\left(\frac{y}{x}\right)$ and $r = \sqrt{x^2 + y^2}$, we obtain the same general solution $3 \arctan\left(\frac{y}{x}\right) - \ln(x^2 + y^2) = C$. □

Problem 1.7.27. $\frac{dy}{dx} = \frac{x(ax^2 + by^2)}{y(bx^2 - ay^2)}$.

Solution. • (Method 1) Let $y = ux$, then $\frac{dy}{dx} = u + x \frac{du}{dx}$. Substituting this into the original equation yields

$$x \frac{du}{dx} = \frac{a(1 + u^4)}{u(b - au^2)}.$$

Integrating both sides yields

$$\frac{b}{2} \arctan(u^2) - \frac{a}{4} \ln |1 + u^4| = a \ln |x| + C.$$

Substitute $u = \frac{y}{x}$:

$$\frac{b}{2} \arctan\left(\frac{y^2}{x^2}\right) - \frac{a}{4} \ln \left| \frac{x^4 + y^4}{x^4} \right| = a \ln |x| + C.$$

The simplified general solution is:

$$\frac{b}{2} \arctan\left(\frac{y^2}{x^2}\right) - \frac{a}{4} \ln(x^4 + y^4) = C.$$

- (Method 2) Let $x = r \cos \theta$ and $y = r \sin \theta$, then:

$$\begin{aligned} dx &= \cos \theta dr - r \sin \theta d\theta, & dy &= \sin \theta dr + r \cos \theta d\theta; \\ \frac{dy}{dx} &= \frac{\sin \theta dr + r \cos \theta d\theta}{\cos \theta dr - r \sin \theta d\theta}. \end{aligned}$$

The right side of the original equation is transformed as follows:

$$\frac{x(ax^2 + by^2)}{y(bx^2 - ay^2)} = \frac{\cos \theta (a \cos^2 \theta + b \sin^2 \theta)}{\sin \theta (b \cos^2 \theta - a \sin^2 \theta)}.$$

After cross-multiplying and simplifying both sides of the equation, after a series of complex calculations, we obtain:

$$\frac{dr}{r} = \frac{\sin(2\theta)(a \cos(2\theta) + b)}{2a(1 + \cos^2(2\theta))} d\theta$$

By substituting $u := \cos(2\theta)$ for solving, the final result is the same as that obtained from the transformation $y = ux$.

$$\frac{b}{2} \arctan\left(\frac{y^2}{x^2}\right) - \frac{a}{4} \ln(x^4 + y^4) = C. \quad \square$$

Problem 1.7.28. $x^2y' = x^2y^2 + xy + 1$. (Hint: $y = -\frac{1}{x}$ is a solution)

Solution. This equation is a Riccati equation. By making the substitution $y = -\frac{1}{x} + \frac{1}{v}$, the original equation is transformed into a linear differential equation $v' - \frac{v}{x} = -1$. Using the integrating factor method, the solution is obtained as $v = x(C - \ln|x|)$. Substituting back to the original variables, the general solution is $y = -\frac{1}{x} + \frac{1}{x(C - \ln|x|)} = \frac{1}{x} \left(\frac{1}{C - \ln|x|} - 1 \right)$, where C is an arbitrary constant. $\square$

Remark 1.9. Riccati 方程形如

$$\frac{dy}{dx} = p(x)y^2 + q(x)y + r(x), \quad p(x) \not\equiv 0, \quad (1.11)$$

求解 Riccati 方程(1.11)必须先猜出一个特解 $y := \varphi(x)$, 之后再令 $z = y - \varphi(x)$, 将其转化为 $n = 2$ 的 Bernoulli 方程. 而求解 $n = 2$ 的 Bernoulli 方程需要作变换 $y \mapsto y^{-1}$, 所以我们直接令 $y = -\frac{1}{x} + \frac{1}{v}$ 一步到位.

Problem 1.7.29. Show that the second-order differential equation

$$y'' + p(x)y' + q(x)y = 0$$

can be transformed into the Riccati's equation.

Proof. Let $w := \frac{y'}{y}$, then $w' = \frac{y''y - (y')^2}{y^2}$, and further $\frac{y''}{y} = w' + \frac{(y')^2}{y^2}$. Substituting this into the original equation and simplifying leads to $w' = -q(x) - p(x)w - w^2$, which is a Riccati equation. $\square$

Problem 1.7.30. Show that the Riccati's equation

$$y' + p(x)y + q(x)y^2 = h(x)$$

with $p \in C^0(I)$, $q \in C^1(I)$, $q(x) \neq 0$ in I , can be transformed into a second-order linear differential equation

$$u'' + u' \left(p - \frac{q'}{q} \right) - hqu = 0$$

using the transformation

$$u(x) = \exp \left[\int q(x)y(x) dx \right].$$

Proof. Taking the logarithmic derivative¹ gives $\frac{u'}{u} = qy$, so $y = \frac{u'}{qu}$. Differentiating again yields $y' = \frac{u''}{qu} - \frac{u'q'}{q^2u} - \frac{(u')^2}{qu^2}$. Substituting these expressions into the Riccati's equation and simplifying (noting that the $(u')^2$ terms cancel), we obtain $\frac{u''}{qu} - \frac{u'q'}{q^2u} + \frac{pu'}{qu} = h$. Multiplying through by qu and rearranging terms gives the second-order linear differential equation $u'' + u' \left(p - \frac{q'}{q} \right) - hqu = 0$. $\square$

¹对数求导法.

Exercise 1.8

Find the general solutions of the following equations.

Problem 1.8.1. $(y')^2 - 2y^2 = 0$.

Solution. Obviously, $y = e^{\sqrt{2}x} + c$ or $y = e^{-\sqrt{2}x} + c$. □

Problem 1.8.2. $3x^2 - (y')^2 = 1$.

Solution. Factorization yields $(y' + \sqrt{3x^2 - 1})(y' - \sqrt{3x^2 - 1}) = 0$. Then the general solution is $y = \pm \left(\frac{1}{2} \sqrt{3x^2 - 1} x - \frac{1}{2\sqrt{3}} \ln \left| \frac{\sqrt{3x^2 - 1} + \sqrt{3}x + 1}{\sqrt{3x^2 - 1} - \sqrt{3}x - 1} \right| + C \right)$. □

Problem 1.8.3. $y^2 + 2(y')^2 = 3$.

Solution. • (Method 1) Consider the differential equation $y^2 + 2(y')^2 = 3$. Rewriting it as

$(y')^2 = \frac{3 - y^2}{2}$, we obtain $y' = \pm \sqrt{\frac{3 - y^2}{2}}$. Separating variables and integrating:

$$\sqrt{2} \int \frac{dy}{\sqrt{3 - y^2}} = \pm \int dx.$$

The integral on the left side yields $\sqrt{2} \arcsin \left(\frac{y}{\sqrt{3}} \right)$, and the right side is $\pm x + C$. Thus, the solution is:

$$y = \sqrt{3} \sin \left(\frac{x}{\sqrt{2}} + C \right).$$

Here, C is an arbitrary constant. Additionally, when $y^2 = 3$, $y' = 0$, we obtain the particular solutions $y = \sqrt{3}$ and $y = -\sqrt{3}$. Therefore, the general solution of the equation is $y = \sqrt{3} \sin \left(\frac{x}{\sqrt{2}} + C \right)$, along with the particular solutions $y = \pm \sqrt{3}$.

- (Method 2) Let $y' = p$, then the original equation becomes $y^2 + 2p^2 = 3$, and the solution is $p = \pm \sqrt{\frac{3 - y^2}{2}}$. Since $p = \frac{dy}{dx}$, we obtain $\frac{dy}{dx} = \pm \sqrt{\frac{3 - y^2}{2}}$. After separation of variables and integration:

$$\sqrt{2} \int \frac{dy}{\sqrt{3 - y^2}} = \pm \int dx,$$

we get $\sqrt{2} \arcsin \left(\frac{y}{\sqrt{3}} \right) = \pm x + C$. After simplification, the general solution is $y = \sqrt{3} \sin \left(\frac{x}{\sqrt{2}} + C \right)$, where C is an arbitrary constant. Additionally, when $y^2 = 3$, $p = 0$, resulting in the particular solution $y = \pm \sqrt{3}$. □

Remark 1.10. 容易看出上面两种方法没有本质区别.

Remark 1.11. 对于此类方程

$$x = f(y, y'). \quad (1.12)$$

令 $\frac{dy}{dx} = p$, 等号两边对 y 求导, 并将 $\frac{dx}{dy} = \frac{1}{p}$ 代入, 得

$$\frac{1}{p} = \frac{\partial f}{\partial y} + \frac{\partial f}{\partial p} \cdot \frac{dp}{dy}.$$

这是关于 y, p 的一阶微分方程.

Problem 1.8.4. $2y - x^2 + (y')^2 = 0$.

Solution. Let $y' = p$, and differentiate both sides of the equation with respect to x . This transforms the original equation into $p \frac{dp}{dx} = x - p$, which is a homogeneous equation. Using the substitution $v = \frac{p}{x}$, substituting it in gives the separable form $\frac{v}{1-v-v^2} dv = \frac{1}{x} dx$. After integrating both sides and restoring the variables, the final implicit solution is $-\frac{1}{2} \ln |x^2 - px - p^2| - \frac{1}{2\sqrt{5}} \left(\frac{2p+x}{\sqrt{5}x} \right) = C$, where C is the integration constant. Therefore, we have $y = \frac{x^2 - p^2}{2}$. $\square$

Remark 1.12. 实际上可以直接写出隐式通解:

$$\begin{aligned} & -\ln(\sqrt{x^2 - 2y} - x) - \frac{\sqrt{5} - 5}{10} \ln(x\sqrt{x^2 - 2y} - x^2 + (\sqrt{5} - 1)y) \\ & \quad + \frac{5 + \sqrt{5}}{10} \ln(x\sqrt{x^2 - 2y} - x^2 - (1 + \sqrt{5})y) = C; \\ & -\ln(\sqrt{x^2 - 2y} - x) + \frac{\sqrt{5} + 5}{10} \ln(x\sqrt{x^2 - 2y} - x^2 + (3 - \sqrt{5})y) \\ & \quad + \frac{5 - \sqrt{5}}{10} \ln(x\sqrt{x^2 - 2y} - x^2 + (3 + \sqrt{5})y) = C. \end{aligned}$$

Remark 1.13. 方程

$$y = f(x, y'). \quad (1.13)$$

令 $p := y'$, 两边对 x 求导, 并将 $p = \frac{dy}{dx}$ 代入, 得

$$p = \frac{\partial f}{\partial x} + \frac{\partial f}{\partial p} \cdot \frac{dp}{dx}, \quad (1.14)$$

这是关于 x, p 的一阶微分方程.

Problem 1.8.5. $y - (y')^3 = 0$.

Solution. Obviously, we have $y = \frac{2}{3} \sqrt{\frac{2}{3}} (x + c)^{3/2}$. $\square$

Problem 1.8.6. $y = (y' - 2)e^{y'}$.

Solution. Let $p = y'$, then the equation becomes $y = (p - 2)e^p$. Differentiate with respect to p and use $dy = p dx$ to obtain $\frac{dx}{dp} = \left(1 - \frac{1}{p}\right) e^p$. Integrate to get $x = e^p - \text{Ei}(p) + C$ (where

$\text{Ei}(p)$ is the exponential integral function²). Therefore, the general solution of the equation in parametric form is: $x = e^p - \text{Ei}(p) + C$, $y = (p - 2)e^p$, where C is an arbitrary constant. $\square$

Problem 1.8.7. $y^2(y' - 2) = (4 - y')^2$.

Solution. Let $p = y'$, we obtain $y^2(p - 2) = (4 - p)^2$. Solving this, we get $y = \pm \frac{4 - p}{\sqrt{p - 2}}$ ($p > 2$). Using the parameter method, we find $x = \pm \frac{1}{\sqrt{p - 2}} + C$. After eliminating the parameter p , the general solution is obtained: $y = 2(x - C) - \frac{1}{x - C}$ and $y = -2(x - C) + \frac{1}{x - C}$, where C is an arbitrary constant. $\square$

Problem 1.8.8. $x^3 + (y')^3 - xy' = 0$.

Solution. Let $p = y'$, then the equation becomes

$$x^3 + p^3 - xp = 0. \quad (1.15)$$

Let $p = tx$, then the equation can be expressed as

$$x = \frac{t}{1 + t^3}, \quad p = \frac{t^2}{1 + t^3}.$$

From $dy = p dx$, we obtain

$$dy = \frac{t^2(1 - 2t^3)}{(1 + t^3)^3} dt.$$

Points earned

$$y = \frac{1 + 4t^3}{6(1 + t^3)^2} + C.$$

Thus, the general solution of the original equation is

$$x = \frac{t}{1 + t^3}, \quad y = \frac{1 + 4t^3}{6(1 + t^3)^2} + C,$$

Here, t is a parameter and C is an arbitrary constant. $\square$

Problem 1.8.9. $(y')^2 - 2xy' + y = 0$.

Solution. Let $p := y'$, then the equation becomes $y = 2xp - p^2$. Differentiate with respect to x to obtain $p = 2p + 2x \frac{dp}{dx} - 2p \frac{dp}{dx}$, which simplifies to $0 = p + 2(x - p) \frac{dp}{dx}$. This equation has solutions $p = 0$ (corresponding to $y = 0$) or $\frac{dp}{dx} = \frac{p}{2(p - x)}$. For the latter, let $v = \frac{x}{p}$, then $p \frac{dv}{dp} = 2 - 3v$, separate variables and integrate to get $2 - 3\frac{x}{p} = Kp^{-3}$ (K is a constant). Solve for $x = \frac{2}{3}p - \frac{K}{3p^2}$, substitute into $y = 2xp - p^2$ to obtain $y = \frac{1}{3}p^2 - \frac{2K}{3p}$. Therefore, the general solution is given by $y = 0$ and the parametric equations $x = \frac{2}{3}t - \frac{C}{3t^2}$, $y = \frac{1}{3}t^2 - \frac{2C}{3t}$ (t is the parameter, C is an arbitrary constant). $\square$

²它的定义见(1.10)

Problem 1.8.10. $y = xy' + x^2(y')^4$.

Solution. Let $p := y'$, then we have $y = xp + x^2p^4$. Differentiate with respect to x and simplify. Introduce the parameter $u = xp^3$, and simplify to obtain a separable variable equation. The solution is $xp^3 = Cp + \frac{1}{2}$. Substitute this into the original equation to obtain the parameter solution:

$$x = \frac{C}{p^2} + \frac{1}{2p^3}, \quad y = \frac{2C}{p} + \frac{3}{4p^2} + C^2.$$

Here, C is an arbitrary constant and p is a parameter. Additionally, $y = 0$ is a singular solution of the equation. Verification shows that all these solutions satisfy the original equation. $\square$

Problem 1.8.11. Determine all the solutions for the following Clairaut's differential equations in explicit form.

(1) $y = xy' + (y')^2$.

(2) $y = 2xy' - \sqrt{y' - 1}$.

Solution. (1) The general solution is $y = Cx + C^2$. The singular solution is $y = -\frac{x^2}{4}$.

(2) 错题. 这不是 Clairaut 方程. 不过它可以写成参数的形式. We solve the differential equation

$y = 2xy' - \sqrt{y' - 1}$ using the parameter method. First, let $p = y' = \frac{dy}{dx}$ (note that $p \geq 1$ to ensure the square root is non-negative), and rewrite the original equation as $y = 2xp - \sqrt{p - 1}$. Differentiate with respect to x and use $\frac{dy}{dx} = p$, resulting in $p =$

$2p + 2x \frac{dp}{dx} - \frac{1}{2\sqrt{p-1}} \frac{dp}{dx}$, which simplifies to $-p = \left(2x - \frac{1}{2\sqrt{p-1}}\right) \frac{dp}{dx}$. Considering this equation as a function of x with respect to p , it transforms into the standard linear form $\frac{dx}{dp} + \frac{2}{p}x = \frac{1}{2p\sqrt{p-1}}$. Calculate the integrating factor $\mu(p) = e^{\int \frac{2}{p} dp} = p^2$, and

multiply both sides by it to obtain $\frac{d}{dp}(xp^2) = \frac{p}{2\sqrt{p-1}}$. Integrate the right side, setting

$t = \sqrt{p-1}$ (i.e., $p = t^2 + 1$, $dp = 2t dt$), then $\int \frac{p}{2\sqrt{p-1}} dp = \int (t^2 + 1) dt = \frac{t^3}{3} + t + C =$

$\frac{(p-1)^{3/2}}{3} + \sqrt{p-1} + C$. Therefore, $xp^2 = \frac{(p+2)\sqrt{p-1}}{3} + C$, and solving for x gives

$x = \frac{(p+2)\sqrt{p-1} + 3C}{3p^2}$. Substitute x back into the original equation $y = 2xp - \sqrt{p-1}$,

simplifying to $y = \frac{(4-p)\sqrt{p-1} + 6C}{3p}$. Finally, we obtain the general solution in terms of

the parameter p , where $p \geq 1$ and C is an arbitrary constant. $\square$

Remark 1.14. Clairaut 微分方程

$$y = xy' + f(y') \quad (1.16)$$

的通解为

$$y = Cx + f(C). \quad (1.17)$$

当 $x + f'(p) = 0$ 时, 得到 Clairaut 方程的一个奇异解

$$x = -f'(p), \quad y = -f'(p)p + f(p), \quad (1.18)$$

其中 p 当作参数.

Problem 1.8.12. Prove that all solutions of the differential equation

$$y = xy' + ay' + b \quad (a, b \text{ are constants})$$

are lines through a fixed point.

Proof. Setting $p := y'$ gives $y = (x + a)p + b$, and differentiating both sides with respect to x yields $p = p + (x + a)\frac{dp}{dx}$, simplifying to $(x + a)\frac{dp}{dx} = 0$. This implies either $\frac{dp}{dx} = 0$ or $x = -a$. When $\frac{dp}{dx} = 0$, we have $p = c$ (constant), leading to the solution $y = cx + ac + b$, which represents a family of straight lines. Substituting $x = -a$ into this solution gives $y = c(-a) + ac + b = b$, showing that all such lines pass through the fixed point $(-a, b)$. The case $x = -a$ directly yields $y = b$, confirming this same point. Thus, all solutions are straight lines passing through the fixed point $(-a, b)$. $\square$

Problem 1.8.13. Solve Lagrange's equation $y = xf(y') + g(y')$, where $f(p), g(p) \in C^1$.

Solution. Let $p := y'$, then the original equation becomes $y = xf(p) + g(p)$.

Taking the derivative with respect to x yields $p = f(p) + (xf'(p) + g'(p))\frac{dp}{dx}$. After rearrangement, we obtain:

$$(p - f(p)) = (xf'(p) + g'(p))\frac{dp}{dx}.$$

When $p \neq f(p)$, it can be rewritten as a linear equation in terms of $x(p)$:

$$\frac{dx}{dp} - \frac{f'(p)}{p - f(p)}x = \frac{g'(p)}{p - f(p)}.$$

Using the method of integrating factors to solve, the integrating factor is $\mu(p) = e^{-\int \frac{f'(p)}{p - f(p)} dp}$. After multiplying both sides by $\mu(p)$, the left side becomes $\frac{d}{dp}[x \cdot \mu(p)]$. Integrating this gives the expression of x in terms of p , and then substituting $y = xf(p) + g(p)$ yields the parameter solution $(x(p), y(p))$.

Furthermore, if the equation $p = f(p)$ has a constant solution $p = c$, then the straight line $y = cx + g(c)$ is also a solution to the original equation. $\square$

Problem 1.8.14. Determine the solutions of the following differential equations in parametric form.

$$(1) \quad y = 2xy' + y' - (y')^2.$$

$$(2) \quad x = yy' + \frac{1}{y'}.$$

Solution. (1) Let $p = y'$. The original equation $y = 2xy' + y' - (y')^2$ becomes $y = 2xp + p - p^2$.

Differentiating with respect to x , we get $p = 2p + 2x\frac{dp}{dx} + \frac{dp}{dx} - 2p\frac{dp}{dx}$, which simplifies to $0 = p + (2x + 1 - 2p)\frac{dp}{dx}$. When $\frac{dp}{dx} \neq 0$, it can be rewritten as $\frac{dx}{dp} - \frac{2}{p}x = 2 - \frac{1}{p}$, which is a first-order linear differential equation. The integrating factor is p^2 , and the solution

is $xp^2 = \frac{2}{3}p^3 - \frac{1}{2}p^2 + C$, or $x = \frac{2}{3}p - \frac{1}{2} + \frac{C}{p^2}$. Substituting $y = 2xp + p - p^2$, we get $y = \frac{1}{3}p^2 + \frac{2C}{p}$. When $\frac{dp}{dx} = 0$, $p = 0$ gives the particular solution $y = 0$ (which can be regarded as the limit case when $C = 0$); thus, the general solution is given by the parametric equations $x = \frac{2}{3}t - \frac{1}{2} + \frac{C}{t^2}$, $y = \frac{1}{3}t^2 + \frac{2C}{t}$, where t is the parameter and C is an arbitrary constant.

(2) Let $p = y'$, then the original equation $x = yy' + \frac{1}{y'}$ becomes $x = yp + \frac{1}{p}$; Differentiating with

respect to x gives $1 = p^2 + \left(y - \frac{1}{p^2}\right) \frac{dp}{dx}$, which can be rearranged to $\frac{dy}{dp} = \frac{p\left(y - \frac{1}{p^2}\right)}{1 - p^2}$;

This is a first-order linear differential equation, and the integrating factor is $\sqrt{|1 - p^2|}$ (with $|p| < 1$), and the solution is $y\sqrt{1 - p^2} = \ln \left| \frac{1 + \sqrt{1 - p^2}}{p} \right| + C$ (where C is a constant);

Combining with $x = yp + \frac{1}{p}$, the general solution is given by the parametric equations:

$$x = p \cdot \frac{\ln \left| \frac{1 + \sqrt{1 - p^2}}{p} \right| + C}{\sqrt{1 - p^2}} + \frac{1}{p}, \quad y = \frac{\ln \left| \frac{1 + \sqrt{1 - p^2}}{p} \right| + C}{\sqrt{1 - p^2}}.$$

Here, p is a parameter (with $0 < |p| < 1$), and C is an arbitrary constant. □

Exercise 1.9

Solve the following differential equations.

Problem 1.9.1. $(xy''' - y'')^2 = (y''')^2 + 1$.

Solution. Let $p = y''$. Then the original equation $(xy''' - y'')^2 = (y''')^2 + 1$ becomes $(xp' - p)^2 = (p')^2 + 1$. Using the hyperbolic identity $\cosh^2 \theta - \sinh^2 \theta = 1$, we make the substitution $xp' - p = \cosh \theta$ and $p' = \sinh \theta$, resulting in $p = x \sinh \theta - \cosh \theta$. Taking the derivative and combining $p' = \sinh \theta$, we get $(x \cosh \theta - \sinh \theta) \frac{d\theta}{dx} = 0$. Taking $\frac{d\theta}{dx} = 0$ implies $\theta = c$ (a constant), then $p = x \sinh c - \cosh c$. Integrating twice gives $y = \frac{1}{6}x^3 \sinh c - \frac{1}{2}x^2 \cosh c + C_1x + C_2$. Letting $a = \frac{1}{6} \sinh c$ and $b = -\frac{1}{2} \cosh c$, it satisfies $4b^2 = 36a^2 + 1$. Therefore, the general solution is $y = ax^3 + bx^2 + cx + d$, where $4b^2 = 36a^2 + 1$ and c, d are arbitrary constants. □

Problem 1.9.2. $(y'')^2 = (1 + (y')^2)^3$.

Solution. Let $p := y'$, then the equation becomes $\left(\frac{dp}{dx}\right)^2 = (1 + p^2)^3$. Separating variables gives $\frac{dp}{(1 + p^2)^{3/2}} = \pm dx$, and after integration, we obtain $\frac{p}{\sqrt{1 + p^2}} = \pm x + C_1$. Substituting $p = y'$ and integrating yields $y = \mp \sqrt{1 - (x + C_1)^2} + C_2$. After rearrangement, we get $(x + C_1)^2 + (y - C_2)^2 = 1$, which is the general solution $(x - a)^2 + (y - b)^2 = 1$, where a and b are arbitrary constants. □

Problem 1.9.3. $yy'' - (y')^2 = 0$.

Solution. $\frac{y'}{y} = \frac{y''}{y'} \implies \ln y = \ln y' - C_1 \implies y = e^{-C_1} y' \implies y = C_2 e^{C_1 x}$. □

Problem 1.9.4. $yy'' - (y')^2 - y^2 y' = 0$.

Solution. Substituting $y' = u$, $y'' = u' = \frac{du}{dx} = u \frac{du}{dy}$ in the equation yields

$$uy \frac{du}{dy} - u^2 - uy^2 = 0 \iff \frac{ydu - udy}{y^2} = dy \iff \frac{u}{y} = y + c_1.$$

Therefore, we have

$$\frac{dy}{dx} = y(y + c_1) \iff \frac{y}{y + c_1} = c_2 e^{c_1 x} \iff y = -\frac{c_1 e^{c_1 x + c_2 c_1}}{-1 + e^{c_1 x + c_2 c_1}}. \quad \square$$

Problem 1.9.5. $yy'' - (y')^2 - 4xy^2 = 0$.

Solution. When $y \neq 0$, we note that $\left(\frac{y'}{y}\right)' = 4x \iff \frac{y'}{y} = 2x^2 + C_1 \iff \ln y = \frac{2}{3}x^3 + C_1 x + C_2 \iff y = \exp\left(\frac{2}{3}x^3 + C_1 x + C_2\right)$. □

Problem 1.9.6. $y'' - (y')^2 = e^{-y}$.

Solution. 错题. Let us introduce the variable substitution $z = e^{-y}$, from which we obtain $z' = -e^{-y} y'$ and $z'' = e^{-y} (y')^2 - e^{-y} y''$. Substituting the original equation $y'' = (y')^2 + e^{-y}$ into the expression for z'' , we simplify to get $z'' = -z^2$. Performing the first integration of this equation: multiplying both sides by $2z'$ gives $2z'z'' = -2z^2z'$, which simplifies to $\frac{d}{dx}[(z')^2] = -\frac{d}{dx}\left[\frac{2}{3}z^3\right]$. After integration, we obtain $(z')^2 = K - \frac{2}{3}z^3$, where K is the integration constant. From this, we solve for the relationship between x and z : $\frac{dz}{dx} = \pm\sqrt{K - \frac{2}{3}z^3}$, which gives $x = \pm \int \frac{dz}{\sqrt{K - \frac{2}{3}z^3}} + D$. To simplify the constant, let $K = \frac{2}{3}C$ ($C > 0$), then $x = \pm \sqrt{\frac{3}{2}} \int \frac{dz}{\sqrt{C - z^3}} + D$. Combining $y = -\ln z$, taking the positive sign, we obtain the solution with $z > 0$ as a parameter:

$$\begin{cases} x = \sqrt{\frac{3}{2}} \int \frac{dz}{\sqrt{C - z^3}} + D, \\ y = -\ln z, \end{cases}$$

where C and D are arbitrary constants, and it must be that $C > z^3$ to ensure the square root is non-negative. □

Problem 1.9.7. $y''' + \ln y''' = x + y''$.

Solution. Let $u = y''$, then $u' = y'''$, and the original equation becomes $u' + \ln u' = x + u$; let $v = u' > 0$ (ensuring that $\ln v$ is defined), then the equation is written as $v + \ln v = x + u$ and

$v = \frac{du}{dx}$; taking the derivative with respect to x on both sides, we get $\frac{dv}{dx} + \frac{1}{v} \frac{dv}{dx} = 1 + v$, which simplifies to $\frac{dv}{dx} \cdot \frac{v+1}{v} = v+1$; since $v > 0$ and $v \neq -1$, dividing both sides by $v+1$ gives $\frac{1}{v} \frac{dv}{dx} = 1$, which means $\frac{dv}{v} = dx$; integrating yields $\ln v = x + C_1$, so $v = Ce^x$ ($C = e^{C_1} > 0$); substituting $v = \frac{du}{dx}$ gives $\frac{du}{dx} = Ce^x$, integrating gives $u = Ce^x + D$; substituting v and u back into the original equation $v + \ln v = x + u$, we get $Ce^x + \ln(Ce^x) = x + Ce^x + D$, which simplifies to $\ln C + x = x + D$, so $D = \ln C$; therefore, $y'' = u = Ce^x + \ln C$; integrating once gives $y' = Ce^x + (\ln C)x + E$; integrating again gives $y = Ce^x + \frac{1}{2}(\ln C)x^2 + Ex + F$, where $C > 0$, E, F are arbitrary constants, and it can be verified that this solution satisfies the original equation. $\square$

Problem 1.9.8. $2x(yy'' + (y')^2) + 3yy' = 4x^3$. (Hint: $u = y^2$)

Solution. Let $u := y^2$, then $u' = 2yy'$, $u'' = 2(y')^2 + 2yy''$, so we have $xu'' + \frac{3}{2}u' = 4x^3$. Let $z = u'$, $z' = u''$, therefore,

$$z' + \frac{3}{2x}z = 4x^2.$$

This is a first-order non-homogeneous linear differential equation, so by (1.2), we have

$$z = \frac{8}{9}x^3 + c_1x^{-\frac{3}{2}}.$$

Therefore,

$$u = \frac{2}{9}x^4 + c_1x^{-\frac{1}{2}} + c_2.$$

$$\text{So } y = \pm \sqrt{\frac{2}{9}x^4 + c_1x^{-\frac{1}{2}} + c_2}.$$

$\square$

Problem 1.9.9. $2x^2yy'' + (xy' - y)^2 = 0$.

Solution. Let $u := y'/y$, then $y'' = (u' + u^2)y$. Then the original equation is transformed into

$$3x^2u^2 + 2x^2u' + 1 - 2xu = 0 \quad (y \neq 0). \quad (1.19)$$

Note that $y = 0$ is also a solution. (1.19) is a Ricatti equation. Note that $u = \frac{1}{x}$ is a particular solution. Let $u := m + \frac{1}{x}$ be its solution. Substituting in (1.19) gives

$$2x^2m' + 4xm + 3x^2m^2 = 0.$$

Let $z := 1/m$, then

$$-2x^2 \frac{dz}{dx} + 4xz + 3x^2 = 0.$$

So we get

$$z' - \frac{2}{x}z - \frac{3}{2} = 0.$$

By (1.2), we have

$$z = -\frac{3}{2}x + c_1x^2 \implies u = \frac{1}{c_1x^2 - \frac{3}{2}x} + \frac{1}{x}.$$

Hence,

$$y = c_2 \exp \left\{ \int \left(\frac{1}{c_1 x^2 - \frac{3}{2}x} + \frac{1}{x} \right) dx \right\} = c_2 \sqrt[3]{x(x - c_1)^2}. \quad \square$$

Problem 1.9.10. $xy'' + (x^2 - 1)(y' - 1) = 0$.

Solution. Setting $y' - 1 := u$ gives that $xu' + (x^2 - 1)u = 0$, which can be solved easily that $u(x) = c_1 x e^{-\frac{x^2}{2}}$. Therefore, $y = -C_1 e^{-\frac{x^2}{2}} + C_2$. $\square$

Problem 1.9.11. $x^3 y'' + 2xyy' - x^2(y')^2 - y^2 = 0$.

Solution. First, set $y = zx$, resulting in $y' = z'x + z$ and $y'' = z''x + 2z'$. Substitute these into the original equation and simplify. After eliminating the intermediate terms, a simplified equation $xz'' + 2z' - x(z')^2 = 0$ is obtained. Let $w := z'$. Then the equation becomes $xw' + 2w - xw^2 = 0$, or $w' = w^2 - \frac{2w}{x}$. This is a Bernoulli's equation. By transforming $u := w^{-1}$, it can be converted into a linear equation. After substitution, we obtain $u' - \frac{2u}{x} = -1$, and its integrating factor is x^{-2} .

Multiply both sides of the equation by the integrating factor, resulting in $\frac{d}{dx}(ux^{-2}) = -x^{-2}$. Integrate to obtain $ux^{-2} = x^{-1} + C$, which simplifies to $u = x + Cx^2$. Substitute $w = u^{-1} = \frac{1}{x(1 + Cx)}$ and integrate to find z :

$$z = \int \frac{1}{x(1 + Cx)} dx = \int \left(\frac{1}{x} - \frac{C}{1 + Cx} \right) dx = \ln \left| \frac{x}{1 + Cx} \right| + D.$$

Finally, the general solution of the original equation is $y = xz = x \ln \left| \frac{x}{1 + Cx} \right| + D$, where C and D are arbitrary constants. $\square$

Problem 1.9.12. $x^2 yy'' - x^2(y')^2 - 4xyy' + 8y^2 = 0$.

Solution. Let $z = \frac{y'}{y}$ (that is, $z = (\ln y)'$), then $y' = zy$ and $y'' = z'y + z^2y$; substituting these expressions into the original equation $x^2 yy'' - x^2(y')^2 - 4xyy' + 8y^2 = 0$, expanding and simplifying gives $x^2 y^2 z' + x^2 y^2 z^2 - x^2 y^2 z^2 - 4xy^2 z + 8y^2 = 0$, which simplifies to $x^2 y^2 z' - 4xy^2 z + 8y^2 = 0$; since $y \neq 0$, dividing both sides by y^2 gives $x^2 z' - 4xz + 8 = 0$, which can be rewritten as the standard first-order linear differential equation $\frac{dz}{dx} - \frac{4}{x}z = -\frac{8}{x^2}$; calculating the integrating factor $\mu(x) = e^{\int -\frac{4}{x} dx} = x^{-4}$ (assuming $x > 0$), multiplying both sides of the equation by $\mu(x)$ gives $(x^{-4}z)' = -8x^{-6}$; integrating yields $x^{-4}z = \frac{8}{5}x^{-5} + C_1$, from which we obtain $z = \frac{8}{5x} + C_1 x^4$; integrating $z = (\ln y)'$ gives $\ln y = \frac{8}{5} \ln |x| + \frac{C_1}{5} x^5 + C_2$, so the general solution is $y = C|x|^{8/5} e^{Dx^5}$, where $C = e^{C_2}$, $D = \frac{C_1}{5}$ are arbitrary constants, and it can be verified that this solution satisfies the original equation. $\square$

Chapter 2

Linear Systems of Differential Equations

Exercise 2.1

Exercise 2.2

Exercise 2.3

教材的 Theorem 2.3.1 非常重要, 它实际上是存在唯一性定理的特例, 我们这里拿出来强调, 也方便后面引用.

Theorem 2.3.1 (一阶线性系统的存在唯一性定理). 设 $\mathbf{A}(x)$ 和 $\mathbf{f}(x)$ 在区间 $a < x < b$ 上连续, 则初值问题

$$(E): \quad \frac{d\mathbf{y}}{dx} = \mathbf{A}(x)\mathbf{y} + \mathbf{f}(x), \quad \mathbf{y}(x_0) = \mathbf{y}_0$$

的解 $\mathbf{y} = \mathbf{y}(x)$ 在区间 $a < x < b$ 上存在而且唯一, 其中 $a < x_0 < b$, $\mathbf{y}_0 \in \mathbb{R}^n$.

Problem 2.3.1. If $\mathbf{x} = \boldsymbol{\varphi}(t)$ is a solution of $\frac{d\mathbf{x}}{dt} = \mathbf{A}(t)\mathbf{x}$ and there exists some $t_0 \in I$ such that $\boldsymbol{\varphi}(t_0) = 0$, then show that

$$\boldsymbol{\varphi}(t) \equiv 0, \quad t \in I.$$

Proof. • (Method 1) Noticing that $\boldsymbol{\varphi}(t) \equiv 0$ is one of the solutions, then by Theorem 2.3.1, we know that the solution is always 0.

- (Method 2) Consider the Picard iterates:

$$\boldsymbol{\varphi}_0(t) = \boldsymbol{\varphi}(t_0) = 0,$$

$$\boldsymbol{\varphi}_1(t) = \boldsymbol{\varphi}(t_0) + \int_{t_0}^t \mathbf{A}(s)\boldsymbol{\varphi}_0(s) ds,$$

.....

$$\boldsymbol{\varphi}_n(t) = \boldsymbol{\varphi}(t_0) + \int_{t_0}^t \mathbf{A}(s)\boldsymbol{\varphi}_{n-1}(s) ds.$$

For any $k \in \mathbb{N}$, we have $\varphi_k(t) \equiv 0 \quad (t \in I)$. Then by Theorem 2.3.1, we know $\varphi(t) = \lim_{n \rightarrow \infty} \varphi_n(t) = 0, \quad (t \in I)$. $\square$

Problem 2.3.2. Show that the initial value problem

$$x''(t) = f(t, x), \quad x(t_0) = x_0, \quad x'(t_0) = x_1,$$

where $f(t, x)$ is continuous in a domain D containing the point (t_0, x_0) , is equivalent to the integral equation

$$x(t) = x_0 + (t - t_0)x_1 + \int_{t_0}^t (t - s)f[s, x(s)] \, ds.$$

Proof. • (Method 1) The sufficiency can be verified by yourself. By the Newton-Leibniz formula, we integrate by parts (IBP),

$$\begin{aligned} x(t) &= x_0 + \int_{t_0}^t x'(s) \, ds \\ &= x_0 + sx'(s) \Big|_{t_0}^t - \int_{t_0}^t sx''(s) \, ds \\ &= x_0 + tx'(t) - t_0x'(t_0) - \int_{t_0}^t sf(s, x) \, ds \\ &= x_0 + (t - t_0)x'(t_0) + t(x'(t) - x'(t_0)) - \int_{t_0}^t sf(s, x) \, ds \\ &= x_0 + (t - t_0)x'(t_0) + t \int_{t_0}^t x''(s) \, ds - \int_{t_0}^t sf(s, x) \, ds \\ &= x_0 + (t - t_0)x_1 + \int_{t_0}^t (t - s)f[s, x(s)] \, ds. \end{aligned} \quad \square$$

• (Method 2) The sufficiency can be verified by yourself. Note that

$$\begin{aligned} x'(t) &= x'(t_0) + \int_{t_0}^t x''(s) \, ds = x_1 + \int_{t_0}^t f(s, x(s)) \, ds \\ x(t) &= x(t_0) + \int_{t_0}^t x'(u) \, du \\ &= x_0 + \int_{t_0}^t x_1 \, du + \int_{t_0}^t \int_{t_0}^u f(s, x(s)) \, ds \, du. \end{aligned}$$

Because $f(t, x(t))$ is continuous in a domain D we have

$$\int_{t_0}^t \int_{t_0}^u f(s, x(s)) \, ds \, du = \int_{t_0}^t \int_s^t f(s, x(s)) \, du \, ds$$

So,

$$x(t) = x_0 + (t - t_0)x_1 + \int_{t_0}^t (t - s)f(s, x(s)) \, ds.$$

Problem 2.3.3. Suppose that $y = \varphi(x)$ is a solution of the initial value problem

$$\begin{cases} y' = yg(x, y), \\ y(0) = 1 \end{cases}$$

on the interval $[0, 1]$, where $g(x, y)$ is a bounded and continuous function in the (x, y) plane. Show that there exists a constant c such that

$$|\varphi(x)| \leq e^{cx}, \quad \forall x \in [0, 1].$$

Proof. • (Method 1) Let $M := \sup_{(x,y)} |g(x, y)|$. From the given conditions of the problem, we have

$$|\varphi'(x)| = |\varphi(x)g(x, \varphi(x))| = |\varphi(x)||g(x, \varphi(x))| \leq M|\varphi(x)|.$$

According to the fundamental theorem of calculus, $\varphi(x)$ can be expressed as

$$\varphi(x) = \varphi(0) + \int_0^x \varphi'(t) dt = 1 + \int_0^x \varphi'(t) dt.$$

Taking the absolute value and using the triangle inequality, we get $|\varphi(x)| \leq 1 + \int_0^x |\varphi'(t)| dt$, substituting $|\varphi'(t)| \leq M|\varphi(t)|$ gives

$$|\varphi(x)| \leq 1 + M \int_0^x |\varphi(t)| dt. \quad (2.1)$$

Let $u(x) = |\varphi(x)|$, then

$$u(x) \leq 1 + M \int_0^x u(t) dt,$$

which is the standard form of Grönwall's inequality. Applying Grönwall's inequality, if $u(x) \leq a + \int_0^x b(t)u(t) dt$, where $a \geq 0$ and $b(t) \geq 0$, then $u(x) \leq a \exp\left(\int_0^x b(t) dt\right)$, here $a = 1$ and $b(t) = M$, so $u(x) \leq \exp\left(\int_0^x M dt\right) = e^{Mx}$. Therefore, $|\varphi(x)| \leq e^{Mx}$, setting $c = M$, we obtain $|\varphi(x)| \leq e^{cx}$ for all $x \in [0, 1]$.

Remark 2.1. 到这个题为止, 教材正文和练习尚未证明 Grönwall 不等式. 因此我们给出一个不依赖 Grönwall 不等式的做法.

- (Method 2) We directly prove a more general proposition (It is not difficult to see from equation 2.1):

Proposition 2.3.1. Let f be a continuous function on the interval $[0, a]$, and suppose there exist positive constants M and c such that

$$|f(x)| \leq M + c \int_0^x |f(t)| dt,$$

Show that $|f(x)| \leq Me^{cx}$ ($\forall x \in [0, a]$).

Proof. Let

$$F(x) := \int_0^x |f(t)| dt,$$

Then the inequality in the condition is

$$F'(x) \leq M + cF(x).$$

Let

$$G(x) := F(x)e^{-cx} + \frac{M}{c}e^{-cx},$$

Then there is

$$\begin{aligned} G'(x) &= F'(x)e^{-cx} - cF(x)e^{-cx} - Me^{-cx} \\ &= |f(x)|e^{-cx} - cF(x)e^{-cx} - Me^{-cx} \\ &\leq (M + cF(x))e^{-cx} - cF(x)e^{-cx} - Me^{-cx} = 0. \end{aligned}$$

This indicates that G is monotonically decreasing on the interval $[0, a]$. Since $G(0) = \frac{M}{c}$, it follows that $G \leq \frac{M}{c}$. Thus

$$F(x) + \frac{M}{c} \leq \frac{M}{c}e^{cx}.$$

Combining the conditions, we can obtain $|f(x)| \leq M + cF(x) \leq Me^{cx}$. □

For this problem, simply set $a = 1$ and $M = 1$. □

Problem 2.3.4. Let $\varphi(x), p(x)$ and $q(x)$ be nonnegative continuous functions on the interval $[x_0 - a, x_0 + a]$ and

$$\varphi(x) \leq p(x) + \left| \int_{x_0}^x q(t)\varphi(t) dt \right| \quad \text{for } |x - x_0| \leq a. \quad (2.2)$$

Then the following inequality holds

$$\varphi(x) \leq p(x) + \left| \int_{x_0}^x p(t)q(t) \exp \left[\int_t^x q(s) ds \right] dt \right| \quad \text{for } |x - x_0| \leq a.$$

Remark 2.2. 这就是著名的 Gronwall 不等式.

Proof. We only prove the case where $x \geq x_0$, the case where $x \leq x_0$ is similar. Solving the differential equation yields the construction function. Considering $F(x) = \int_a^x q(t)\varphi(t) dt$, then

$$F'(x) = q(x)\varphi(x) \leq q(x)p(x) + q(x)F(x).$$

So, the consideration of differential equations was made.

$$y' = q(x)p(x) + q(x)y \implies y = c \exp \left(\int_a^x q(t) dt \right) + \int_a^x q(t)p(t) \exp \left(\int_t^x q(u) du \right) dt.$$

Therefore, the constructor is obtained.

$$\begin{aligned} c(x) &= \left[F(x) - \int_a^x q(t)p(t) \exp \left(\int_t^x q(u) du \right) dt \right] \cdot \exp \left(- \int_a^x q(t) dt \right) \\ &= F(x) \exp \left(- \int_a^x q(t) dt \right) - \int_a^x q(t)p(t) \exp \left(\int_t^x q(u) du \right) dt \quad (x \in [a, b]). \end{aligned}$$

Construct

$$c(x) := F(x) \exp \left(- \int_a^x q(t) dt \right) - \int_a^x q(t)p(t) \exp \left(\int_t^x q(u) du \right) dt \quad (x \in [a, b]),$$

Here, $F(x) := \int_a^x q(t)\varphi(t) dt$. According to the inequality (2.2), we have

$$F'(x) \leq p(x)q(x) + F(x)q(x), \forall t \in [a, b]. \quad (2.3)$$

Thus, from (2.3), we have

$$c'(x) = [F'(x) - p(x)q(x) - q(x)F(x)] \exp \left(\int_t^a q(t) dt \right) \leq 0,$$

Therefore, $c(x) \leq c(a) = 0$, and this leads to

$$F(x) \exp \left(- \int_a^x q(t) dt \right) \leq \int_a^x q(t)p(t) \exp \left(\int_t^x q(u) du \right) dt.$$

This still has a certain gap from what we aim to prove. Let's use the given inequality (2.2) again, and we will obtain

$$\varphi(x) \leq p(x) + \int_a^x q(t)p(t) \exp \left(\int_t^x q(u) du \right) dt \quad (\forall x \in [a, b]). \quad \square$$

Problem 2.3.5. Show that the initial value problem

$$\begin{cases} y' + p(x)y = q(x) \\ x(t_0) = y_0 \end{cases}$$

possesses a unique solution for all real x , where $p(x), q(x)$ are continuous in $\mathbb{R}$.

Proof. For the first-order linear differential equation $y' + p(x)y = q(x)$, under the condition that $p(x)$ and $q(x)$ are continuous, it can be solved using the integrating factor $\mu(x) = e^{\int_{x_0}^x p(t) dt}$, resulting in the explicit solution

$$y(x) = e^{-\int_{x_0}^x p(t) dt} \left[y_0 + \int_{x_0}^x e^{\int_{x_0}^t p(s) ds} q(t) dt \right],$$

which is defined for all real numbers x . If there are two solutions y_1 and y_2 , then their difference $z = y_1 - y_2$ satisfies $z' + p(x)z = 0$ and $z(x_0) = 0$. By using the same integrating factor, it can be proved that $z(x) \equiv 0$, thus ensuring the uniqueness of the solution. $\square$

Problem 2.3.6. Suppose that $\mathbf{x} = \varphi(t)$ is a solution of

$$\frac{d\mathbf{x}}{dt} = \mathbf{A}(t)\mathbf{x} + \mathbf{f}(t), \quad t \in \mathbb{R},$$

where $\mathbf{A}(t) = \mathbf{A}(t + \omega)$ and $\mathbf{f}(t) = \mathbf{f}(t + \omega)$, $\omega > 0$. Then $\mathbf{x} = \varphi(t)$ is a periodic function with period ω if and only if $\varphi(0) = \varphi(\omega)$.

Proof. Obviously, if $\mathbf{x} = \varphi(t)$ is a periodic function with period ω , then there must be $\varphi(0) = \varphi(\omega)$. On the other hand, if $\varphi(0) = \varphi(\omega)$, consider $\frac{d(\varphi(t + \omega) - \varphi(t))}{dt} = \mathbf{A}(t)(\varphi(t + \omega) - \varphi(t))$. Denote $\varphi(t + \omega) - \varphi(t)$ by $\psi(t)$, then

$$\frac{d\psi(t)}{dt} = \mathbf{A}(t)\psi(t), \quad \psi(0) = 0.$$

So by Problem 2.3.1, we know $\psi(t) \equiv 0$, which implies $\varphi(t)$ is a periodic function with period ω . □

Problem 2.3.7. Let $\varphi(t, t_0, \mathbf{x}_0)$ be a solution of

$$\frac{d\mathbf{x}}{dt} = \mathbf{A}(t)\mathbf{x} \tag{2.4}$$

in the interval I . Show that

$$\varphi(t, t_0, \alpha\mathbf{x}_1 + \beta\mathbf{x}_2) = \alpha\varphi(t, t_0, \mathbf{x}_1) + \beta\varphi(t, t_0, \mathbf{x}_2),$$

where α, β are arbitrary constants, $t \in I$.

Proof. Let $\mathbf{z}(t) := \alpha\varphi(t, t_0, \mathbf{x}_1) + \beta\varphi(t, t_0, \mathbf{x}_2)$, where $\varphi(t, t_0, \mathbf{x}_1)$ and $\varphi(t, t_0, \mathbf{x}_2)$ are the solutions of the system. Firstly, $\mathbf{z}(t)$ satisfies the differential equation:

$$\begin{aligned} \frac{d\mathbf{z}}{dt} &= \alpha \frac{d\varphi(t, t_0, \mathbf{x}_1)}{dt} + \beta \frac{d\varphi(t, t_0, \mathbf{x}_2)}{dt} = \alpha\mathbf{A}(t)\varphi(t, t_0, \mathbf{x}_1) + \beta\mathbf{A}(t)\varphi(t, t_0, \mathbf{x}_2) \\ &= \mathbf{A}(t)(\alpha\varphi(t, t_0, \mathbf{x}_1) + \beta\varphi(t, t_0, \mathbf{x}_2)) = \mathbf{A}(t)\mathbf{z}(t). \end{aligned}$$

Secondly, $\mathbf{z}(t)$ satisfies the initial condition:

$$\mathbf{z}(t_0) = \alpha\varphi(t_0, t_0, \mathbf{x}_1) + \beta\varphi(t_0, t_0, \mathbf{x}_2) = \alpha\mathbf{x}_1 + \beta\mathbf{x}_2.$$

Since $\varphi(t, t_0, \alpha\mathbf{x}_1 + \beta\mathbf{x}_2)$ is the unique solution of the same initial value problem (when $\mathbf{A}(t)$ is continuous on the interval I), we have $\mathbf{z}(t) = \varphi(t, t_0, \alpha\mathbf{x}_1 + \beta\mathbf{x}_2)$, that is

$$\varphi(t, t_0, \alpha\mathbf{x}_1 + \beta\mathbf{x}_2) = \alpha\varphi(t, t_0, \mathbf{x}_1) + \beta\varphi(t, t_0, \mathbf{x}_2). \quad \square$$

Exercise 2.4

Find the general solution of the following systems.

Problem 2.4.1.
$$\begin{cases} \frac{dx}{dt} = \frac{x}{t}, & t \neq 0, \\ \frac{dy}{dt} = \frac{y}{t}. \end{cases}$$

Solution. We have $x = \pm e^{c_1 t}, y = \pm e^{c_2 t}$. Note that $x \equiv 0, y \equiv 0$ is also a solution. Hence, the general solution is

$$\begin{pmatrix} x \\ y \end{pmatrix} = k_1 \begin{pmatrix} t \\ 0 \end{pmatrix} + k_2 \begin{pmatrix} 0 \\ t \end{pmatrix},$$

where k_1, k_2 are arbitrary constants. □

Problem 2.4.2.
$$\begin{cases} \frac{dx}{dt} = x + y, \\ \frac{dy}{dt} = y. \end{cases}$$

Solution. Firstly, we have $y = \pm e^{t+c_1}$. Then solving $\frac{dx}{dt} = x \pm e^{t+c_1}$ yields $x = \pm t e^{c_1+t} + c_2 e^t$. So

$$\begin{pmatrix} x \\ y \end{pmatrix} = c_2 \begin{pmatrix} e^t \\ 0 \end{pmatrix} + \begin{pmatrix} \pm t e^{c_1+t} \\ \pm e^{c_1+t} \end{pmatrix}.$$

□

Problem 2.4.3.
$$\begin{cases} \frac{dr}{dt} = -r(r^2 - 1), \\ \frac{d\theta}{dt} = 2. \end{cases}$$

Solution.

$$\begin{cases} r = \pm \frac{e^{t-C_1}}{\sqrt{1 \pm e^{2t-2C_1}}}, \\ \theta = 2t + C_2. \end{cases} \implies \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \frac{e^t}{\sqrt{e^{2t} \pm e^{2C_1}}} \\ 2t \end{pmatrix} + \begin{pmatrix} 0 \\ C_2 \end{pmatrix}.$$

□

Problem 2.4.4.
$$\begin{cases} \frac{dx}{dt} = p(t)x + q(t)y, \\ \frac{dy}{dt} = q(t)x + p(t)y, \end{cases} \quad \text{where } p(t), q(t) \text{ are continuous on the interval } [a, b].$$

Solution. Add two equations, we get

$$\frac{d(x+y)}{dt} = [p(t) + q(t)](x+y) \implies x+y = c_1 e^{\int [p(t)+q(t)] dt}.$$

Subtract two equations, we get

$$\frac{d(x-y)}{dt} = [p(t) - q(t)](x-y) \implies x-y = c_2 e^{\int [p(t)-q(t)] dt}$$

So

$$\begin{cases} x = \frac{1}{2} c_1 e^{\int [p(t)+q(t)] dt} + \frac{1}{2} c_2 e^{\int [p(t)-q(t)] dt} \\ y = \frac{1}{2} c_1 e^{\int [p(t)+q(t)] dt} - \frac{1}{2} c_2 e^{\int [p(t)-q(t)] dt} \end{cases}$$

□

is a general solution.

Problem 2.4.5.
$$\begin{cases} \frac{dx}{dt} = \frac{2t}{1+t^2}x, \\ \frac{dy}{dt} = -\frac{1}{t}y + x + t \quad (t > 0), \\ x(1) = 0, \\ y(1) = \frac{4}{3}. \end{cases}$$

Solution. First, solve the first equation $\frac{dx}{dt} = \frac{2t}{1+t^2}x$. This is a separable variable equation, and its general solution is $x(t) = C(1+t^2)$. Using the initial condition $x(1) = 0$, we get $C = 0$, so $x(t) = 0$.

Substituting $x(t) = 0$ into the second equation $\frac{dy}{dt} = -\frac{1}{t}y + t$, this is a first-order linear differential equation. It can be solved using the integrating factor method (with the integrating factor being t), resulting in $y(t) = \frac{t^2}{3} + \frac{D}{t}$. Given the initial condition $y(1) = \frac{4}{3}$, we find that $D = 1$.

Therefore, the solution that satisfies the initial conditions is:

$$\begin{cases} x(t) = 0, \\ y(t) = \frac{t^2}{3} + \frac{1}{t}. \end{cases} \quad \square$$

Problem 2.4.6. Show that the linear differential system

$$\frac{d\mathbf{y}}{dx} = \mathbf{A}(x)\mathbf{y} + \mathbf{f}(x)$$

remains linear after the change of the independent variable $x = p(t)$.

Proof. Under the variable substitution $x = p(t)$, by the chain rule, we have $\frac{d\mathbf{y}}{dx} = \frac{d\mathbf{y}}{dt} \cdot \frac{1}{p'(t)}$ (where $p'(t) \neq 0$). Substituting this into the original equation and simplifying, we obtain $\frac{d\mathbf{y}}{dt} = p'(t)\mathbf{A}(p(t))\mathbf{y} + p'(t)\mathbf{f}(p(t))$. Let $\mathbf{B}(t) := p'(t)\mathbf{A}(p(t))$ and $\mathbf{g}(t) = p'(t)\mathbf{f}(p(t))$, then the new system $\frac{d\mathbf{y}}{dt} = \mathbf{B}(t)\mathbf{y} + \mathbf{g}(t)$ is still linear because $\mathbf{B}(t)$ and $\mathbf{g}(t)$ only depend on t and not on $\mathbf{y}$, and the equation maintains a linear relationship with respect to $\mathbf{y}$ and its derivatives. $\square$

Problem 2.4.7. Show that the vector-valued functions

$$\varphi_1(t) = \begin{pmatrix} t \\ 1 \end{pmatrix} \quad \text{and} \quad \varphi_2(t) = \begin{pmatrix} t^2 \\ t \end{pmatrix}$$

are linearly independent in any interval I , and

$$W(\varphi_1(t), \varphi_2(t)) \equiv 0 \text{ in } I.$$

What does the fact show?

Proof. If there are non-zero constants k_1, k_2 such that $k_1\varphi_1(t) + k_2\varphi_2(t) = \mathbf{0}$, then $k_1 + k_2t = 0$. But $1, t$ are linearly independent, so $k_1 = k_2 = 0$, implying that $\varphi_1(t), \varphi_2(t)$ are linearly independent in any interval I . Obviously,

$$W(\varphi_1(t), \varphi_2(t)) = \begin{vmatrix} t & t^2 \\ 1 & t \end{vmatrix} = 0.$$

This fact indicates the following content which we will elaborate in the notes. $\square$

Remark 2.3. 我们来强调一下 Wronskian 行列式: 在不考虑线性微分方程组的解的时候, 任给向量值函数, 我们有如下的定理:

Theorem 2.4.1. 如果向量值函数 $f_1(x), f_2(x), \dots, f_n(x)$ 在区间 $a < x < b$ 上线性相关, 那么它们的 Wronskian 行列式 $W(x) \equiv 0, x \in (a, b)$.

Corollary 2.4.1. 若 $W[f_1, f_2, \dots, f_n](x) \not\equiv 0, x \in (a, b)$, 则 $f_1(x), f_2(x), \dots, f_n(x)$ 在 (a, b) 上线性无关.

但注意定理的逆命题不真. 比如 Problem 2.4.7. 再如

$$f_1(x) = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, f_2(x) = \begin{pmatrix} x \\ 0 \\ 0 \end{pmatrix}, f_3(x) = \begin{pmatrix} x^2 \\ 0 \\ 0 \end{pmatrix}$$

在 $a < x < b$ 上线性无关, 但

$$W[f_1, f_2, f_3](x) = \begin{vmatrix} 1 & x & x^2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{vmatrix} \equiv 0.$$

特别地, 对于齐次方程组

$$\frac{dy}{dx} = A(x)y, \quad (2.5)$$

我们有

Theorem 2.4.2. 设 $y_1(x), y_2(x), \dots, y_n(x)$ 是 (2.5) 的解. 如果 $y_1(x), y_2(x), \dots, y_n(x)$ 是线性无关的, 那么它们的 Wronskian 行列式 $W(x) \not\equiv 0, a < x < b$.

也就是说, 对于齐次方程组 (2.5), Theorem 2.4.1 是充分必要的. 这是教材的 Theorem 2.4.4 的内容.

Problem 2.4.8. If the following systems

$$\frac{dx}{dt} = A_1(t)x \quad \text{and} \quad \frac{dx}{dt} = A_2(t)x$$

have the same fundamental matrix $\Phi(t)$, then show that

$$A_1(t) \equiv A_2(t),$$

where $A_1(t), A_2(t)$ are continuous $n \times n$ matrices.

Proof. Since $\Phi(t)$ is the fundamental solution matrix of the system given in the problem, we have $A_1(t)\Phi(t) = A_2(t)\Phi(t)$. Noting that $\Phi(t)$ is non-singular, we can multiply both sides of the equation by $\Phi^{-1}(t)$ simultaneously to obtain the conclusion. $\square$

Problem 2.4.9. The fundamental matrix $\Phi(t)$ of $\frac{dx}{dt} = A(t)x$ is called a transition matrix provided $\Phi(t)$ satisfies the initial condition $\Phi(t_0) = E$ (identity matrix), where $t_0 \in I$. Obviously, for any one fundamental matrix $\Phi(t)$ of $\frac{dx}{dt} = A(t)x$, we get

$$\Psi(t, t_0) = \Phi(t)\Phi^{-1}(t_0) \quad (2.6)$$

is a transition matrix. Show the following:

- (1) The transition matrix $\Psi(t, t_0)$ is independent of the choice of the fundamental matrix.
- (2) $\Psi(t, t_0)$ with respect to t is a unique solution of matrix equation

$$\frac{dY}{dt} = A(t)Y$$

which satisfies the initial condition $Y(t_0) = E$.

- (3) For any $t, t_0 \in I$, $\Psi(t, t_0)$ is nonsingular and

$$\Psi^{-1}(t, t_0) = \Psi(t_0, t). \quad (2.7)$$

- (4) For any $t, t_0, t_1 \in I$,

$$\Psi(t, t_0) = \Psi(t, t_1)\Psi(t_1, t_0). \quad (2.8)$$

- (5) The solution of initial value problem

$$\begin{cases} \frac{dx}{dt} = A(t)x \\ x(t_0) = x_0 \end{cases}$$

can be written as

$$x(t, t_0, x_0) = \Psi(t, t_0)x_0.$$

Proof. (1) By Theorem 2.4.5, $\Phi_1(t) = \Phi(t)P$ is another fundamental matrix of $\frac{dx}{dt} = A(t)x$, where P is a constant nonsingular $n \times n$ matrix. Therefore, $\Psi(t, t_0) = \Phi_1(t)\Phi_1^{-1}(t_0) = \Phi(t)PP^{-1}\Phi^{-1}(t_0) = \Phi(t)\Phi^{-1}(t_0)$. Thus, $\Psi(t, t_0)$ is independent of the choice of the fundamental matrix.

(2) Substitute $\mathbf{Y} := \Psi(t, t_0)$ into the equation, we get

$$\frac{d\Psi(t, t_0)}{dt} = \frac{d\Phi(t)\Phi^{-1}(t_0)}{dt} = \mathbf{A}(t)\Phi(t)\Phi^{-1}(t_0) = \mathbf{A}(t)\Psi(t, t_0).$$

Hence, $\Psi(t, t_0)$ is a solution of $\frac{d\mathbf{Y}}{dt} = \mathbf{A}(t)\mathbf{Y}$. Note that when $t = t_0$, $\Psi(t_0, t_0) = \mathbf{E}$, thus $\Psi(t, t_0)$ is a unique solution of $\frac{d\mathbf{Y}}{dt} = \mathbf{A}(t)\mathbf{Y}$.

(3) Note that

$$\Psi(t, t_0)\Psi(t_0, t) = \Phi(t)\Phi^{-1}(t_0)\Phi(t_0)\Phi^{-1}(t) = \mathbf{E},$$

so $\Psi^{-1}(t, t_0) = \Psi(t_0, t)$.

(4) $\Psi(t, t_1)\Psi(t_1, t_0) = \Phi(t)\Phi^{-1}(t_1)\Phi(t_1)\Phi^{-1}(t_0) = \Phi(t)\Phi^{-1}(t_0) = \Psi(t, t_0)$.

(5) Substitute $\mathbf{x}(t, t_0, \mathbf{x}_0) = \Psi(t, t_0)\mathbf{x}_0$ into $\frac{d\mathbf{x}}{dt} = \mathbf{A}(t)\mathbf{x}$, we get

$$\begin{aligned} \frac{d\mathbf{x}(t, t_0, \mathbf{x}_0)}{dt} &= \frac{d\Psi(t, t_0)\mathbf{x}_0}{dt} = \frac{d\Phi(t)\Phi^{-1}(t_0)\mathbf{x}_0}{dt} \\ &= \mathbf{A}(t)\Phi(t)\Phi^{-1}(t_0)\mathbf{x}_0 = \mathbf{A}(t)\Psi(t, t_0)\mathbf{x}_0 = \mathbf{A}(t)\mathbf{x}(t, t_0, \mathbf{x}_0). \end{aligned}$$

When $t = t_0$, $\mathbf{x}(t_0, t_0, \mathbf{x}_0) = \mathbf{x}_0$, $\mathbf{x}(t, t_0, \mathbf{x}_0) = \Psi(t, t_0)\mathbf{x}_0$ is the solution of the equation. By Theorem 2.3.1, the solution of the equation can be written as $\mathbf{x}(t, t_0, \mathbf{x}_0) = \Psi(t, t_0)\mathbf{x}_0$. $\square$

Problem 2.4.10. Suppose that $\Phi(t)$ is a fundamental matrix of $\frac{d\mathbf{x}}{dt} = \mathbf{A}(t)\mathbf{x}$. Show that $\Phi(t)$ determines the matrix $\mathbf{A}(t)$ uniquely by the relation $\mathbf{A}(t) = \Phi'(t)\Phi^{-1}(t)$.

Proof. Let $\Phi(t)$ be the fundamental solution matrix of the differential system $\frac{d\mathbf{x}}{dt} = \mathbf{A}(t)\mathbf{x}$. Then each column of $\Phi(t)$ is a linearly independent solution of this system. Therefore, $\Phi(t)$ satisfies the matrix differential equation:

$$\Phi'(t) = \mathbf{A}(t)\Phi(t).$$

Therefore, multiplying both sides of the equation by $\Phi^{-1}(t)$, we obtain:

$$\Phi'(t)\Phi^{-1}(t) = \mathbf{A}(t).$$

Therefore, $\mathbf{A}(t) = \Phi'(t)\Phi^{-1}(t)$. If there exists another matrix $\mathbf{B}(t)$ that also satisfies $\Phi'(t) = \mathbf{B}(t)\Phi(t)$, then it can be concluded that $\mathbf{B}(t) = \Phi'(t)\Phi^{-1}(t) = \mathbf{A}(t)$, which indicates that $\mathbf{A}(t)$ is uniquely determined by the basic solution matrix $\Phi(t)$. In conclusion, $\mathbf{A}(t)$ is uniquely determined by the relationship $\mathbf{A}(t) = \Phi'(t)\Phi^{-1}(t)$. $\square$

Problem 2.4.11. The system

$$\frac{d\mathbf{x}}{dt} = -\mathbf{A}^\top(t)\mathbf{x}(t) \tag{2.9}$$

is called the adjoint system to the differential system

$$\frac{d\mathbf{x}}{dt} = \mathbf{A}(t)\mathbf{x}(t), \tag{2.10}$$

where $\mathbf{A}^\top(t)$ is a transpose of $\mathbf{A}(t)$. This relationship is symmetric in the sense that the system (2.10) is the adjoint system to (2.9) and vice versa¹. Show the following:

- (1) Let $\Phi(t)$ be a fundamental matrix of (2.10). Then $[\Phi^{-1}(t)]^\top$ is a fundamental matrix of its adjoint system (2.9).
- (2) If $\Phi(t)$ is a fundamental matrix of the differential system (2.10), then $\Psi(t)$ is a fundamental matrix of its adjoint system (2.9) if and only if

$$\Psi^\top(t)\Phi(t) = \mathbf{C}, \quad (2.11)$$

where $\mathbf{C}$ is a constant nonsingular $(n \times n)$ matrix.

Proof. (1) The verification can be done by directly using the derivative formula of the inverse matrix function (2.13).

- (2) On the one hand, let $\psi(t)$ be any solution of the adjoint system (2.9), then $\frac{d\psi(t)}{dt} = -\mathbf{A}^\top(t)\psi(t)$. It follows that $\frac{d\psi^\top(t)}{dt} = -\psi^\top(t)\mathbf{A}(t)$. Let $\phi(t)$ be a solution to (2.10), then

$$\frac{d}{dt}(\psi^\top(t)\phi(t)) = \frac{d\psi^\top(t)}{dt}\phi(t) + \psi^\top(t)\frac{d\phi(t)}{dt} \equiv 0.$$

Therefore, $\psi^\top(t)\phi(t) = C$, where C is a constant number. Since $\Psi(t)$ is the fundamental matrix of the equation $\frac{d\mathbf{y}}{dt} = -\mathbf{A}^\top(t)\mathbf{y}$, it follows that there exists a matrix $\mathbf{C}$ such that $\psi^\top(t)\Phi(t) = \mathbf{C}$. Moreover, since $\det \Phi(t) \neq 0$ and $\det \Psi(t) \neq 0$, it follows that $\det \mathbf{C} \neq 0$, which means $\mathbf{C}$ is non-singular.

On the other hand, suppose there exists a non-singular constant matrix $\mathbf{C}$ such that $\Psi^\top(t)\Phi(t) = \mathbf{C}$. Then it is obvious that $\det \Psi(t) \neq 0$. Moreover, by

$$\mathbf{0} = \frac{d}{dt}(\Psi^\top(t)\Phi(t)) = \frac{d\Psi^\top(t)}{dt}\Phi(t) + \Psi^\top(t)\frac{d\Phi(t)}{dt},$$

we know

$$\frac{d\Psi^\top(t)}{dt}\Phi(t) = -\Psi^\top(t)\mathbf{A}(t)\Phi(t).$$

Since $\det \Phi(t) \neq 0$, we have $\frac{d\Psi^\top(t)}{dt} = -\Psi^\top(t)\mathbf{A}(t)$, that is

$$\frac{d\Psi(t)}{dt} = -\mathbf{A}^\top(t)\Psi(t).$$

Therefore, $\Psi(t)$ is the fundamental solution matrix of the equation $\frac{d\mathbf{y}}{dt} = -\mathbf{A}^\top(t)\mathbf{y}$. □

Remark 2.4. (1) 用到了矩阵函数求导公式:

¹反之亦然.

- 若矩阵函数 $A(x)$ 与 $B(x)$ 可以相乘, 且均可导, 则

$$\frac{d}{dx}[A(x)B(x)] = \frac{dA(x)}{dx}B(x) + A(x)\frac{dB(x)}{dx}; \quad (2.12)$$

- 若矩阵 (方阵) 函数 $A(x)$ 可逆且可导, 则 $A^{-1}(x)$ 可导, 并且

$$\frac{d}{dx}[A^{-1}(x)] = -A^{-1}(x)\frac{dA(x)}{dx}A^{-1}(x). \quad (2.13)$$

Problem 2.4.12. The differential system (2.10) is said to be self-adjoint when $A(t) = -A^\top(t)$ for all t in I . Show the following:

- (1) If $\Phi(t)$ is a fundamental matrix of the self-adjoint system (2.10), then

$$\Phi^\top(t)\Phi(t) = C,$$

where C is a constant nonsingular $n \times n$ matrix.

- (2) Let $\Phi(t, t_0)$ be the transition matrix of the system (2.10), and $\Psi(t, t_0)$ be the transition matrix of the adjoint system (2.9). If the differential system (2.10) is self-adjoint, then

$$\Phi(t, t_0) = \Psi^\top(t_0, t).$$

Proof. (1) Since Φ is the fundamental matrix of (2.10), we know $(\Phi^{-1})^\top(t)$ is also a fundamental matrix of (2.10) due to the self-adjointness. Therefore, by (1), we can know the conclusion.

- (2) By the definition of transition matrix in (2.6), suppose that $F(t)$ is a fundamental matrix of the system (2.10), we know $\Phi(t, t_0) = F(t)F^{-1}(t_0)$. Due to the self-adjointness, we know $(\Phi^{-1})^\top(t)$ is also a fundamental matrix of (2.10). Therefore, we have $\Psi(t, t_0) = (F^{-1})^\top(t)F^\top(t_0) \implies \Psi(t_0, t) = \Psi^{-1}(t, t_0) = (F^{-1})^\top(t_0)F^\top(t) = (F(t)F^{-1}(t_0))^\top = \Phi^\top(t, t_0)$. The proof is completed. □

Exercise 2.5

Problem 2.5.1. Find a solution of the nonhomogeneous differential system

$$\begin{cases} \frac{dx_1}{dt} = \frac{1}{t}x_1 - x_2 + t, \\ \frac{dx_2}{dt} = \frac{1}{t^2}x_1 + \frac{2}{t}x_2 - t^2. \end{cases}$$

The known fundamental matrix is given by

$$\Phi(t) = \begin{pmatrix} t^2 & -t^2 \ln t \\ -t & t + t \ln t \end{pmatrix}.$$

Solution. Firstly, the system can be expressed as

$$\frac{d\mathbf{x}}{dt} = \mathbf{A}(t)\mathbf{x} + \mathbf{f}(t),$$

where

$$\mathbf{x} = \begin{pmatrix} x \\ y \end{pmatrix}, \quad \mathbf{A}(t) = \begin{pmatrix} \frac{1}{t} & -1 \\ \frac{1}{t^2} & \frac{2}{t} \end{pmatrix}, \quad \mathbf{f}(t) = \begin{pmatrix} t \\ -t^2 \end{pmatrix}.$$

We calculate the determinant of the known fundamental matrix $\Phi(t) = \begin{pmatrix} t^2 & -t^2 \ln t \\ -t & t + t \ln t \end{pmatrix}$, obtaining $\det(\Phi(t)) = t^3$, and subsequently find its inverse matrix $\Phi^{-1}(t) = \begin{pmatrix} \frac{1 + \ln t}{t^2} & \frac{\ln t}{t} \\ \frac{1}{t^2} & \frac{1}{t} \end{pmatrix}$.

Next, we calculate $\Phi^{-1}(t)\mathbf{f}(t) = \begin{pmatrix} \frac{1 + \ln t}{t} - t \ln t \\ \frac{1}{t} - t \end{pmatrix}$, and integrate this result to obtain

$$\int \Phi^{-1}(t)\mathbf{f}(t)dt = \begin{pmatrix} \ln t + \frac{(\ln t)^2}{2} - \frac{t^2}{2} \ln t + \frac{t^2}{4} \\ \ln t - \frac{t^2}{2} \end{pmatrix}.$$

Finally, through matrix multiplication $\mathbf{x}_p(t) = \Phi(t) \int \Phi^{-1}(t)\mathbf{f}(t)dt$, we obtain a particular solution

$$\mathbf{x}_p(t) = \begin{pmatrix} t^2 \ln t - \frac{t^2(\ln t)^2}{2} + \frac{t^4}{4} \\ \frac{t(\ln t)^2}{2} - \frac{3t^3}{4} \end{pmatrix}$$

for the non-homogeneous system. □

Remark 2.5. 我们有如下定理:

Theorem 2.5.1 (教材 65 页 Theorem 2.5.4). 设 $\Phi(x)$ 是

$$\frac{d\mathbf{x}}{dt} = \mathbf{A}(t)\mathbf{x}, \quad a < t < b \tag{2.14}$$

的一个基解矩阵, 则

$$\frac{d\mathbf{x}}{dt} = \mathbf{A}(t)\mathbf{x} + \mathbf{f}(t), \quad a < t < b \tag{2.15}$$

在区间 $a < t < b$ 上的通解可以表示为

$$\mathbf{x} = \Phi(t) \left[\mathbf{C} + \int \Phi^{-1}(t)\mathbf{f}(t) dt \right], \tag{2.16}$$

其中 $\mathbf{C}$ 是任意的 n 维常数列向量, 而且(2.15)满足初值条件 $\mathbf{x}(t_0) = \mathbf{x}_0$ 的解可以表示为

$$\mathbf{x} = \Phi(t)\Phi^{-1}(t_0)\mathbf{x}_0 + \Phi(t) \int_{t_0}^t \Phi(s)^{-1} \mathbf{f}(s) ds, \quad (2.17)$$

其中 $t_0 \in (a, b)$.

Problem 2.5.2. Determine the general solution of the system

$$\begin{cases} \frac{dx}{dt} = (3t-1)x - (1-t)y + te^{t^2}, \\ \frac{dy}{dt} = -(t+2)x + (t-2)y - e^{t^2}. \end{cases}$$

Hint. The homogeneous system has a solution of the form $[x(t), y(t)] = [\varphi(t), -\varphi(t)]$.

Solution. From the given information, the homogeneous system has a solution form $[x, y] = [\varphi(t), -\varphi(t)]$. Substituting $y = -x$ into the first equation of the homogeneous system: $\frac{dx}{dt} = 2tx$, this is a separable variable equation, and the solution is $x = c_1 e^{t^2}$, taking $c_1 = 1$ gives the basic solution $x = e^{t^2}$, $y = -e^{t^2}$, that is, the first homogeneous solution $\mathbf{x}_1(t) = [e^{t^2}, -e^{t^2}]^\top$. To find the second linearly independent solution, apply Liouville's identity: for the system $\frac{d\mathbf{x}}{dt} =$

$\mathbf{A}(t)\mathbf{x}$, the Wronskian $W(t)$ satisfies $\frac{dW}{dt} = \text{tr}(\mathbf{A}(t))W(t)$. Here, $\mathbf{A}(t) := \begin{pmatrix} 3t-1 & t-1 \\ -t-2 & t-2 \end{pmatrix}$,

the trace $\text{tr}(\mathbf{A}(t)) = 4t - 3$, solving $\frac{dW}{dt} = (4t - 3)W$ gives $W(t) = Ce^{2t^2-3t}$. By directly calculating the Wronskian of $\mathbf{x}_1$ and the assumed second solution $\mathbf{x}_2 = [p(t), q(t)]^\top$, we get

$$W(t) = \det \begin{pmatrix} e^{t^2} & p \\ -e^{t^2} & q \end{pmatrix} = e^{t^2}(p+q), \text{ setting it equal to } 9e^{2t^2-3t} \text{ (taking } C = 9 \text{ to simplify}$$

subsequent calculations), we obtain $p+q = 9e^{t^2-3t}$. Substituting $q = 9e^{t^2-3t} - p$ into the first equation of the homogeneous system $p' = (3t-1)p + (t-1)q$, simplification yields $p' - 2tp = 9(t-1)e^{t^2-3t}$. The integrating factor of this first-order linear ODE is e^{-t^2} . Multiplying both sides by it gives $\frac{d}{dt}(pe^{-t^2}) = 9(t-1)e^{-3t}$. Integrating this yields

$$pe^{-t^2} = 9 \int (t-1)e^{-3t} dt = 9e^{-3t} \left(-\frac{t}{3} + \frac{2}{9} \right) + C.$$

Taking $C = 0$ to obtain the particular solution, we get $p = e^{t^2} - 3t(-3t+2)$, and subsequently $q = e^{t^2} - 3t(3t+7)$. Thus, the second homogeneous solution is $\mathbf{x}_2(t) = e^{t^2-3t}[-3t+2, 3t+7]^\top$.

Constructing the fundamental matrix $\Phi(t) = \begin{pmatrix} e^{t^2} & (-3t+2)e^{t^2-3t} \\ -e^{t^2} & (3t+7)e^{t^2-3t} \end{pmatrix}$, its determinant

$$\det \Phi(t) = 9e^{t^2} - 3t, \text{ and its inverse } \Phi^{-1}(t) = \frac{1}{9e^{t^2} - 3t} \begin{pmatrix} (3t+7)e^{t^2-3t} - 3t & (3t-2)e^{t^2-3t} \\ e^{t^2} & e^{t^2} \end{pmatrix}.$$

For the non-homogeneous term $\mathbf{f}(t) = [te^{t^2}, -e^{t^2}]^\top$, calculating

$$\Phi^{-1}(t)\mathbf{f}(t) = \begin{pmatrix} \frac{1}{9}(3t^2 + 4t + 2) \\ \frac{1}{9}e^t 3t(t-1) \end{pmatrix},$$

integrating gives

$$\int \Phi^{-1}(t)\mathbf{f}(t) dt = \begin{pmatrix} \frac{1}{9}(t^3 + 2t^2 + 2t) \\ \frac{1}{81}e^t 3t(3t-4) \end{pmatrix}.$$

Applying the parameter change method $\mathbf{x}_p = \Phi(t) \int \Phi^{-1}(t)\mathbf{f}(t) dt$ yields the particular solution

$$\mathbf{x}_p = \frac{e^{t^2}}{81} \begin{pmatrix} 9t^3 + 9t^2 + 36t - 8 \\ -9t^3 - 9t^2 - 9t - 28 \end{pmatrix}.$$

The final general solution is the sum of the homogeneous solution and the particular solution:

$$\begin{cases} x(t) = c_1 e^{t^2} + c_2 (2 - 3t) e^{t^2} - 3t + \frac{(9t^3 + 9t^2 + 36t - 8)e^{t^2}}{81}, \\ y(t) = -c_1 e^{t^2} + c_2 (3t + 7) e^{t^2} - 3t + \frac{(-9t^3 - 9t^2 - 9t - 28)e^{t^2}}{81}. \end{cases} \quad \square$$

Problem 2.5.3. Let $\Phi(t, t_0)$ be a transition matrix of the associated homogeneous system of the system $\frac{d\mathbf{x}}{dt} = \mathbf{A}(t)\mathbf{x}$. Show that the initial value problem

$$\begin{cases} \frac{d\mathbf{x}}{dt} = \mathbf{A}(t)\mathbf{x} + \mathbf{f}(t) \\ \mathbf{x}(t_0) = \mathbf{x}_0 \end{cases}$$

has a unique solution of the form

$$\mathbf{x}(t) = \Phi(t, t_0)\mathbf{x}_0 + \int_{t_0}^t \Phi(t, s)\mathbf{f}(s) ds.$$

Proof. By (2.17), we know the unique solution of the system can be written as

$$\mathbf{x}(t) = \Phi(t)\Phi^{-1}(t_0)\mathbf{x}_0 + \Phi(t) \int_{t_0}^t \Phi^{-1}(s)\mathbf{f}(s) ds.$$

Observe that $\Phi(t, t_0)$ is a transition matrix, so by (2.6), $\Phi(t)\Phi^{-1}(t_0) = \Phi(t, t_0)$. Then by (2.7)

and (2.8) in Problem 2.4.9, we have

$$\begin{aligned} \mathbf{x}(t) &= \Phi(t)\Phi^{-1}(t_0)\mathbf{x}_0 + \Phi(t) \int_{t_0}^t \Phi^{-1}(s)\mathbf{f}(s) \, ds \\ &= \Phi(t, t_0)\mathbf{x}_0 + \int_{t_0}^t \Phi(t, t_0)\Phi^{-1}(s, t_0)\mathbf{f}(s) \, ds \\ &= \Phi(t, t_0)\mathbf{x}_0 + \int_{t_0}^t \Phi(t, t_0)\Phi(t_0, s)\mathbf{f}(s) \, ds \\ &= \Phi(t, t_0)\mathbf{x}_0 + \int_{t_0}^t \Phi(t, s)\mathbf{f}(s) \, ds. \end{aligned}$$

Problem 2.5.4. Show that the nonhomogeneous differential system

$$\frac{d\mathbf{x}}{dt} = \mathbf{A}(t)\mathbf{x}(t) + \mathbf{f}(t) \quad (\mathbf{f}(t) \not\equiv 0)$$

has $n + 1$ linearly independent solutions.

Proof. By Theorem 2.4.2, $\frac{d\mathbf{x}}{dt} = \mathbf{A}(t)\mathbf{x}(t)$ has n linearly independent solutions, $\varphi_1(t), \varphi_2(t), \dots, \varphi_n(t)$. Let $\psi(t)$ be a particular solution of $\frac{d\mathbf{x}}{dt} = \mathbf{A}(t)\mathbf{x}(t) + \mathbf{f}(t)$. We claim that $\psi(t), \varphi_1(t) + \psi(t), \dots, \varphi_n(t) + \psi(t)$ are linearly independent.

Assume there exist non-zero numbers $c_0, c_1, \dots, c_n$ such that

$$c_0\psi(t) + c_1(\varphi_1(t) + \psi(t)) + \dots + c_n(\varphi_n(t) + \psi(t)) = 0,$$

That is,

$$(c_0 + c_1 + \dots + c_n)\psi(t) + c_1\varphi_1(t) + \dots + c_n\varphi_n(t) = 0.$$

If $c_0 + c_1 + \dots + c_n \neq 0$, then $\psi(t) = -\frac{c_1\varphi_1(t) + \dots + c_n\varphi_n(t)}{c_0 + c_1 + \dots + c_n}$, which shows that $\psi(t)$ is a solution of $\frac{d\mathbf{x}}{dt} = \mathbf{A}(t)\mathbf{x}(t)$, contradicting with the definition of $\psi(t)$. So $c_0 + c_1 + \dots + c_n = 0$. Furthermore, we get $c_0 = c_1 = c_2 = \dots = c_n = 0$, which implies that $\psi(t), \varphi_1(t) + \psi(t), \dots, \varphi_n(t) + \psi(t)$ are linearly independent.

If $\frac{d\mathbf{x}}{dt} = \mathbf{A}(t)\mathbf{x}(t) + \mathbf{f}(t)$ has $n + 2$ linearly independent solutions

$$\psi_1(t), \psi_2(t), \dots, \psi_{n+2}(t),$$

then

$$\psi_{n+2}(t) - \psi_1(t), \quad \psi_{n+1}(t) - \psi_1(t), \quad \dots, \quad \psi_2(t) - \psi_1(t)$$

are $n + 1$ linearly independent solutions of $\frac{d\mathbf{x}}{dt} = \mathbf{A}(t)\mathbf{x}(t)$, which contradicts with the theorem. □

Problem 2.5.5. Let $\varphi_1(t), \varphi_2(t), \dots, \varphi_{n+1}(t)$ be $n + 1$ linearly independent solutions of the nonhomogeneous system (2.15) in the interval I . Give the expression of general solution of (2.15).

Solution. Let $\tilde{x}(t)$ be any solution and particular solution of (2.15), and $\phi_i(t) (i = 1, 2, \dots, n)$

be the n linearly independent solutions of the corresponding homogeneous system. Without loss of generality, let $\varphi_{n+1}(t)$ be a particular solution of (2.15). Then, $\phi_i + \varphi_{n+1} = \varphi_i$ ($i = 1, \dots, n$), thus there exist non-zero constants $C_1, C_2, \dots, C_n$, such that

$$\begin{aligned}\tilde{x}(t) &= \varphi_{n+1}(t) + C_1\phi_1(t) + \dots + C_n\phi_n(t) \\ &= \varphi_{n+1}(t) + C_1(\varphi_1(t) - \varphi_{n+1}(t)) + \dots + C_n(\varphi_n(t) - \varphi_{n+1}(t)) \\ &= (1 - C_1 - \dots - C_n)\varphi_{n+1}(t) + C_1\varphi_1(t) + \dots + C_n\varphi_n(t),\end{aligned}$$

which is the expression of general solution of (2.15). □

Problem 2.5.6. If $\Phi(t)$ is a fundamental matrix of $\frac{dx}{dt} = A(t)x$, and n -dimensional vector-valued function $f(t, x)$ is continuous in the domain $R: a < t < b, |x| < \infty$, then the initial value problem

$$\begin{cases} \frac{dx}{dt} = A(t)x + f(t, x), \\ x(t_0) = x_0, \end{cases}$$

is equivalent to the integral equation

$$x(t) = \Phi(t)\Phi^{-1}(t_0)x_0 + \int_{t_0}^t \Phi(t)\Phi^{-1}(s)f(s, x(s))ds,$$

where $t_0 \in (a, b)$.

Proof. Since $\Phi^{-1}(t)\Phi(t) = E$, taking the derivative on both sides gives

$$\frac{d\Phi^{-1}(t)}{dt}\Phi(t) + \Phi^{-1}(t)\frac{d\Phi(t)}{dt} = 0.$$

Furthermore, since $\frac{d\Phi(t)}{dt} = A(t)\Phi(t)$, it follows that

$$\frac{d\Phi^{-1}(t)}{dt} + \Phi^{-1}(t)A(t) = 0.$$

Let $x = x(t)$ be the solution of the initial value problem, then

$$\begin{aligned}\frac{dx(t)}{dt} &= A(t)x(t) + f(t, x(t)), \quad x(t_0) = x_0 \\ \iff \Phi^{-1}(t)\frac{dx(t)}{dt} &= \Phi^{-1}(t)A(t)x(t) + \Phi^{-1}(t)f(t, x(t)), \quad x(t_0) = x_0 \\ \iff \Phi^{-1}(t)\frac{dx(t)}{dt} + \frac{d\Phi^{-1}(t)}{dt}x(t) &= \Phi^{-1}(t)f(t, x(t)), \quad x(t_0) = x_0 \\ \iff \frac{d}{dt}(\Phi^{-1}(t)x(t)) &= \Phi^{-1}(t)f(t, x(t)), \quad x(t_0) = x_0 \\ \iff \Phi^{-1}(t)x(t) &= \Phi^{-1}(t_0)x_0 + \int_{t_0}^t \Phi^{-1}(s)f(s, x(s))ds \\ \iff x(t) &= \Phi(t)\Phi^{-1}(t_0)x_0 + \int_{t_0}^t \Phi(t)\Phi^{-1}(s)f(s, x(s))ds,\end{aligned}$$

That is, $x = x(t)$ is the solution to the integral equation. □

Exercise 2.6

Find general solutions of the homogeneous and inhomogeneous systems below.

Problem 2.6.1. $\frac{d\mathbf{x}}{dt} = \begin{pmatrix} -2 & 1 \\ -1 & -4 \end{pmatrix} \mathbf{x}.$

Solution. Let $\mathbf{A} := \begin{pmatrix} -2 & 1 \\ -1 & -4 \end{pmatrix}$. Firstly, we can easily figure out that $\lambda = -3$ are its 2-fold eigenvalues.

For the 2-fold eigenvalues $\lambda = -3$, consider the generalized characteristic equation system $(\mathbf{A} + 3\mathbf{E})^2 \mathbf{r} = \mathbf{0}$. We can easily figure out $(\mathbf{A} + 3\mathbf{E})^2 = \mathbf{0}$. Obviously, two linearly independent generalized eigenvectors are

$$\mathbf{r}_{10} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \mathbf{r}_{20} = \begin{pmatrix} 0 \\ 1 \end{pmatrix},$$

respectively. Therefore,

$$\begin{aligned} \mathbf{r}_{11} &= (\mathbf{A} + 3\mathbf{E})\mathbf{r}_{10} = \begin{pmatrix} 1 & 1 \\ -1 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \\ \mathbf{r}_{21} &= (\mathbf{A} + 3\mathbf{E})\mathbf{r}_{20} = \begin{pmatrix} 1 & 1 \\ -1 & -1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}. \end{aligned}$$

Therefore, for the 2-fold eigenvalue $\lambda = -3$, two linearly independent solutions can be obtained, namely

$$e^{-3t}(\mathbf{r}_{10} + t\mathbf{r}_{11}) = e^{-3t} \begin{pmatrix} 1+t \\ -t \end{pmatrix}, \quad e^{-3t}(\mathbf{r}_{20} + t\mathbf{r}_{21}) = e^{-3t} \begin{pmatrix} t \\ 1-t \end{pmatrix}.$$

From this, we obtain the fundamental matrix

$$\Phi(t) = \begin{pmatrix} e^{-3t}(1+t) & e^{-3t}t \\ e^{-3t}(-t) & e^{-3t}(1-t) \end{pmatrix}.$$

Therefore, the general solution of the system of equations can be written as

$$\mathbf{x} = e^{t\mathbf{A}_1} \mathbf{c} = e^{-3t} \begin{pmatrix} c_1(t+1) + c_2t \\ -c_1t - c_2(t-1) \end{pmatrix},$$

where c_1, c_2 are arbitrary constants. □

Problem 2.6.2. $\frac{d\mathbf{x}}{dt} = \begin{pmatrix} 3 & -1 \\ 1 & 1 \end{pmatrix} \mathbf{x}.$

Solution. Let $\mathbf{A} := \begin{pmatrix} 3 & -1 \\ 1 & 1 \end{pmatrix}$. Firstly, we can easily figure out that $\lambda = 2$ are its 2-fold eigenvalues.

For the 2-fold eigenvalues $\lambda = 2$, consider the generalized characteristic equation system $(\mathbf{A} - 2\mathbf{E})^2 \mathbf{r} = \mathbf{0}$. We can easily figure out $(\mathbf{A} - 2\mathbf{E})^2 = \mathbf{O}$. Obviously, two linearly independent generalized eigenvectors are

$$\mathbf{r}_{10} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \mathbf{r}_{20} = \begin{pmatrix} 0 \\ 1 \end{pmatrix},$$

respectively. Therefore,

$$\begin{aligned} \mathbf{r}_{11} &= (\mathbf{A} - 2\mathbf{E})\mathbf{r}_{10} = \begin{pmatrix} 1 & -1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \\ \mathbf{r}_{21} &= (\mathbf{A} - 2\mathbf{E})\mathbf{r}_{20} = \begin{pmatrix} 1 & -1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ -1 \end{pmatrix}. \end{aligned}$$

Therefore, for the 2-fold eigenvalue $\lambda = 2$, two linearly independent solutions can be obtained, namely

$$e^{2t}(\mathbf{r}_{10} + t\mathbf{r}_{11}) = e^{2t} \begin{pmatrix} 1+t \\ t \end{pmatrix}, \quad e^{2t}(\mathbf{r}_{20} + t\mathbf{r}_{21}) = e^{2t} \begin{pmatrix} -t \\ 1-t \end{pmatrix}.$$

From this, we obtain the fundamental matrix

$$\Phi(t) = \begin{pmatrix} e^{2t}(t+1) & -e^{2t}t \\ e^{2t}t & e^{2t}(1-t) \end{pmatrix}.$$

Therefore, the general solution of the system of equations can be written as

$$\mathbf{x} = e^{t\mathbf{A}_1} \mathbf{c} = e^{2t} \begin{pmatrix} c_1(t+1) - c_2t \\ (c_1 - c_2)t + c_2 \end{pmatrix},$$

where c_1, c_2 are arbitrary constants. □

Problem 2.6.3. $\frac{d\mathbf{x}}{dt} = \begin{pmatrix} 2 & -3 \\ 1 & -2 \end{pmatrix} \mathbf{x}.$

Solution. Let $\mathbf{A} := \begin{pmatrix} 2 & -3 \\ 1 & -2 \end{pmatrix}$. Firstly, we can easily figure out that $\lambda_1 = -1, \lambda_2 = 1$ are its eigenvalues.

It is easy to obtain that the eigenvector corresponding to the eigenvalue $\lambda_1 = -1$ is $(-1, -1)^\top$, and the eigenvector corresponding to the eigenvalue $\lambda_2 = 1$ is $(3, 1)^\top$. From this, we obtain the fundamental matrix

$$\Phi(t) = \begin{pmatrix} -e^{-t} & 3e^t \\ -e^{-t} & e^t \end{pmatrix}.$$

Therefore, the general solution of the system of equations can be written as

$$\mathbf{x} = e^{t\mathbf{A}_1} \mathbf{c} = c_1 e^{-t} \begin{pmatrix} -1 \\ -1 \end{pmatrix} + c_2 e^t \begin{pmatrix} 3 \\ 1 \end{pmatrix},$$

where c_1, c_2 are arbitrary constants. □

Problem 2.6.4. $\frac{d\mathbf{x}}{dt} = \begin{pmatrix} 7 & 1 \\ -4 & 3 \end{pmatrix} \mathbf{x}.$

Solution. Let $\mathbf{A} := \begin{pmatrix} 7 & 1 \\ -4 & 3 \end{pmatrix}$. Firstly, we can easily figure out that $\lambda = 5$ are its 2-fold eigenvalues.

For the 2-fold eigenvalues $\lambda = 5$, consider the generalized characteristic equation system $(\mathbf{A} - 5\mathbf{E})^2 \mathbf{r} = \mathbf{0}$. We can easily figure out $(\mathbf{A} - 5\mathbf{E})^2 = \mathbf{O}$. Obviously, two linearly independent generalized eigenvectors are

$$\mathbf{r}_{10} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \mathbf{r}_{20} = \begin{pmatrix} 0 \\ 1 \end{pmatrix},$$

respectively. Therefore,

$$\begin{aligned} \mathbf{r}_{11} &= (\mathbf{A} - 5\mathbf{E})\mathbf{r}_{10} = \begin{pmatrix} 2 & 1 \\ -4 & -2 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 2 \\ -4 \end{pmatrix}, \\ \mathbf{r}_{21} &= (\mathbf{A} - 5\mathbf{E})\mathbf{r}_{20} = \begin{pmatrix} 2 & 1 \\ -4 & -2 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ -2 \end{pmatrix}. \end{aligned}$$

Therefore, for the 2-fold eigenvalue $\lambda = 5$, two linearly independent solutions can be obtained,

namely

$$e^{5t}(\mathbf{r}_{10} + t\mathbf{r}_{11}) = e^{5t} \begin{pmatrix} 1 + 2t \\ -4t \end{pmatrix}, \quad e^{5t}(\mathbf{r}_{20} + t\mathbf{r}_{21}) = e^{5t} \begin{pmatrix} t \\ 1 - 2t \end{pmatrix}.$$

From this, we obtain the fundamental matrix

$$\Phi(t) = \begin{pmatrix} e^{5t}(1 + 2t) & e^{5t}t \\ -4e^{5t}t & e^{5t}(1 - 2t) \end{pmatrix}.$$

Therefore, the general solution of the system of equations can be written as

$$\mathbf{x} = e^{t\mathbf{A}}\mathbf{c} = c_1 e^{5t} \begin{pmatrix} 1 + 2t \\ -4t \end{pmatrix} + c_2 e^{5t} \begin{pmatrix} t \\ 1 - 2t \end{pmatrix},$$

where c_1, c_2 are arbitrary constants. □

Problem 2.6.5. $\frac{d\mathbf{x}}{dt} = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & -1 \\ 1 & -1 & 0 \end{pmatrix} \mathbf{x}.$

Solution. Let $\mathbf{A} := \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & -1 \\ 1 & -1 & 0 \end{pmatrix}$. Firstly, we can easily figure out that $\lambda_1 = -2, \lambda_2 = \lambda_3 = 1$ are its eigenvalues.

For the single eigenvalue $\lambda = -2$, we can easily determine that its eigenvector is $(-1, 1, 1)^\top$. For the 2-fold eigenvalues $\lambda = 1$, consider the generalized characteristic equation system $(\mathbf{A} -$

$\mathbf{E})^2 \mathbf{r} = \mathbf{0}$. We can easily figure out $(\mathbf{A} - \mathbf{E})^2 = \begin{pmatrix} 3 & -3 & -3 \\ -3 & 3 & 3 \\ -3 & 3 & 3 \end{pmatrix}$, with $\text{rank}((\mathbf{A} - \mathbf{E})^2) = 1$.

Therefore, we can figure out two linearly independent generalized eigenvectors, namely,

$$\mathbf{r}_{10} = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \quad \mathbf{r}_{20} = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix},$$

respectively. Therefore,

$$\begin{aligned} \mathbf{r}_{11} = (\mathbf{A} - \mathbf{E})\mathbf{r}_{10} &= \begin{pmatrix} -1 & 1 & 1 \\ 1 & -1 & -1 \\ 1 & -1 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}, \\ \mathbf{r}_{21} = (\mathbf{A} - 2\mathbf{E})\mathbf{r}_{20} &= \begin{pmatrix} -1 & 1 & 1 \\ 1 & -1 & -1 \\ 1 & -1 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}. \end{aligned}$$

Therefore, for the 2-fold eigenvalue $\lambda = 2$, two linearly independent solutions can be obtained, namely

$$\mathbf{e}^t \mathbf{r}_{10} = \mathbf{e}^t \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \quad \mathbf{e}^t \mathbf{r}_{20} = \mathbf{e}^t \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}.$$

From this, we obtain the fundamental matrix

$$\Phi(t) = \begin{pmatrix} -\mathbf{e}^{-2t} & \mathbf{e}^t & \mathbf{e}^t \\ \mathbf{e}^{-2t} & \mathbf{e}^t & 0 \\ \mathbf{e}^{-2t} & 0 & \mathbf{e}^t \end{pmatrix}.$$

Therefore, the general solution of the system of equations can be written as

$$\mathbf{x} = \Phi(t)\mathbf{c} = c_1 \mathbf{e}^{-2t} \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix} + c_2 \mathbf{e}^t \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} + c_3 \mathbf{e}^t \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix},$$

where c_1, c_2, c_3 are arbitrary constants. □

Problem 2.6.6. $\frac{d\mathbf{x}}{dt} = \begin{pmatrix} 2 & 0 & 0 \\ -7 & 9 & 7 \\ 0 & 0 & 2 \end{pmatrix} \mathbf{x}.$

Solution. Let $\mathbf{A} := \begin{pmatrix} 2 & 0 & 0 \\ -7 & 9 & 7 \\ 0 & 0 & 2 \end{pmatrix}$. Firstly, we can easily figure out that $\lambda_1 = 9, \lambda_2 = \lambda_3 = 2$ are

its eigenvalues.

For the single eigenvalue $\lambda = 9$, we can easily determine that its eigenvector is $(0, 1, 0)^\top$. For the 2-fold eigenvalues $\lambda = 2$, consider the generalized characteristic equation system $(\mathbf{A} - 2\mathbf{E})^2 \mathbf{r} = \mathbf{0}$. We can easily figure out $(\mathbf{A} - 2\mathbf{E})^2 = \begin{pmatrix} 0 & 0 & 0 \\ -49 & 49 & 49 \\ 0 & 0 & 0 \end{pmatrix}$, with $\text{rank}((\mathbf{A} - 2\mathbf{E})^2) =$

1. Therefore, we can figure out two linearly independent generalized eigenvectors, namely,

$$\mathbf{r}_{10} = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \quad \mathbf{r}_{20} = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix},$$

respectively. Therefore,

$$\begin{aligned} \mathbf{r}_{11} &= (\mathbf{A} - \mathbf{E})\mathbf{r}_{10} = \begin{pmatrix} 0 & 0 & 0 \\ -49 & 49 & 49 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}, \\ \mathbf{r}_{21} &= (\mathbf{A} - 2\mathbf{E})\mathbf{r}_{20} = \begin{pmatrix} 0 & 0 & 0 \\ -49 & 49 & 49 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}. \end{aligned}$$

Therefore, for the 2-fold eigenvalue $\lambda = 2$, two linearly independent solutions can be obtained, namely

$$\mathbf{e}^{2t}\mathbf{r}_{10} = \mathbf{e}^{2t} \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \quad \mathbf{e}^{2t}\mathbf{r}_{20} = \mathbf{e}^{2t} \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}.$$

From this, we obtain the fundamental matrix

$$\Phi(t) = \begin{pmatrix} \mathbf{e}^{9t} & \mathbf{e}^{2t} & \mathbf{e}^{2t} \\ 0 & \mathbf{e}^{2t} & 0 \\ \mathbf{e}^{9t} & 0 & \mathbf{e}^{2t} \end{pmatrix}.$$

Therefore, the general solution of the system of equations can be written as

$$\mathbf{x} = \Phi(t)\mathbf{c} = c_1 e^{9t} \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + c_2 e^{2t} \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} + c_3 e^{2t} \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix},$$

where c_1, c_2, c_3 are arbitrary constants. □

Problem 2.6.7. $\frac{d\mathbf{x}}{dt} = \begin{pmatrix} 0 & 1 & -1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix} \mathbf{x}.$

Solution. Let $\mathbf{A} := \begin{pmatrix} 0 & 1 & -1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}$. Firstly, we can easily figure out that $\lambda_1 = \lambda_2 = 1, \lambda_3 = 0$ are its eigenvalues.

For the single eigenvalue $\lambda = 0$, we can easily determine that its eigenvector is $(-1, 1, 1)^\top$. For the 2-fold eigenvalues $\lambda = 1$, consider the generalized characteristic equation system $(\mathbf{A} - \mathbf{E})^2 \mathbf{r} = \mathbf{0}$. We can easily figure out $(\mathbf{A} - \mathbf{E})^2 = \begin{pmatrix} 1 & -1 & 1 \\ -1 & 1 & -1 \\ -1 & 1 & -1 \end{pmatrix}$, with $\text{rank}((\mathbf{A} - \mathbf{E})^2) = 1$.

Therefore, we can figure out two linearly independent generalized eigenvectors, namely,

$$\mathbf{r}_{10} = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \quad \mathbf{r}_{20} = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix},$$

respectively. Therefore,

$$\begin{aligned} \mathbf{r}_{11} &= (\mathbf{A} - \mathbf{E})\mathbf{r}_{10} = \begin{pmatrix} -1 & 1 & -1 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, \\ \mathbf{r}_{21} &= (\mathbf{A} - \mathbf{E})\mathbf{r}_{20} = \begin{pmatrix} -1 & 1 & -1 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ -1 \\ -1 \end{pmatrix}. \end{aligned}$$

Therefore, for the 2-fold eigenvalue $\lambda = 1$, two linearly independent solutions can be obtained, namely

$$e^t(\mathbf{r}_{10} + t\mathbf{r}_{11}) = e^t \begin{pmatrix} 1 \\ 1+t \\ t \end{pmatrix}, \quad e^t(\mathbf{r}_{20} + t\mathbf{r}_{21}) = e^t \begin{pmatrix} -1 \\ -t \\ 1-t \end{pmatrix}.$$

From this, we obtain the fundamental matrix

$$\Phi(t) = \begin{pmatrix} 1 & e^t t & -e^t t \\ 1 & e^t(1+t) & -e^t t \\ 0 & e^t t & (1-t)e^t \end{pmatrix}.$$

Therefore, the general solution of the system of equations can be written as

$$\mathbf{x} = \Phi(t)\mathbf{c} = c_1 \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix} + c_2 e^t \begin{pmatrix} 1 \\ 1+t \\ t \end{pmatrix} + c_3 e^t \begin{pmatrix} -1 \\ -t \\ 1-t \end{pmatrix},$$

where c_1, c_2, c_3 are arbitrary constants. □

Problem 2.6.8. $\frac{d\mathbf{x}}{dt} = \begin{pmatrix} -3 & 0 & -4 \\ -1 & -1 & -1 \\ 1 & 0 & 1 \end{pmatrix} \mathbf{x}.$

Solution. Let $\mathbf{A} := \begin{pmatrix} -3 & 0 & -4 \\ -1 & -1 & -1 \\ 1 & 0 & 1 \end{pmatrix}$. Firstly, we can easily figure out that $\lambda_1 = \lambda_2 = \lambda_3 = -1$ are its eigenvalues.

For the 3-fold eigenvalues $\lambda = -1$, consider the generalized characteristic equation system $(\mathbf{A} + \mathbf{E})^3 \mathbf{r} = \mathbf{0}$. We can easily figure out $(\mathbf{A} + \mathbf{E})^3 = \mathbf{O}$. Obviously, three linearly independent generalized eigenvectors,

$$\mathbf{r}_{10} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad \mathbf{r}_{20} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad \mathbf{r}_{30} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

respectively. Therefore,

$$\mathbf{r}_{11} = (\mathbf{A} + \mathbf{E})\mathbf{r}_{10} = \begin{pmatrix} -2 & 0 & -4 \\ -1 & 0 & -1 \\ 1 & 0 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} -2 \\ -1 \\ 1 \end{pmatrix},$$

$$\mathbf{r}_{12} = (\mathbf{A} + \mathbf{E})\mathbf{r}_{11} = \begin{pmatrix} -2 & 0 & -4 \\ -1 & 0 & -1 \\ 1 & 0 & 2 \end{pmatrix} \begin{pmatrix} -2 \\ -1 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix},$$

$$\mathbf{r}_{21} = (\mathbf{A} + \mathbf{E})\mathbf{r}_{20} = \begin{pmatrix} -2 & 0 & -4 \\ -1 & 0 & -1 \\ 1 & 0 & 2 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix},$$

$$\mathbf{r}_{22} = (\mathbf{A} + \mathbf{E})\mathbf{r}_{21} = \begin{pmatrix} -2 & 0 & -4 \\ -1 & 0 & -1 \\ 1 & 0 & 2 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix},$$

$$\mathbf{r}_{31} = (\mathbf{A} + \mathbf{E})\mathbf{r}_{30} = \begin{pmatrix} -2 & 0 & -4 \\ -1 & 0 & -1 \\ 1 & 0 & 2 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} -4 \\ -1 \\ 2 \end{pmatrix},$$

$$\mathbf{r}_{32} = (\mathbf{A} + \mathbf{E})\mathbf{r}_{31} = \begin{pmatrix} -2 & 0 & -4 \\ -1 & 0 & -1 \\ 1 & 0 & 2 \end{pmatrix} \begin{pmatrix} -4 \\ -1 \\ 2 \end{pmatrix} = \begin{pmatrix} 0 \\ 2 \\ 0 \end{pmatrix}.$$

Therefore, for the 3-fold eigenvalue $\lambda = -1$, two linearly independent solutions can be obtained,

namely

$$\begin{aligned} e^{-t} \left(\mathbf{r}_{10} + t\mathbf{r}_{11} + \frac{t^2}{2}\mathbf{r}_{12} \right) &= e^{-t} \begin{pmatrix} 1 - 2t \\ -t + \frac{t^2}{2} \\ t \end{pmatrix}, \\ e^{-t} \left(\mathbf{r}_{20} + t\mathbf{r}_{21} + \frac{t^2}{2}\mathbf{r}_{22} \right) &= e^{-t} \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \\ e^{-t} \left(\mathbf{r}_{30} + t\mathbf{r}_{31} + \frac{t^2}{2}\mathbf{r}_{32} \right) &= e^{-t} \begin{pmatrix} -4t \\ t^2 - t \\ 2t + 1 \end{pmatrix}, \end{aligned}$$

From this, we obtain the fundamental matrix

$$\Phi(t) = \begin{pmatrix} e^{-t}(1 - 2t) & 0 & -4e^{-t}t \\ e^{-t} \left(\frac{t^2}{2} - t \right) & e^{-t} & e^{-t}(t^2 - t) \\ e^{-t}t & 0 & e^{-t}(2t + 1) \end{pmatrix}.$$

Therefore, the general solution of the system of equations can be written as

$$\mathbf{x} = \Phi(t)\mathbf{c} = c_1 e^{-t} \begin{pmatrix} 1 - 2t \\ \frac{t^2}{2} - t \\ t \end{pmatrix} + c_2 e^{-t} \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + c_3 e^{-t} \begin{pmatrix} -4t \\ t^2 - t \\ 2t + 1 \end{pmatrix},$$

where c_1, c_2, c_3 are arbitrary constants. □

Problem 2.6.9. $\frac{d\mathbf{x}}{dt} = \begin{pmatrix} 0 & 0 & 1 \\ -5 & -1 & -5 \\ 4 & 1 & -2 \end{pmatrix} \mathbf{x}.$

Solution. Let $\mathbf{A} := \begin{pmatrix} 0 & 0 & 1 \\ -5 & -1 & -5 \\ 4 & 1 & -2 \end{pmatrix}$. Firstly, we can easily figure out that $\lambda_1 = \lambda_2 = \lambda_3 = -1$

are its eigenvalues.

For the 3-fold eigenvalues $\lambda = -1$, consider the generalized characteristic equation system $(\mathbf{A} + \mathbf{E})^3 \mathbf{r} = \mathbf{0}$. We can easily figure out $(\mathbf{A} + \mathbf{E})^3 = \mathbf{O}$. Obviously, three linearly independent generalized eigenvectors,

$$\mathbf{r}_{10} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad \mathbf{r}_{20} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad \mathbf{r}_{30} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

respectively. Therefore,

$$\begin{aligned} \mathbf{r}_{11} &= (\mathbf{A} + \mathbf{E})\mathbf{r}_{10} = \begin{pmatrix} 1 & 0 & 1 \\ -5 & 0 & -5 \\ 4 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ -5 \\ 4 \end{pmatrix}, \\ \mathbf{r}_{12} &= (\mathbf{A} + \mathbf{E})\mathbf{r}_{11} = \begin{pmatrix} 1 & 0 & 1 \\ -5 & 0 & -5 \\ 4 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ -5 \\ 4 \end{pmatrix} = \begin{pmatrix} 5 \\ -25 \\ -5 \end{pmatrix}, \\ \mathbf{r}_{21} &= (\mathbf{A} + \mathbf{E})\mathbf{r}_{20} = \begin{pmatrix} 1 & 0 & 1 \\ -5 & 0 & -5 \\ 4 & 1 & -1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}, \\ \mathbf{r}_{22} &= (\mathbf{A} + \mathbf{E})\mathbf{r}_{21} = \begin{pmatrix} 1 & 0 & 1 \\ -5 & 0 & -5 \\ 4 & 1 & -1 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ -5 \\ -1 \end{pmatrix}, \\ \mathbf{r}_{31} &= (\mathbf{A} + \mathbf{E})\mathbf{r}_{30} = \begin{pmatrix} 1 & 0 & 1 \\ -5 & 0 & -5 \\ 4 & 1 & -1 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ -5 \\ -1 \end{pmatrix}, \\ \mathbf{r}_{32} &= (\mathbf{A} + \mathbf{E})\mathbf{r}_{31} = \begin{pmatrix} 1 & 0 & 1 \\ -5 & 0 & -5 \\ 4 & 1 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ -5 \\ -1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}. \end{aligned}$$

Therefore, for the 3-fold eigenvalue $\lambda = -1$, two linearly independent solutions can be obtained,

namely

$$\begin{aligned} e^{-t} \left(\mathbf{r}_{10} + t\mathbf{r}_{11} + \frac{t^2}{2}\mathbf{r}_{12} \right) &= e^{-t} \begin{pmatrix} 1 + t + 5/2t^2 \\ -5t - 25/2t^2 \\ 4t - 5/2t^2 \end{pmatrix}, \\ e^{-t} \left(\mathbf{r}_{20} + t\mathbf{r}_{21} + \frac{t^2}{2}\mathbf{r}_{22} \right) &= e^{-t} \begin{pmatrix} t^2/2 \\ 1 - 5/2t^2 \\ t - t^2/2 \end{pmatrix}, \\ e^{-t} \left(\mathbf{r}_{30} + t\mathbf{r}_{31} + \frac{t^2}{2}\mathbf{r}_{32} \right) &= e^{-t} \begin{pmatrix} t \\ -5t \\ 1 - t \end{pmatrix}, \end{aligned}$$

From this, we obtain the fundamental matrix

$$\Phi(t) = \begin{pmatrix} e^{-t} \left(\frac{5t^2}{2} + t + 1 \right) & \frac{1}{2}e^{-t}t^2 & e^{-t}t \\ e^{-t} \left(-\frac{25t^2}{2} - 5t \right) & e^{-t} \left(1 - \frac{5t^2}{2} \right) & -5e^{-t}t \\ e^{-t} \left(4t - \frac{5t^2}{2} \right) & e^{-t} \left(t - \frac{t^2}{2} \right) & e^{-t}(1 - t) \end{pmatrix}.$$

Therefore, the general solution of the system of equations can be written as

$$\mathbf{x} = \Phi(t)\mathbf{c} = c_1 e^{-t} \begin{pmatrix} 1 + t + 5/2t^2 \\ -5t - 25/2t^2 \\ 4t - 5/2t^2 \end{pmatrix} + c_2 e^{-t} \begin{pmatrix} t^2/2 \\ 1 - 5/2t^2 \\ t - t^2/2 \end{pmatrix} + c_3 e^{-t} \begin{pmatrix} t \\ -5t \\ 1 - t \end{pmatrix},$$

where c_1, c_2, c_3 are arbitrary constants. □

Problem 2.6.10. $\frac{d\mathbf{x}}{dt} = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 3 & 1 \\ -2 & -4 & -1 \end{pmatrix} \mathbf{x}.$

Solution. Let $\mathbf{A} := \begin{pmatrix} 1 & 0 & 0 \\ 1 & 3 & 1 \\ -2 & -4 & -1 \end{pmatrix}$. Firstly, we can easily figure out that $\lambda_1 = \lambda_2 = \lambda_3 = 1$

are its eigenvalues.

For the 3-fold eigenvalues $\lambda = 1$, consider the generalized characteristic equation system $(\mathbf{A} - \mathbf{E})^3 \mathbf{r} = \mathbf{0}$. We can easily figure out $(\mathbf{A} - \mathbf{E})^3 = \mathbf{O}$. Obviously, three linearly independent generalized eigenvectors,

$$\mathbf{r}_{10} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad \mathbf{r}_{20} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad \mathbf{r}_{30} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

respectively. Therefore,

$$\mathbf{r}_{11} = (\mathbf{A} - \mathbf{E})\mathbf{r}_{10} = \begin{pmatrix} 0 & 0 & 0 \\ 1 & 2 & 1 \\ -2 & -4 & -2 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ -2 \end{pmatrix},$$

$$\mathbf{r}_{12} = (\mathbf{A} - \mathbf{E})\mathbf{r}_{11} = \begin{pmatrix} 0 & 0 & 0 \\ 1 & 2 & 1 \\ -2 & -4 & -2 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ -2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix},$$

$$\mathbf{r}_{21} = (\mathbf{A} - \mathbf{E})\mathbf{r}_{20} = \begin{pmatrix} 0 & 0 & 0 \\ 1 & 2 & 1 \\ -2 & -4 & -2 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 2 \\ -4 \end{pmatrix},$$

$$\mathbf{r}_{22} = (\mathbf{A} - \mathbf{E})\mathbf{r}_{21} = \begin{pmatrix} 0 & 0 & 0 \\ 1 & 2 & 1 \\ -2 & -4 & -2 \end{pmatrix} \begin{pmatrix} 0 \\ 2 \\ -4 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix},$$

$$\mathbf{r}_{31} = (\mathbf{A} - \mathbf{E})\mathbf{r}_{30} = \begin{pmatrix} 0 & 0 & 0 \\ 1 & 2 & 1 \\ -2 & -4 & -2 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ -2 \end{pmatrix},$$

$$\mathbf{r}_{32} = (\mathbf{A} - \mathbf{E})\mathbf{r}_{31} = \begin{pmatrix} 0 & 0 & 0 \\ 1 & 2 & 1 \\ -2 & -4 & -2 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ -2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}.$$

Therefore, for the 3-fold eigenvalue $\lambda = 1$, two linearly independent solutions can be obtained,

namely

$$\begin{aligned} e^t \left(\mathbf{r}_{10} + t\mathbf{r}_{11} + \frac{t^2}{2}\mathbf{r}_{12} \right) &= e^t \begin{pmatrix} 1 \\ t \\ -2t \end{pmatrix}, \\ e^t \left(\mathbf{r}_{20} + t\mathbf{r}_{21} + \frac{t^2}{2}\mathbf{r}_{22} \right) &= e^t \begin{pmatrix} 0 \\ 1+2t \\ -4t \end{pmatrix}, \\ e^t \left(\mathbf{r}_{30} + t\mathbf{r}_{31} + \frac{t^2}{2}\mathbf{r}_{32} \right) &= e^t \begin{pmatrix} 0 \\ t \\ 1-2t \end{pmatrix}, \end{aligned}$$

From this, we obtain the fundamental matrix

$$\Phi(t) = \begin{pmatrix} e^t & 0 & 0 \\ te^t & e^t(1+2t) & te^t \\ -2te^t & -4te^t & (1-2t)e^t \end{pmatrix}.$$

Therefore, the general solution of the system of equations can be written as

$$\mathbf{x} = \Phi(t)\mathbf{c} = c_1 e^t \begin{pmatrix} 1 \\ t \\ -2t \end{pmatrix} + c_2 e^t \begin{pmatrix} 0 \\ 1+2t \\ -4t \end{pmatrix} + c_3 e^t \begin{pmatrix} 0 \\ t \\ 1-2t \end{pmatrix},$$

where c_1, c_2, c_3 are arbitrary constants. □

Problem 2.6.11. $\frac{d\mathbf{x}}{dt} = \begin{pmatrix} 0 & 1 & -5 \\ -5 & 3 & 0 \\ 1 & 0 & -3 \end{pmatrix} \mathbf{x}.$

Solution. Let $\mathbf{A} := \begin{pmatrix} 0 & 1 & -5 \\ -5 & 3 & 0 \\ 1 & 0 & -3 \end{pmatrix}$. Firstly, we can easily figure out that $\lambda_1 = i, \lambda_2 = -i, \lambda_3 = 0$

are its eigenvalues.

For the singular eigenvalue $\lambda_1 = 0$, the corresponding eigenvector is $(3, 5, 1)^\top$; for the singular eigenvalue $\lambda_2 = -i$, the corresponding eigenvector is $(3 - i, 4 - 3i, 1)^\top$; for the singular eigenvalue $\lambda_3 = i$, the corresponding eigenvector is $(3 + i, 4 + 3i, 1)^\top$. Therefore, the fundamental matrix is

$$\Phi(t) = \begin{pmatrix} 3 & 3 \cos t - \sin t & 3 \sin t + \cos t \\ 5 & 4 \cos t - 3 \sin t & 4 \sin t + 3 \cos t \\ 1 & \cos t & \sin t \end{pmatrix}.$$

Therefore, the general solution is

$$\mathbf{x}(t) = \Phi(t)\mathbf{c} = c_1 \begin{pmatrix} 3 \\ 5 \\ 1 \end{pmatrix} + c_2 \begin{pmatrix} 3 \cos t - \sin t \\ 4 \cos t - 3 \sin t \\ \cos t \end{pmatrix} + c_3 \begin{pmatrix} 3 \sin t + \cos t \\ 4 \sin t + 3 \cos t \\ \sin t \end{pmatrix},$$

where c_1, c_2, c_3 are arbitrary constants. □

Problem 2.6.12. $\frac{d\mathbf{x}}{dt} = \begin{pmatrix} -1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{pmatrix} \mathbf{x}.$

Solution. Let $\mathbf{A} := \begin{pmatrix} -1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{pmatrix}$. Firstly, we can easily figure out that $\lambda_1 = \lambda_2 = -2, \lambda_3 = 1$ are its eigenvalues.

For the single eigenvalue $\lambda = 1$, we can easily determine that its eigenvector is $(1, 1, 1)^\top$. For the 2-fold eigenvalues $\lambda = -2$, consider the generalized characteristic equation system

$$(\mathbf{A} + 2\mathbf{E})^2 \mathbf{r} = \mathbf{0}. \text{ We can easily figure out } (\mathbf{A} + 2\mathbf{E})^2 = \begin{pmatrix} 3 & 3 & 3 \\ 3 & 3 & 3 \\ 3 & 3 & 3 \end{pmatrix}, \text{ with rank } ((\mathbf{A} + 2\mathbf{E})^2) =$$

1. Therefore, we can figure out two linearly independent generalized eigenvectors, namely,

$$\mathbf{r}_{10} = \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}, \quad \mathbf{r}_{20} = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix},$$

respectively. Therefore,

$$\begin{aligned} \mathbf{r}_{11} = (\mathbf{A} + 2\mathbf{E})\mathbf{r}_{10} &= \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}, \\ \mathbf{r}_{21} = (\mathbf{A} + 2\mathbf{E})\mathbf{r}_{20} &= \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}. \end{aligned}$$

Therefore, for the 2-fold eigenvalue $\lambda = -2$, two linearly independent solutions can be obtained, namely

$$e^{-2t}\mathbf{r}_{10} = e^{-2t} \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}, \quad e^{-2t}\mathbf{r}_{20} = e^{-2t} \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}.$$

From this, we obtain the fundamental matrix

$$\Phi(t) = \begin{pmatrix} e^t & -e^{-2t} & -e^{-2t} \\ e^t & e^{2t} & 0 \\ e^t & 0 & e^{-2t} \end{pmatrix}.$$

Therefore, the general solution of the system of equations can be written as

$$\mathbf{x} = \Phi(t)\mathbf{c} = c_1 e^t \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + c_2 e^{-2t} \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} + c_3 e^{-2t} \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix},$$

where c_1, c_2, c_3 are arbitrary constants. □

Find general solutions of the nonhomogeneous systems in Problems 15 through 20.

Problem 2.6.13. $\frac{d\mathbf{x}}{dt} = \begin{pmatrix} 0 & -n^2 \\ -n^2 & 0 \end{pmatrix} \mathbf{x} + \begin{pmatrix} \cos nt \\ \sin nt \end{pmatrix}.$

Solution. Step 1 **Solve the fundamental matrix corresponding to the homogeneous linear differential equation system.**

Let $\mathbf{A} := \begin{pmatrix} 0 & -n^2 \\ -n^2 & 0 \end{pmatrix}$. The eigenvalues of $\mathbf{A}$ can be easily obtained as $\lambda_1 = n^2$ and

$\lambda_2 = -n^2$, and the corresponding eigenvectors are $(-1, 1)^\top$ and $(1, 1)^\top$. Therefore, its fundamental solution matrix is

$$\Phi(t) = \begin{pmatrix} -e^{n^2 t} & e^{-n^2 t} \\ e^{n^2 t} & e^{-n^2 t} \end{pmatrix}.$$

Step 2 Write the expression of the general solution. The general solution is

$$\begin{aligned} \mathbf{x}(t) &= \Phi(t) \left[\mathbf{C} + \int \Phi^{-1}(t) \mathbf{f}(t) dt \right] \\ &= \begin{pmatrix} -e^{n^2 t} & e^{-n^2 t} \\ e^{n^2 t} & e^{-n^2 t} \end{pmatrix} \left[\begin{pmatrix} c_1 \\ c_2 \end{pmatrix} + \int \begin{pmatrix} -e^{n^2 t} & e^{-n^2 t} \\ e^{n^2 t} & e^{-n^2 t} \end{pmatrix}^{-1} \begin{pmatrix} \cos nt \\ \sin nt \end{pmatrix} dt \right] \\ &= \begin{pmatrix} \frac{(n+1)\sin(nt)}{n^3+n} + c_1(-e^{n^2 t}) + c_2 e^{-n^2 t} \\ \frac{(n-1)\cos(nt)}{n^3+n} + c_1 e^{n^2 t} + c_2 e^{-n^2 t} \end{pmatrix}. \end{aligned} \quad \square$$

Problem 2.6.14. $\frac{d\mathbf{x}}{dt} = \begin{pmatrix} 2 & -1 \\ 1 & 0 \end{pmatrix} \mathbf{x} + \begin{pmatrix} 0 \\ 2e^t \end{pmatrix}.$

Solution. **Step 1 Solve the fundamental matrix corresponding to the homogeneous linear differential equation system.**

Let $\mathbf{A} := \begin{pmatrix} 2 & -1 \\ -1 & 0 \end{pmatrix}$. The eigenvalues of $\mathbf{A}$ can be easily obtained as $\lambda_1 = \lambda_2 = 1$.

Consider finding the generalized eigenvectors. Noting that $(\mathbf{A} - \mathbf{E})^2 = \mathbf{O}$, we can obtain two linearly independent generalized eigenvectors as follows:

$$\mathbf{r}_{10} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \mathbf{r}_{20} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

Therefore,

$$\begin{aligned} \mathbf{r}_{11} &= (\mathbf{A} - \mathbf{E})\mathbf{r}_{10} = \begin{pmatrix} 1 & -1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \\ \mathbf{r}_{21} &= (\mathbf{A} - \mathbf{E})\mathbf{r}_{20} = \begin{pmatrix} 1 & -1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ -1 \end{pmatrix}. \end{aligned}$$

Therefore, for the 2-fold eigenvalue $\lambda = 1$, two linearly independent solutions can be

obtained, namely

$$e^t(\mathbf{r}_{10} + t\mathbf{r}_{11}) = e^t \begin{pmatrix} 1+t \\ t \end{pmatrix}, \quad e^t(\mathbf{r}_{20} + t\mathbf{r}_{21}) = e^t \begin{pmatrix} -t \\ 1-t \end{pmatrix}.$$

From this, we obtain the fundamental matrix

$$\Phi(t) = \begin{pmatrix} e^t(1+t) & -e^t t \\ e^t t & e^t(1-t) \end{pmatrix}.$$

Step 2 Write the expression of the general solution. The general solution is

$$\begin{aligned} \mathbf{x}(t) &= \Phi(t) \left[\mathbf{C} + \int \Phi^{-1}(t) \mathbf{f}(t) dt \right] \\ &= \begin{pmatrix} e^t(1+t) & -e^t t \\ e^t t & e^t(1-t) \end{pmatrix} \left[\begin{pmatrix} c_1 \\ c_2 \end{pmatrix} + \int \begin{pmatrix} e^t(1+t) & -e^t t \\ e^t t & e^t(1-t) \end{pmatrix}^{-1} \begin{pmatrix} 0 \\ 2e^t \end{pmatrix} dt \right] \\ &= \begin{pmatrix} -e^t(t^2 + (c_2 - c_1)t - c_1) \\ -e^t(t^2 + (-2 - c_1 + c_2)t - c_2) \end{pmatrix}. \end{aligned} \quad \square$$

Problem 2.6.15. $\frac{d\mathbf{x}}{dt} = \begin{pmatrix} -1 & -1 & 0 \\ 0 & -1 & -1 \\ 0 & 0 & -1 \end{pmatrix} \mathbf{x} + \begin{pmatrix} t^2 \\ 2t \\ t \end{pmatrix}.$

Solution. **Step 1 Solve the fundamental matrix corresponding to the homogeneous linear differential equation system.**

Let $\mathbf{A} := \begin{pmatrix} -1 & -1 & 0 \\ 0 & -1 & -1 \\ 0 & 0 & -1 \end{pmatrix}$. Obviously, $\mathbf{A}$ have 3-fold eigenvalues $\lambda = -1$. Consider

finding the generalized eigenvectors. Noting that $(\mathbf{A} + \mathbf{E})^3 = \mathbf{O}$, we can obtain two linearly independent generalized eigenvectors as follows:

$$\mathbf{r}_{10} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad \mathbf{r}_{20} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad \mathbf{r}_{30} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}.$$

Therefore,

$$\begin{aligned} \mathbf{r}_{11} &= (\mathbf{A} + \mathbf{E})\mathbf{r}_{10} = \begin{pmatrix} 0 & -1 & 0 \\ 0 & 0 & -1 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}, \\ \mathbf{r}_{21} &= (\mathbf{A} + \mathbf{E})\mathbf{r}_{20} = \begin{pmatrix} 0 & -1 & 0 \\ 0 & 0 & -1 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} -1 \\ 0 \\ 0 \end{pmatrix}, \\ \mathbf{r}_{22} &= (\mathbf{A} + \mathbf{E})\mathbf{r}_{20} = \begin{pmatrix} 0 & -1 & 0 \\ 0 & 0 & -1 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} -1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}, \\ \mathbf{r}_{31} &= (\mathbf{A} + \mathbf{E})\mathbf{r}_{30} = \begin{pmatrix} 0 & -1 & 0 \\ 0 & 0 & -1 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ -1 \\ 0 \end{pmatrix}, \\ \mathbf{r}_{32} &= (\mathbf{A} + \mathbf{E})\mathbf{r}_{31} = \begin{pmatrix} 0 & -1 & 0 \\ 0 & 0 & -1 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ -1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \end{aligned}$$

Therefore, for the 3-fold eigenvalue $\lambda = -1$, three linearly independent solutions can be

obtained, namely

$$\begin{aligned} e^{-t} \mathbf{r}_{10} &= e^{-t} \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \\ e^{-t} (\mathbf{r}_{20} + t\mathbf{r}_{21}) &= e^{-t} \begin{pmatrix} -t \\ 1 \\ 0 \end{pmatrix}, \\ e^{-t} \left(\mathbf{r}_{30} + t\mathbf{r}_{31} + \frac{t^2}{2}\mathbf{r}_{32} \right) &= e^{-t} \begin{pmatrix} t^2/2 \\ -t \\ 1 \end{pmatrix}, \end{aligned}$$

From this, we obtain the fundamental matrix

$$\Phi(t) = \begin{pmatrix} e^{-t} & -e^{-t}t & e^{-t}t^2/2 \\ 0 & e^{-t} & -e^{-t}t \\ 0 & 0 & e^{-t} \end{pmatrix}.$$

Step 2 Write the expression of the general solution. The general solution is

$$\begin{aligned} \mathbf{x}(t) &= \Phi(t) \left[\mathbf{C} + \int \Phi^{-1}(t) \mathbf{f}(t) dt \right] \\ &= \begin{pmatrix} e^{-t} & -e^{-t}t & e^{-t}t^2/2 \\ 0 & e^{-t} & -e^{-t}t \\ 0 & 0 & e^{-t} \end{pmatrix} \left[\begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix} + \int \begin{pmatrix} e^{-t} & -e^{-t}t & e^{-t}t^2/2 \\ 0 & e^{-t} & -e^{-t}t \\ 0 & 0 & e^{-t} \end{pmatrix}^{-1} \begin{pmatrix} t^2 \\ 2t \\ t \end{pmatrix} dt \right] \\ &= \begin{pmatrix} t^2 - 3t + \frac{1}{2}e^{-t}t(c_3t - 2c_2) + c_1e^{-t} + 3 \\ e^{-t}(e^t t - c_3t + c_2) \\ t + c_3e^{-t} - 1 \end{pmatrix}. \end{aligned}$$

□

Problem 2.6.16. $\frac{d\mathbf{x}}{dt} = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix} \mathbf{x} + \begin{pmatrix} 3 \\ -2 \end{pmatrix}.$

Solution. $\mathbf{x}(t) = \begin{pmatrix} \frac{1}{6} (3(c_1 - c_2)e^{-t} + 3(c_1 + c_2)e^{3t} + 14) \\ \frac{1}{6}e^{-t} (-16e^t + 3(c_1 + c_2)e^{4t} - 3c_1 + 3c_2) \end{pmatrix}.$ □

Problem 2.6.17. $\frac{d\mathbf{x}}{dt} = \begin{pmatrix} 2 & 3 \\ 2 & 1 \end{pmatrix} \mathbf{x} + \begin{pmatrix} 5 \\ -2t \end{pmatrix}.$

Solution. $\mathbf{x}(t) = \begin{pmatrix} \frac{1}{40} (60t + 8(2c_1 - 3c_2)e^{-t} + 24(c_1 + c_2)e^{4t} + 5) \\ \frac{1}{20}e^{-t} (-5e^t(4t + 5) + 8(c_1 + c_2)e^{5t} - 8c_1 + 12c_2) \end{pmatrix}.$ □

Problem 2.6.18. $\frac{d\mathbf{x}}{dt} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \mathbf{x} + \begin{pmatrix} 2e^t \\ -3e^t \end{pmatrix}.$

Solution. $\mathbf{x}(t) = \begin{pmatrix} \frac{1}{4}e^t (10t + 2c_1(e^{2t} + 1) + 2c_2e^{2t} + 1 - 2c_2) \\ \frac{1}{4}e^t (-10t + 2c_1(e^{2t} - 1) + 2c_2e^{2t} + 1 + 2c_2) \end{pmatrix}.$ □

Apply the method of undetermined coefficients to solve the initial value problems 21 through 25.

Problem 2.6.19. $\frac{d\mathbf{x}}{dt} = \begin{pmatrix} 3 & 4 \\ 3 & 2 \end{pmatrix} \mathbf{x} + \begin{pmatrix} 0 \\ t^2 \end{pmatrix}, \begin{pmatrix} x_1(0) \\ x_2(0) \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$

Solution. Let $\mathbf{A} := \begin{pmatrix} 3 & 4 \\ 3 & 2 \end{pmatrix}$. The eigenvalues of $\mathbf{A}$ can be easily obtained as $\lambda_1 = 6, \lambda_2 = -1$.

For the single root $\lambda_1 = 6$, the solution of the system of equations is of the form $\mathbf{x}(t) = e^{6t} \begin{pmatrix} u_1 \\ u_2 \end{pmatrix}$, where u_1, u_2 , and u_3 are undetermined constants. Substituting this into the original differential equation system yields the linear algebraic system

$$\begin{cases} 6u_1 = 3u_1 + 4u_2, \\ 6u_2 = 3u_1 + 2u_2. \end{cases}$$

We obtain $3u_1 = 4u_2$, so for the single root $\lambda_1 = 6$, we get the solution of the system of equations containing any constant C_1 as $\mathbf{x}(t) = C_1 e^{6t} \begin{pmatrix} 4 \\ 3 \end{pmatrix}$. Similarly, for the single root $\lambda_2 = -1$, we obtain the solution of the system of equations containing any constant C_1 as

$\mathbf{x}(t) = C_2 e^{-t} \begin{pmatrix} -1 \\ 1 \end{pmatrix}$. Therefore, the fundamental matrix of the homogeneous system is

$$\Phi(t) = \begin{pmatrix} 4e^{6t} & -e^{-t} \\ 3e^{6t} & e^{-t} \end{pmatrix},$$

and thus, the solution of the system is

$$\begin{aligned} \mathbf{x}(t) &= \Phi(t)\Phi^{-1}(0)\mathbf{y}_0 + \Phi(t) \int_0^t \Phi^{-1}(s)\mathbf{f}(s) ds \\ &= \begin{pmatrix} 4e^{6t} & -3e^{-t} \\ 3e^{6t} & e^{-t} \end{pmatrix} \begin{pmatrix} 4e^{6t} & -3e^{-t} \\ 3e^{6t} & e^{-t} \end{pmatrix}^{-1} \begin{pmatrix} 0 \\ 0 \end{pmatrix} + \begin{pmatrix} 4e^{6t} & -3e^{-t} \\ 3e^{6t} & e^{-t} \end{pmatrix} \int_0^t \begin{pmatrix} 4e^{6s} & -3e^{-s} \\ 3e^{6s} & e^{-s} \end{pmatrix}^{-1} \begin{pmatrix} 0 \\ s^2 \end{pmatrix} ds \\ &= \begin{pmatrix} \frac{1}{189}e^{-t}(-7e^t(18t^2 - 30t + 31) + e^{7t} + 216) \\ \frac{1}{252}e^{-t}(7e^t(18t^2 - 42t + 41) + e^{7t} - 288) \end{pmatrix}. \end{aligned}$$

Problem 2.6.20. $\frac{d\mathbf{x}}{dt} = \begin{pmatrix} 4 & 1 \\ 6 & -1 \end{pmatrix} \mathbf{x} + \begin{pmatrix} e^t \\ -e^t \end{pmatrix}, \begin{pmatrix} x_1(0) \\ x_2(0) \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$

Solution. $\mathbf{x}(t) = \begin{pmatrix} \frac{1}{84}e^{-2t}(-7e^{3t} + 99e^{7t} - 8) \\ \frac{4e^{-2t}}{7} - \frac{3e^t}{4} + \frac{33e^{5t}}{28} \end{pmatrix}$ □

Problem 2.6.21. $\frac{d\mathbf{x}}{dt} = \begin{pmatrix} 2 & 4 \\ 1 & 2 \end{pmatrix} \mathbf{x} + \begin{pmatrix} 2 \\ 3 \end{pmatrix}, \begin{pmatrix} x_1(0) \\ x_2(0) \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$

Solution. $\mathbf{x}(t) = \begin{pmatrix} \frac{1}{2}(-4t + e^{4t} + 1) \\ \frac{1}{4}(4t + e^{4t} - 5) \end{pmatrix}.$ □

Problem 2.6.22. $\frac{d\mathbf{x}}{dt} = \begin{pmatrix} 4 & -3 \\ 2 & -1 \end{pmatrix} \mathbf{x} + \begin{pmatrix} \sin t \\ -2\cos t \end{pmatrix}, \begin{pmatrix} x_1(0) \\ x_2(0) \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$

Solution. $\mathbf{x}(t) = \begin{pmatrix} e^t(3e^t - 4) - 2\sin(t) + \cos(t) \\ 2(e^t(e^t - 2) - \sin(t) + \cos(t)) \end{pmatrix}.$ □

Problem 2.6.23. $\frac{d\mathbf{x}}{dt} = \begin{pmatrix} 0 & -2 \\ 2 & 0 \end{pmatrix} \mathbf{x} + \begin{pmatrix} 3t \\ 2 \end{pmatrix}, \begin{pmatrix} x_1(0) \\ x_2(0) \end{pmatrix} = \begin{pmatrix} 2 \\ 3 \end{pmatrix}.$

Solution. $\mathbf{x}(t) = \begin{pmatrix} 3 \sin(2t) + \frac{9}{4} \cos(2t) - \frac{1}{4} \\ \frac{3}{4}(2t + 3 \sin(2t) + 4 \cos(2t)) \end{pmatrix}.$ □

Problem 2.6.24. If $\mathbf{A} = \begin{pmatrix} \alpha & \beta \\ -\beta & \alpha \end{pmatrix}$, show that

$$e^{\mathbf{A}t} = e^{\alpha t} \begin{pmatrix} \cos \beta t & \sin \beta t \\ -\sin \beta t & \cos \beta t \end{pmatrix}.$$

Proof. The eigenvalues of $\mathbf{A}$ are $\lambda_1 = \alpha - i\beta, \lambda_2 = \alpha + i\beta$, then its Jordan canonical form is

$\mathbf{J} := \text{diag}\{\alpha - i\beta, \alpha + i\beta\}$, where the transition matrix $\mathbf{P}$ is $\begin{bmatrix} i & -i \\ 1 & 1 \end{bmatrix}$. Therefore,

$$e^{t\mathbf{A}} = \mathbf{P}e^{t\mathbf{J}}\mathbf{P}^{-1} = e^{\alpha t} \begin{pmatrix} \cos \beta t & \sin \beta t \\ -\sin \beta t & \cos \beta t \end{pmatrix}. \quad \square$$

Problem 2.6.25. If $\mathbf{A}^2 = \alpha\mathbf{A}$, show that $e^{\mathbf{A}t} = \mathbf{E} + \left[\frac{e^{\alpha t} - 1}{\alpha} \right] \mathbf{A}$.

Proof. Note that $\mathbf{A}^n = \alpha^{n-1}\mathbf{A} (n \geq 2)$, we have

$$e^{t\mathbf{A}} = \sum_{i=0}^{\infty} \frac{\mathbf{A}^i}{i!} = \mathbf{E} + \mathbf{A} + \sum_{i=2}^{\infty} \frac{\mathbf{A}^i}{i!} = \mathbf{E} + \mathbf{A} + \sum_{i=2}^{\infty} \frac{\alpha^{i-1}\mathbf{A}}{i!} = \mathbf{E} + \left(\frac{e^{\alpha t} - 1}{\alpha} \right) \mathbf{A}. \quad \square$$

Problem 2.6.26. Show the following:

- (1) Every solution of the differential system $\frac{d\mathbf{x}}{dt} = \mathbf{A}\mathbf{x}$ tends to zero as $t \rightarrow +\infty$, if and only if the real parts of the eigenvalues of the matrix $\mathbf{A}$ are negative.
- (2) Every solution of the differential system $\frac{d\mathbf{x}}{dt} = \mathbf{A}\mathbf{x}$ is bounded in the interval $[0, \infty)$ if and only if the real parts of the multiple eigenvalues of the matrix $\mathbf{A}$ are negative, and the real parts of the simple eigenvalues of the matrix $\mathbf{A}$ are nonpositive.

Proof. (1) • (Method 1) We can write $\mathbf{y}$ as

$$\mathbf{y}(t) = e^{t\mathbf{A}}\mathbf{y}_0.$$

Using the Jordan decomposition, we have

$$e^{t\mathbf{A}} = \mathbf{P}e^{t\mathbf{D}} \left(1 + t\mathbf{N} + \cdots + \frac{t^n \mathbf{N}^n}{n!} \right) \mathbf{P}^{-1}.$$

Here

$$e^{tD} = \begin{pmatrix} e^{t\lambda_1} & & & \\ & e^{t\lambda_2} & & \\ & & \ddots & \\ & & & e^{t\lambda_n} \end{pmatrix}$$

where $\lambda_1, \lambda_2, \dots, \lambda_n$ are all eigenvalues (could repeat) of A . By assumption, $\operatorname{Re}(\lambda_i) < 0$ for all λ_i , so

$$\lim_{t \rightarrow +\infty} e^{\operatorname{Re}(\lambda_i)t} t^m = \mathbf{O} \quad \text{for all } \operatorname{Re}(\lambda_i) < 0, m \in \mathbb{Z},$$

it follows that

$$\lim_{t \rightarrow +\infty} e^{tD} \left(1 + tN + \dots + \frac{t^n N^n}{n!} \right) = \mathbf{O}.$$

We conclude that

$$\lim_{t \rightarrow +\infty} e^{tA} = \mathbf{O}, \quad \text{hence} \quad \lim_{t \rightarrow +\infty} e^{tA} \mathbf{y}_0 = \mathbf{0}.$$

- (Method 2) From the general solution expression of the system of equations, it can be seen that all its solutions are

$$\mathbf{x}(t) = \Phi(t)\mathbf{c},$$

Here, $\mathbf{c}$ represents an arbitrary n -dimensional constant vector. Suppose the eigenvalues of $\mathbf{A}$ are $\lambda_1, \lambda_2, \dots, \lambda_s$, and their multiplicities are $n_1, n_2, \dots, n_s$ (where $n_1 + n_2 + \dots + n_s = n$). Then

$$\Phi(t) = \left(e^{\lambda_1 t} \mathbf{P}_1^{(1)}(t), \dots, e^{\lambda_1 t} \mathbf{P}_{n_1}^{(1)}(t), \dots, e^{\lambda_s t} \mathbf{P}_1^{(s)}(t), \dots, e^{\lambda_s t} \mathbf{P}_{n_s}^{(s)}(t) \right),$$

Here, $\mathbf{P}_j^{(i)}(t) (i = 1, 2, \dots, s; j = 1, 2, \dots, n_i)$ are vector polynomials of t , with the degree not exceeding n_i .

If $\operatorname{Re} \lambda_i \leq \alpha < 0 (i = 1, 2, \dots, s)$, since $\mathbf{P}_j^{(i)}(x)$ is a vector polynomial, there exists a constant $M > 0$ such that for all i and j , there is

$$\|\mathbf{P}_j^{(i)}(t)\| \leq M e^{\frac{\alpha}{2}t}, \quad t \geq 0.$$

So

$$\|\Phi(t)\| \leq n^2 M e^{\frac{\alpha}{2}t}, \quad t \geq 0.$$

Therefore

$$\lim_{t \rightarrow +\infty} \Phi(t) = \mathbf{O}.$$

Thus, for any n -dimensional constant vector $\mathbf{c}$, there is

$$\lim_{t \rightarrow +\infty} \Phi(t)\mathbf{c} = \mathbf{0}.$$

Thus, all the solutions of the system of equations $\frac{d\mathbf{x}}{dt} = \mathbf{A}\mathbf{x}$ tend to zero as t approaches positive infinity.

- Conversely, if all the solutions of the system $\frac{d\mathbf{x}}{dt} = \mathbf{A}\mathbf{x}$ tend to zero as $t \rightarrow +\infty$, then

$$\lim_{t \rightarrow +\infty} \Phi(t) = \mathbf{0}.$$

Suppose that the real part of a certain eigenvalue of $\mathbf{A}$ is not less than zero, and let this eigenvalue be λ_1 . Then, in the expression of $\Phi(t)$, the first n_1 columns do not tend to zero as $t \rightarrow +\infty$. It is noted that these first n_1 columns are all solutions to the system of equations $\frac{d\mathbf{x}}{dt} = \mathbf{A}\mathbf{x}$, and thus a contradiction is derived.

- (2) From the general solution expression of the system of equations, it can be seen that all its solutions are

$$\mathbf{x}(t) = \Phi(t)\mathbf{c},$$

Here, $\mathbf{c}$ represents an arbitrary n -dimensional constant vector. Suppose the multiple eigenvalues of $\mathbf{A}$ are $\lambda_1, \lambda_2, \dots, \lambda_s$, their multiplicities are $n_1, n_2, \dots, n_s$ (where $n_1 + n_2 + \dots + n_s < n$); the simple eigenvalues are $\lambda_{s+1}, \dots, \lambda_l$. Then

$$\Phi(t) = \left(e^{\lambda_1 t} \mathbf{P}_1^{(1)}(t), \dots, e^{\lambda_1 t} \mathbf{P}_{n_1}^{(1)}(t), \dots, e^{\lambda_s t} \mathbf{P}_1^{(s)}(t), \dots, e^{\lambda_s t} \mathbf{P}_{n_s}^{(s)}(t), e^{\lambda_{s+1} t} \mathbf{r}_{s+1}, \dots, e^{\lambda_l t} \mathbf{r}_l \right),$$

Here, $\mathbf{P}_j^{(i)}(t) (i = 1, 2, \dots, s; j = 1, 2, \dots, n_i)$ are vector polynomials of t , with the degree not exceeding n_i ; $\mathbf{r}_k (k = s+1, \dots, l)$ are constant vectors.

If $\operatorname{Re} \lambda_i \leq \alpha < 0 (i = 1, 2, \dots, s)$, since $\mathbf{P}_j^{(i)}(x)$ is a vector polynomial, there exists a constant $M > 0$ such that for all i and j , there is

$$\|\mathbf{P}_j^{(i)}(t)\| \leq M e^{\frac{\alpha}{2} t}, \quad t \geq 0.$$

So

$$\|\Phi(t)\| \leq n^2 M e^{\frac{\alpha}{2} t} + n^2 K, \quad t \geq 0.$$

Every solution of the differential system $\frac{d\mathbf{x}}{dt} = \mathbf{A}\mathbf{x}$ is bounded in the interval $[0, \infty)$

- (3) The proof by contradiction is sufficient. The details is left to the readers. □

Problem 2.6.27. If the homogeneous system $\frac{d\mathbf{x}}{dt} = \mathbf{A}\mathbf{x}$ has a solution of the form $\mathbf{x} = \mathbf{P}(t)e^{\lambda t}$, where λ is a constant, $\mathbf{P}(t)$ is an n -dimensional vector valued function in which every component is a polynomial of degree $\leq m$, then show that

$$e^{\lambda t} \mathbf{P}(t), e^{\lambda t} \frac{d\mathbf{P}(t)}{dt}, \dots, e^{\lambda t} \frac{d^m \mathbf{P}(t)}{dt^m}$$

are $(m+1)$ linearly independent solutions of $\frac{d\mathbf{x}}{dt} = \mathbf{A}\mathbf{x}$.

Proof. First, substitute the solution into the system to obtain:

$$\frac{d}{dt} (e^{\lambda t} \mathbf{P}(t)) = \mathbf{A} e^{\lambda t} \mathbf{P}(t) \implies e^{\lambda t} (\lambda \mathbf{P}(t) + \mathbf{P}'(t)) = \mathbf{A} e^{\lambda t} \mathbf{P}(t),$$

After eliminating $e^{\lambda t}$, we have $\lambda \mathbf{P}(t) + \mathbf{P}'(t) = \mathbf{A} \mathbf{P}(t)$. For $k = 1, 2, \dots, m$, by induction, it can be proved that:

$$\lambda \mathbf{P}^{(k)}(t) + \mathbf{P}^{(k+1)}(t) = \mathbf{A} \mathbf{P}^{(k)}(t),$$

(The base step $k = 0$ has been established; when inductive reasoning, take the derivative on both sides of the equation and utilize the fact that $\mathbf{A}$ is a constant matrix). When $k = m$, $\mathbf{P}^{(m+1)}(t) = \mathbf{0}$, but $\mathbf{P}^{(m)}(t)$ is a non-zero constant vector, and the equation still holds. Therefore, each function $e^{\lambda t} \mathbf{P}^{(k)}(t)$ satisfies the original system.

Secondly, verify the linear independence. Let the linear combination be:

$$\sum_{k=0}^m c_k e^{\lambda t} \mathbf{P}^{(k)}(t) = \mathbf{0}, \quad \forall t.$$

Since $e^{\lambda t} \neq 0$, this is equivalent to $\sum_{k=0}^m c_k \mathbf{P}^{(k)}(t) = \mathbf{0}$. Substituting $\mathbf{P}(t) = \sum_{j=0}^m \mathbf{a}_j t^j$ (where $\mathbf{a}_m \neq \mathbf{0}$): The t^m term appears only in the $k = 0$ term, with a coefficient of $c_0 \mathbf{a}_m$. Given $\mathbf{a}_m \neq \mathbf{0}$, we have $c_0 = 0$. After substituting $c_0 = 0$, the t^{m-1} term appears only in the $k = 1$ term, with a coefficient of $c_1 \cdot m \mathbf{a}_m$. Since $m \neq 0$ and $\mathbf{a}_m \neq \mathbf{0}$, we get $c_1 = 0$. Continuing in this manner, for the t^{m-k} term, the coefficient is $c_k \cdot \frac{m!}{(m-k)!} \mathbf{a}_m$ is given by $\frac{m!}{(m-k)!}$. Given that $\neq 0$ and $\mathbf{a}_m \neq \mathbf{0}$, we have $c_k = 0$. Finally, the constant term (when $k = m$) gives $c_m \cdot m! \mathbf{a}_m = \mathbf{0}$, thus $c_m = 0$.

Therefore, all the coefficients $c_k = 0$, and these $m+1$ solutions are linearly independent. $\square$

Problem 2.6.28. If $\Phi(t)$ is a fundamental matrix of the system $\frac{d\mathbf{x}}{dt} = \mathbf{A}\mathbf{x}$ with $\Phi(0) = \mathbf{E}$, then show that for any $\tau \in \mathbb{R}$

$$\Phi(t)\Phi^{-1}(\tau) = \Phi(t - \tau).$$

Proof. For any $\mathbf{x}_0 \in \mathbb{R}^n$, both $\Phi(t)\Phi^{-1}(t_0)\mathbf{x}_0$ and $\Phi(t - \tau)\mathbf{x}_0$ are solutions to the initial value problem

$$\frac{d\mathbf{x}}{dt} = \mathbf{A}\mathbf{x}, \quad \mathbf{x}(t_0) = \mathbf{x}_0.$$

According to the existence and uniqueness theorem of solutions, $\Phi(t)\Phi^{-1}(t_0)\mathbf{x}_0 = \Phi(t - \tau)\mathbf{x}_0$. Then, by the arbitrariness of $\mathbf{x}_0$, it follows that $\Phi(t)\Phi^{-1}(\tau) = \Phi(t - \tau)$. $\square$

Problem 2.6.29. Show that the vector valued function

$$\mathbf{x}(t) = \exp \left[\int_{t_0}^t \mathbf{A}(s) ds \right] \mathbf{x}_0$$

is not a solution of the differential system

$$\frac{d\mathbf{x}}{dt} = \mathbf{A}(t)\mathbf{x}, \tag{2.18}$$

unless $\mathbf{A}(t)$ and $\int_{t_0}^t \mathbf{A}(s) \, ds$ commute for all t , where $\mathbf{x}(t_0) = \mathbf{x}_0$.

Proof. Consider the contrapositive proposition². By definition,

$$\exp \left[\int_{t_0}^t \mathbf{A}(s) \, ds \right] = \mathbf{E} + \sum_{k=1}^{\infty} \frac{1}{k!} \left[\int_{t_0}^t \mathbf{A}(s) \, ds \right]^k,$$

Hence,

$$\begin{aligned} \frac{d}{dt} \left[\exp \left(\int_{t_0}^t \mathbf{A}(s) \, ds \right) \right] &= \sum_{k=1}^{\infty} \frac{k}{k!} \left(\int_{t_0}^t \mathbf{A}(s) \, ds \right)^{k-1} \mathbf{A}(t) \quad (\text{by the commutativity}) \\ &= \left[\mathbf{E} + \sum_{k=1}^{\infty} \frac{1}{k!} \left(\int_{t_0}^t \mathbf{A}(s) \, ds \right)^k \right] \mathbf{A}(t) \\ &= \mathbf{A}(t) \left[\mathbf{E} + \sum_{k=1}^{\infty} \frac{1}{k!} \left(\int_{t_0}^t \mathbf{A}(s) \, ds \right)^k \right] \quad (\text{by the commutativity}) \\ &= \mathbf{A}(t) \exp \left(\int_{t_0}^t \mathbf{A}(s) \, ds \right), \end{aligned}$$

That is, $\mathbf{x}(t) = \exp \left(\int_{t_0}^t \mathbf{A}(s) \, ds \right)$ is the solution matrix of (2.18). Also, since $\mathbf{x}(t_0) = \mathbf{E}$, it follows that $\mathbf{x}(t)$ is the fundamental matrix of (2.18). Therefore, the solution of equation (2.18) is

$$\mathbf{x} = \exp \left(\int_{t_0}^t \mathbf{A}(s) \, ds \right) \cdot \mathbf{x}_0. \quad \square$$

Exercise 2.7

不在考试范围内, 不写.

²逆否命题.

By Fingolfn, Grade 2024

Chapter 3

High-Order Linear Differential Equations

Exercise 3.1

Problem 3.1.1. Define the linear differential operator

$$L[x] := x^{(n)} + p_1(t)x^{(n-1)} + \cdots + p_{n-1}(t)x' + p_n(t)x, \quad (3.1)$$

Show that the equations

$$L[kx] = kL[x], \quad k \text{ is a constant}, \quad (3.2)$$

and

$$L[x_1 + x_2] = L[x_1] + L[x_2] \quad (3.3)$$

hold.

Proof. Obviously. □

Problem 3.1.2. Verify that $x_1(t) = \cos 2t$ and $x_2(t) = \sin 2t$ are two solutions of

$$x'' + 4x = 0.$$

Show that

$$x(t) = c_1 \cos 2t + c_2 \sin 2t$$

is a solution of this equation, where c_1 and c_2 are constants.

Proof. Trivially. □

First verify that x_1 and x_2 are solutions of the differential equation. Then find a particular solution of the form $x = c_1x_1 + c_2x_2$ that satisfies the given initial conditions in Problems 3 through 6.

Problem 3.1.3. $x'' - x = 0; x_1(t) = e^t, x_2(t) = e^{-t}; x(0) = 0, x'(0) = 5.$

Solution. After verifying that both e^t and e^{-t} satisfy the equation $x'' - x = 0$, we assume the solution to be $x(t) = c_1 e^t + c_2 e^{-t}$. Substituting the initial conditions $x(0) = 0$ and $x'(0) = 5$ into the equation yields the system of equations $c_1 + c_2 = 0$ and $c_1 - c_2 = 5$. Solving these equations gives $c_1 = \frac{5}{2}$ and $c_2 = -\frac{5}{2}$, thus the particular solution is $x(t) = \frac{5}{2}e^t - \frac{5}{2}e^{-t}$. $\square$

Problem 3.1.4. $x'' + 25x = 0$; $x_1(t) = \cos 5t$, $x_2(t) = \sin 5t$; $x(0) = 10$, $x'(0) = -10$.

Solution. After verifying that both $\cos 5t$ and $\sin 5t$ satisfy the equation $x'' + 25x = 0$, we assume the solution to be $x(t) = c_1 \cos 5t + c_2 \sin 5t$. Substituting the initial conditions $x(0) = 10$ and $x'(0) = -10$ yields $c_1 = 10$ and $5c_2 = -10$, resulting in $c_1 = 10$ and $c_2 = -2$. Therefore, the particular solution is $x(t) = 10 \cos 5t - 2 \sin 5t$. $\square$

Problem 3.1.5. $x'' - 3x' = 0$; $x_1 = 1$, $x_2 = e^{3t}$; $x(0) = 4$, $x'(0) = -2$.

Solution. After verifying that both 1 and e^{3t} satisfy the equation $x'' - 3x' = 0$, let the solution be $x(t) = c_1 + c_2 e^{3t}$. Substituting the initial conditions $x(0) = 4$ and $x'(0) = -2$ into the equation gives the system of equations $c_1 + c_2 = 4$ and $3c_2 = -2$. Solving these equations yields $c_1 = \frac{14}{3}$, $c_2 = -\frac{2}{3}$, thus the particular solution is $x(t) = \frac{14}{3} - \frac{2}{3}e^{3t}$. $\square$

Problem 3.1.6. $x'' - 10x' + 25x = 0$; $x_1(t) = e^{5t}$, $x_2(t) = te^{5t}$; $x(0) = 3$, $x'(0) = 13$.

Solution. After verifying that both e^{5t} and te^{5t} satisfy the equation $x'' - 10x' + 25x = 0$, let the solution be $x(t) = c_1 e^{5t} + c_2 te^{5t}$. Substituting the initial conditions $x(0) = 3$ and $x'(0) = 13$ yields $c_1 = 3$ and $5c_1 + c_2 = 13$. Solving these equations gives $c_1 = 3$ and $c_2 = -2$. Therefore, the particular solution is $x(t) = 3e^{5t} - 2te^{5t}$. $\square$

Exercise 3.2

Problem 3.2.1. Let $x = \varphi(t)$ be a solution of

$$x^{(n)} + p_1(t)x^{(n-1)} + \cdots + p_{n-1}(t)x' + p_n(t)x = 0, \quad (3.4)$$

satisfying $\varphi(t_0) = \varphi'(t_0) = \cdots = \varphi^{(n-1)}(t_0) = 0$, $t_0 \in I$. Then show that $\varphi(t) \equiv 0$.

Proof. Obviously, $x \equiv 0$ is a solution to (3.4). By Theorem 3.2.1, the initial value problem

$$\begin{cases} x^{(n)} + p_1(t)x^{(n-1)} + \cdots + p_{n-1}(t)x' + p_n(t)x = 0, \\ x(t_0) = x'(t_0) = \cdots = x^{(n-1)}(t_0) = 0, \end{cases}$$

has a unique solution on the entire interval I , i.e., $\varphi(t) \equiv 0$. $\square$

Problem 3.2.2. Verify that $x_1(t) = t^2$, $x_2(t) = t^3$ are two solutions of $t^2 x'' - 4tx' + 6x = 0$ and that both satisfy the initial conditions $x(0) = x'(0) = 0$. Why does this not contradict the uniqueness part of Theorem 3.2.1.

Solution. The verification process is quite simple. Note that the coefficient of the second derivative in this equation is a variable coefficient, while in Theorem 3.2.1, the coefficient of the highest-order term in the equation is 1. $\square$

Exercise 3.3

Determine whether the pairs of functions in Problems 1 through 4 are linear independent or linearly dependent on the real line.

Problem 3.3.1. $t^3, t^2|t|$.

Solution. The functions t^3 and $t^2|t|$ are linearly independent **on the real line**. $\square$

Problem 3.3.2. $\sin^2 t, 1 - \cos 2t$.

Solution. The functions $\sin^2 t$ and $1 - \cos 2t$ are linearly dependent on the real line. $\square$

Problem 3.3.3. $e^t \sin t, e^t \cos t$.

Solution. The functions $e^t \sin t$ and $e^t \cos t$ are linearly independent on the real line as demonstrated by their Wronskian $W = e^{2t}$, which is never zero for any real t ; calculating the Wronskian explicitly: $W = -e^{2t} \neq 0$, confirming linear independence. $\square$

Problem 3.3.4. te^t, t^2e^t .

Solution. The functions te^t and t^2e^t are linearly independent on the real line as shown by their Wronskian $W = t^2e^{2t}$, which is not identically zero on the real line (it equals zero only at $t = 0$ but is nonzero elsewhere); computing the Wronskian: $W = t^2e^{2t}$, and since a non-trivial zero set doesn't make the Wronskian identically zero, the functions are linearly independent. $\square$

In Problems 5 through 8, use the Wronskian to prove that the given functions are linearly independent on the indicated interval.

Problem 3.3.5. e^t, e^{2t}, e^{3t} ; the real line.

Solution. For functions e^t, e^{2t} , and e^{3t} on the real line, the Wronskian is calculated as $W =$

$$\begin{vmatrix} e^t & e^{2t} & e^{3t} \\ e^t & 2e^{2t} & 3e^{3t} \\ e^t & 4e^{2t} & 9e^{3t} \end{vmatrix} = 2e^{6t} \neq 0.$$

Therefore, the functions are linearly independent on the real line. $\square$

Problem 3.3.6. $e^x, x^{-2}, x^{-2} \ln x; x > 0$.

Solution. For functions e^x, x^{-2} , and $x^{-2} \ln x$ on $x > 0$, the Wronskian simplifies to $W = e^x \cdot x^{-7} \cdot (x+1)(x+4) \neq 0$ indicating its linear independence. $\square$

Problem 3.3.7. $e^x, \sin x, \cos x$; the real line.

Solution. For functions $e^x, \sin x$, and $\cos x$ on the real line, the Wronskian computes to $W =$

$$\begin{vmatrix} e^x & \sin x & \cos x \\ e^x & \cos x & -\sin x \\ e^x & -\sin x & -\cos x \end{vmatrix} = -2e^x \neq 0.$$

Hence the functions are linearly independent on the real line. $\square$

Problem 3.3.8. $x, \cos(\ln x), \sin(\ln x); x > 0$.

Solution. For functions $x, \cos(\ln x)$, and $\sin(\ln x)$ on $x > 0$, the Wronskian evaluates to $W = \frac{2}{x^2} \neq 0$, confirming linear independence of these functions. $\square$

In Problems 9 through 12, a third-order homogeneous linear equation and three linearly independent solutions are given. Find a particular solution satisfying the given initial conditions.

Problem 3.3.9. $x''' - 3x'' + 3x' - x = 0; x(0) = 2, x'(0) = 0, x''(0) = 0; x_1 = e^t, x_2 = te^t, x_3 = t^2e^t$.

Solution. The general solution is $x(t) = c_1e^t + c_2te^t + c_3t^2e^t$, and applying initial conditions $x(0) = 2, x'(0) = 0, x''(0) = 0$ yields the system $c_1 = 2, c_1 + c_2 = 0, c_1 + 2c_2 + 2c_3 = 0$, which solves to $c_1 = 2, c_2 = -2, c_3 = 1$, giving the particular solution $x(t) = (t^2 - 2t + 2)e^t$. $\square$

Problem 3.3.10. $t^3x''' - 3t^2x'' + 6tx' - 6x = 0; x(1) = 6, x'(1) = 14, x''(1) = 22; x_1 = t, x_2 = t^2, x_3 = t^3$.

Solution. The general solution is $x(t) = c_1t + c_2t^2 + c_3t^3$, and applying initial conditions at $t = 1$ gives $x(1) = 6, x'(1) = 14, x''(1) = 22$, forming the system $c_1 + c_2 + c_3 = 6, c_1 + 2c_2 + 3c_3 = 14, 2c_2 + 6c_3 = 22$, which solves to $c_1 = 1, c_2 = 2, c_3 = 3$, giving the particular solution $x(t) = t + 2t^2 + 3t^3$. $\square$

Problem 3.3.11. $x''' + 9x' = 0; x(0) = 3, x'(0) = -1, x''(0) = 2; x_1 = 1, x_2 = \cos 3t, x_3 = \sin 3t$.

Solution. The general solution is $x(t) = c_1 + c_2 \cos 3t + c_3 \sin 3t$, and applying initial conditions $x(0) = 3, x'(0) = -1, x''(0) = 2$ yields $c_1 + c_2 = 3, 3c_3 = -1, -9c_2 = 2$, solving to $c_1 = \frac{29}{9}, c_2 = -\frac{2}{9}, c_3 = -\frac{1}{3}$, giving the particular solution $x(t) = \frac{29}{9} - \frac{2}{9} \cos 3t - \frac{1}{3} \sin 3t$. $\square$

Problem 3.3.12. $x''' + 2x'' - x' - 2x = 0; x(0) = 1, x'(0) = 2, x''(0) = 0; x_1 = e^t, x_2 = e^{-t}, x_3 = e^{-2t}$.

Solution. The general solution is $x(t) = c_1e^t + c_2e^{-t} + c_3e^{-2t}$, and applying initial conditions $x(0) = 1, x'(0) = 2, x''(0) = 0$ gives the system $c_1 + c_2 + c_3 = 1, c_1 - c_2 - 2c_3 = 2, c_1 + c_2 + 4c_3 = 0$, which solves to $c_1 = \frac{4}{3}, c_2 = 0, c_3 = -\frac{1}{3}$, giving the particular solution $x(t) = \frac{4}{3}e^t - \frac{1}{3}e^{-2t}$. $\square$

In Problems 13 through 20, a differential equation and one solution $x_1(t)$ are given. Use the method as in Example 3.3.4 to find a second linearly independent solution $x_2(t)$.

Remark 3.1. 对于二阶齐次线性方程, 知道一个非零解后, 可以求出通解.

Proposition 3.3.1 (教材 93 页例 3.3.4). 设 $x = \varphi(t)$ 是二阶齐次线性微分方程

$$x'' + p(t)x' + q(t)x = 0$$

的一个非零解, 其中 $p(t)$ 和 $q(t)$ 是区间 $a < t < b$ 上的连续函数, 则方程的通解为

$$x = \varphi(t) \left[C_1 + C_2 \int \frac{1}{\varphi^2(t)} e^{-\int p(t) dx} dx \right], \quad (3.5)$$

其中 C_1 和 C_2 为任意常数.

Problem 3.3.13. $t^2x'' + tx' - 9x = 0$ ($t > 0$); $x_1(t) = t^3$.

Solution. For the equation $t^2x'' + tx' - 9x = 0$ (where $t > 0$), first transform the equation into the standard form: $x'' + \frac{1}{t}x' - \frac{9}{t^2}x = 0$. Thus, $p(t) = \frac{1}{t}$. Let $x_2(t)$ be another solution which is different with the given solution $x_1(t)$. By Liouville's Formula, we have

$$W'(t) = \begin{vmatrix} x_1 & x_2 \\ x_1' & x_2' \end{vmatrix} = C_1 e^{-\int p(t) dt} = ce^{-\ln t} = \frac{C_1}{t},$$

that is

$$x_1x_2' - x_1'x_2 = \frac{C_1}{t}.$$

Multiplying both sides by $\frac{1}{x_1^2}$ gives

$$\frac{d}{dt} \left(\frac{x_2}{x_1} \right) = \frac{C_1}{tx_1^2} = \frac{C_1}{t^7}.$$

Integrating both sides of the last equation yields

$$x_2 = t^3 \left(\int \frac{C_1}{t^7} dt + C_2 \right) = t^3 \left(-\frac{C_1}{6t^6} + C_2 \right) = -\frac{C_1}{6t^3} + C_2t^3.$$

Ignoring the constant factor, the linearly independent second solution is $x_2(t) = \frac{1}{t^3}$. □

Problem 3.3.14. $t^2x'' - t(t+2)x' + (t+2)x = 0$ ($t > 0$); $x_1(t) = t$.

Solution. For the equation $t^2x'' - t(t+2)x' + (t+2)x = 0$ (where $t > 0$), first transform the equation into the standard form: $x'' - \frac{t+2}{t}x' + \frac{t+2}{t^2}x = 0$. Thus, $p(t) = \frac{t+2}{t}$. Let $x_2(t)$ be another solution which is different with the given solution $x_1(t)$. By Liouville's Formula, we have

$$W'(t) = \begin{vmatrix} x_1 & x_2 \\ x_1' & x_2' \end{vmatrix} = C_1 e^{-\int p(t) dt} = C_1 e^{t+2\ln t} = C_1 t^2 e^t,$$

that is

$$x_1x_2' - x_1'x_2 = C_1 t^2 e^t.$$

Multiplying both sides by $\frac{1}{x_1^2}$ gives

$$\frac{d}{dt} \left(\frac{x_2}{x_1} \right) = \frac{C_1 t^2 e^t}{x_1^2} = C_1 e^t.$$

Integrating both sides of the last equation yields

$$x_2 = t \left(\int C_1 e^t dt + C_2 \right) = t (C_1 e^t + C_2) = C_1 t e^t + C_2 t.$$

Ignoring the constant factor, the linearly independent second solution is $x_2(t) = t e^t$. □

Problem 3.3.15. $(1 - t^2)x'' + 2tx' - 2x = 0$ ($-1 < t < 1$); $x_1(t) = t$.

Solution. For the equation $(1 - t^2)x'' + 2tx' - 2x = 0$ (where $t > 0$), first transform the equation into the standard form: $x'' + \frac{2t}{1 - t^2}x' - \frac{2}{1 - t^2}x = 0$. Thus, $p(t) = \frac{2t}{1 - t^2}$. Let $x_2(t)$ be another solution which is different with the given solution $x_1(t)$. By Liouville's Formula, we have

$$W'(t) = \begin{vmatrix} x_1 & x_2 \\ x_1' & x_2' \end{vmatrix} = C_1 e^{-\int p(t) dt} = C_1 e^{\ln(1-t^2)} = C_1(1 - t^2),$$

that is

$$x_1 x_2' - x_1' x_2 = C_1(1 - t^2).$$

Multiplying both sides by $\frac{1}{x_1^2}$ gives

$$\frac{d}{dt} \left(\frac{x_2}{x_1} \right) = \frac{C_1(1 - t^2)}{x_1^2} = \frac{C_1(1 - t^2)}{t^2}.$$

Integrating both sides of the last equation yields

$$x_2 = t \left(\int \frac{C_1(1 - t^2)}{t^2} dt + C_2 \right) = t \left[-C_1 \left(\frac{1}{t} + t \right) + C_2 \right] = -C_1(1 + t^2) + C_2 t.$$

Ignoring the constant factor, the linearly independent second solution is $x_2(t) = 1 + t^2$. □

Problem 3.3.16. $(t - 1)x'' - tx' + x = 0$; $x_1(t) = t$.

Solution. For the equation $(t - 1)x'' - tx' + x = 0$; $x_1(t) = t$, first transform the equation into the standard form: $x'' - \frac{t}{t-1}x' + \frac{1}{t-1}x = 0$. Thus, $p(t) = -\frac{t}{t-1}$. Let $x_2(t)$ be another solution which is different with the given solution $x_1(t)$. By Liouville's Formula, we have

$$W'(t) = \begin{vmatrix} x_1 & x_2 \\ x_1' & x_2' \end{vmatrix} = C_1 e^{-\int p(t) dt} = C_1 e^{t + \ln(t-1)} = C_1(t - 1)e^t, \quad (t > 1),$$

that is

$$x_1 x_2' - x_1' x_2 = C_1(t - 1)e^t, \quad t > 1.$$

Multiplying both sides by $\frac{1}{x_1^2}$ gives

$$\frac{d}{dt} \left(\frac{x_2}{x_1} \right) = \frac{C_1(t - 1)e^t}{x_1^2} = \frac{C_1(t - 1)e^t}{t^2}, \quad t > 1.$$

Integrating both sides of the last equation yields

$$x_2 = t \left(\int \frac{C_1(t - 1)e^t}{t^2} dt + C_2 \right) = t \left[C_1 \frac{e^t}{t} + C_2 \right] = C_1 e^t + C_2 t, \quad t > 1.$$

Ignoring the constant factor, the linearly independent second solution is $x_2(t) = e^t$. When $t < 1$, it is similar. $\square$

Problem 3.3.17. $(1 - \ln t)x'' + \frac{1}{t}x' - \frac{1}{t^2}x = 0; x_1(t) = \ln t$.

Solution. For the equation $(1 - \ln t)x'' + \frac{1}{t}x' - \frac{1}{t^2}x = 0$, first transform the equation into the standard form: $x'' + \frac{1}{t - t \ln t}x' - \frac{1}{t^2(1 - \ln t)}x = 0$. Thus, $p(t) = \frac{1}{t - t \ln t}$. Let $x_2(t)$ be another solution which is different with the given solution $x_1(t)$. By Liouville's Formula, we have

$$W'(t) = \begin{vmatrix} x_1 & x_2 \\ x_1' & x_2' \end{vmatrix} = C_1 e^{-\int p(t) dt} = C_1 e^{\ln(1 - \ln t)} = C_1(1 - \ln t), \quad (0 < t < e),$$

that is

$$x_1 x_2' - x_1' x_2 = C_1(1 - \ln t), \quad (0 < t < e).$$

Multiplying both sides by $\frac{1}{x_1^2}$ gives

$$\frac{d}{dt} \left(\frac{x_2}{x_1} \right) = \frac{C_1(1 - \ln t)}{x_1^2} = \frac{C_1(1 - \ln t)}{\ln^2 t}, \quad t > 1.$$

Integrating both sides of the last equation yields

$$x_2 = \ln t \left(\int \frac{C_1(1 - \ln t)}{\ln^2 t} dt + C_2 \right) = \ln t \left[-C_1 \frac{t}{\ln t} + C_2 \right] = -C_1 t + C_2 \ln t, \quad 0 < t < e.$$

Ignoring the constant factor, the linearly independent second solution is $x_2(t) = t$. When $t > e$, it is similar. $\square$

Problem 3.3.18. $(t^2 - t)x'' + (3t - 1)x' + x = 0 (x \neq 0, 1); x_1(t) = \frac{1}{t - 1}$.

Solution. For the equation $(t^2 - t)x'' + (3t - 1)x' + x = 0$, first transform the equation into the standard form: $x'' + \frac{3t - 1}{t^2 - t}x' - \frac{1}{t^2 - t}x = 0$. Thus, $p(t) = \frac{3t - 1}{t^2 - t}$. Let $x_2(t)$ be another solution which is different with the given solution $x_1(t)$. By Liouville's Formula, we have

$$W'(t) = \begin{vmatrix} x_1 & x_2 \\ x_1' & x_2' \end{vmatrix} = C_1 e^{-\int p(t) dt} = C_1 e^{-2 \ln(1-t) - \ln t} = C_1 \frac{1}{t(1-t)^2}, \quad (0 < t < 1),$$

that is

$$x_1 x_2' - x_1' x_2 = C_1 \frac{1}{t(1-t)^2}, \quad (0 < t < 1).$$

Multiplying both sides by $\frac{1}{x_1^2}$ gives

$$\frac{d}{dt} \left(\frac{x_2}{x_1} \right) = \frac{C_1}{t(1-t)^2 x_1^2} = \frac{C_1}{t}, \quad 0 < t < 1.$$

Integrating both sides of the last equation yields

$$x_2 = \frac{1}{t-1} \left(\int \frac{C_1}{t} dt + C_2 \right) = \frac{1}{t-1} [C_1 \ln t + C_2] = \frac{C_1 \ln t}{t-1} + \frac{C_2}{t-1}, \quad 0 < t < e.$$

Ignoring the constant factor, the linearly independent second solution is $x_2(t) = \frac{\ln t}{t-1}$. When t takes on other values, it is similar. $\square$

Problem 3.3.19. $tx'' - x' - 4t^3x = 0$ ($x \neq 0$); $x_1(t) = e^{t^2}$.

Solution. For the equation $tx'' - x' - 4t^3x = 0$, first transform the equation into the standard form: $x'' - \frac{1}{t}x' - 4t^2x = 0$. Thus, $p(t) = -\frac{1}{t}$. Let $x_2(t)$ be another solution which is different with the given solution $x_1(t)$. By Liouville's Formula, we have

$$W'(t) = \begin{vmatrix} x_1 & x_2 \\ x_1' & x_2' \end{vmatrix} = C_1 e^{-\int p(t) dt} = C_1 e^{\ln t} = C_1 t,$$

that is

$$x_1 x_2' - x_1' x_2 = C_1 t,$$

Multiplying both sides by $\frac{1}{x_1^2}$ gives

$$\frac{d}{dt} \left(\frac{x_2}{x_1} \right) = \frac{C_1 t}{x_1^2} = \frac{C_1 t}{e^{2t^2}},$$

Integrating both sides of the last equation yields

$$x_2 = e^{t^2} \left(\int \frac{C_1 t}{e^{2t^2}} dt + C_2 \right) = e^{t^2} \left[-\frac{C_1}{4} e^{-2t^2} + C_2 \right] = -\frac{C_1}{4} e^{-t^2} + C_2 e^{t^2}, \quad 0 < t < e.$$

Ignoring the constant factor, the linearly independent second solution is $x_2(t) = e^{-t^2}$. $\square$

Problem 3.3.20. $y'' + \lambda(xy' + y) = 0$; $y_1(x) = \exp\left(-\frac{\lambda x^2}{2}\right)$.

Solution. For the equation $y'' + \lambda(xy' + y) = 0$, it is transformed into the standard form: $y'' + \lambda xy' + \lambda y = 0$. Thus, $p(x) = \lambda x$. By applying the Liouville formula, we calculate $\int p(x) dx = \frac{\lambda x^2}{2}$, so $e^{-\int p(x) dx} = e^{-\frac{\lambda x^2}{2}}$. Substituting the known solution $y_1(x) = e^{-\frac{\lambda x^2}{2}}$, we obtain $y_2(x) = e^{-\frac{\lambda x^2}{2}} \int \frac{1}{e^{-\lambda x^2}} \cdot e^{-\frac{\lambda x^2}{2}} dx = e^{-\frac{\lambda x^2}{2}} \int e^{\frac{\lambda x^2}{2}} dx$. Therefore, the second solution is $y_2(x) = y_1(x) \int y_1(x) dx$ (this integral cannot be expressed in terms of elementary functions). $\square$

Problem 3.3.21. Let $x_1 = t$ and $x_2 = t^2$ be two particular solutions of the linear equation

$$t^3 x''' - 3t^2 x'' + 6tx' - 6x = 0.$$

Find a general solution of this equation.

Solution. It is known that $x_1 = t$ and $x_2 = t^2$ are two linearly independent solutions of the equation $t^3 x''' - 3t^2 x'' + 6tx' - 6x = 0$. The equation is transformed into the standard form $x''' - \frac{3}{t}x'' + \frac{6}{t^2}x' - \frac{6}{t^3}x = 0$, where $p(t) = -\frac{3}{t}$. According to Liouville's formula, the Wronskian of the three solutions satisfies $W(t) = C \exp\left(-\int p(t) dt\right) = C \exp\left(\int \frac{3}{t} dt\right) = Ct^3$. Let the third solution be $x_3(t)$, then the Wronskian is:

$$W(t) = \begin{vmatrix} t & t^2 & x_3 \\ 1 & 2t & x_3' \\ 0 & 2 & x_3'' \end{vmatrix} = t^2 x_3'' - 2tx_3' + 2x_3 = Ct^3.$$

Without loss of generality, taking $C = 2$, we obtain the second-order equation $t^2 x_3'' - 2tx_3' + 2x_3 = 2t^3$. First, solve the corresponding homogeneous equation $t^2 x_3'' - 2tx_3' + 2x_3 = 0$, which has the standard form $x_3'' - \frac{2}{t}x_3' + \frac{2}{t^2}x_3 = 0$. Given one solution $x_{3,1} = t$, again applying the Bôuldoville formula, $p(t) = -\frac{2}{t}$, $x_{3,2} = t \int \frac{1}{t^2} e^{-\int -\frac{2}{t} dt} dt = t \int \frac{1}{t^2} \cdot t^2 dt = t \int 1 dt = t^2$. Using the parameter variation method to find the particular solution of the non-homogeneous equation, let $x_3 = u_1 t + u_2 t^2$, and we obtain $u_1 = -t^2$, $u_2 = 2t$, thus $x_3 = (-t^2)t + (2t)t^2 = t^3$. Verifying that the Wronskian $W(t) = 2t^3 \neq 0$ (for $t \neq 0$), the general solution is $x(t) = c_1 t + c_2 t^2 + c_3 t^3$. $\square$

Problem 3.3.22. Let $\varphi_1(t)$ be a nontrivial solution of homogeneous n -th order linear equation

$$x^{(n)} + p_1(t)x^{(n-1)} + \cdots + p_{n-1}(t)x' + p_n(t)x = 0.$$

Then use the substitution $x = \varphi_1(t)y$ to transform this equation into $n-1$ th-order homogeneous linear equation.

Solution. Firstly, we have

$$\begin{cases} x = \varphi_1 \cdot y, \\ x' = \varphi_1' \cdot y + \varphi_1 y', \\ x'' = \varphi_1'' \cdot y + 2\varphi_1' y' + \varphi_1 y'', \\ \dots\dots\dots, \\ x^{(n)} = \sum_{k=0}^n \binom{n}{k} \cdot \varphi_1^{(k)} \cdot y^{(n-k)}. \end{cases}$$

Set $p_0(t) := 1$, then the original equation is equivalent to

$$\begin{aligned} y \cdot \left[p_0 \cdot \varphi_1^{(n)} + p_1 \cdot \varphi_1^{(n-1)} + p_2 \cdot \varphi_1^{(n-2)} + \cdots + p_n \cdot \varphi_1 \right] + y' \cdot \left[\sum_{k=1}^n k \cdot p_{n-k} \cdot \varphi_1^{(k-1)} \right] \\ + \cdots + y^{(n-1)} \left[p_1 \cdot \varphi_1 + \binom{n}{1} \cdot p_0 \cdot \varphi_1' \right] + y^{(n)} \cdot p_0 \cdot \varphi_1 = 0 \end{aligned}$$

And because $\varphi_1(t)$ be a nontrivial solution of homogeneous n th-order linear equation, we have

$$p_0\varphi^{(n)} + p_1\varphi^{(n-1)} + \cdots + p_{n-1}\varphi' + p_n\varphi = 0.$$

Therefore, we have

$$y' \cdot \left[\sum_{k=1}^n k \cdot p_{n-k} \cdot \varphi_1^{(k-1)} \right] + \cdots + y^{(n-1)} \left[p_1 \cdot \varphi_1 + \binom{n}{1} \cdot p_0 \cdot \varphi_1' \right] + y^{(n)} \cdot p_0 \cdot \varphi_1 = 0.$$

Let

$$\begin{cases} q_0 := p_0 \cdot \varphi_1, \\ q_1 := p_1 \cdot \varphi_1 + \binom{n}{1} \cdot p_0 \cdot \varphi_1', \\ \cdots, \\ q_{n-1} := \sum_{k=1}^n k \cdot p_{n-k} \cdot \varphi_1^{(k-1)}, \\ z := y'. \end{cases}$$

Hence, we have

$$q_0 z^{(n-1)} + q_1 z^{(n-2)} + \cdots + q_{n-1} z = 0, \quad q_0 = p_0 \cdot \varphi_1 \neq 0.$$

Thus,

$$z^{(n-1)} + \frac{q_1}{q_0} z^{(n-2)} + \cdots + \frac{q_{n-1}}{q_0} z = 0$$

is the desired $(n-1)$ th-order homogeneous equation. $\square$

Problem 3.3.23. Verify that $x_1 = t$ and $x_2 = t^2$ are linearly independent solutions on the entire real line of the equation

$$t^2 x'' - 2tx' + 2x = 0,$$

but that $W(t)$ vanishes at $t = 0$. Why do these observations not contradict part (2) of Theorem 3.3.4.

Solution. Upon verification, it can be seen that both $x_1 = t$ and $x_2 = t^2$ satisfy the equation $t^2 x'' - 2tx' + 2x = 0$. Since their ratio $\frac{t^2}{t} = t$ is non-zero, they are linearly independent over

the entire real number line. Their Wronskian is $W(t) = \begin{vmatrix} t & t^2 \\ 1 & 2t \end{vmatrix} = 2t^2 - t^2 = t^2$, and at $t = 0$,

$W(0) = 0$. This indicates that when the coefficients of the equation are discontinuous at a certain point (here, $t = 0$), the Wronskian of the linearly independent solutions may be zero at that point. $\square$

Remark 3.2. 线性微分方程的系数的连续性要求是重要的, 它保证了线性无关特解的个数等于方程的阶, 否则, 结论可能不对. 例如, 函数 $y_1(x) = x$, $y_2(x) = x^5$, $y_3(x) = |x|^5$ 满足二阶齐次

线性微分方程

$$x^2 y'' - 5xy' + 5y = 0, \quad (3.6)$$

但这三个函数 $y_1(x), y_2(x), y_3(x)$ 在区间 $(-1, 1)$ 内线性无关 (请验证). 因为将方程(3.6)变成

$$y'' - \frac{5}{x}y' + \frac{5}{x^2}y = 0,$$

可见系数 $a_1(x) = -\frac{5}{x}, a_2(x) = \frac{5}{x^2}$ 在点 $x = 0$ 处不连续.

Problem 3.3.24. Prove that any two solutions $\varphi_1(t)$ and $\varphi_2(t)$ of the differential equation

$$x'' + p(t)x' + q(t)x = 0$$

satisfying the same initial condition $x(0) = 0$ are linearly dependent, where $p(t)$ and $q(t)$ are continuous on the interval $0 \leq t \leq 1$.

Proof. Note that their Wronskian $W(t) = 0$ at $t = 0$. So $W(t) \equiv 0, t \in [0, 1]$. Therefore, $\varphi_1(t)$ and $\varphi_2(t)$ are two linearly dependent solutions. $\square$

Exercise 3.4

Find a general solution of the following nonhomogeneous differential equations.

Problem 3.4.1. $x'' + 4x = \sin 2t$.

Solution. The characteristic equation is $\lambda^2 + 4 = 0$. Thus, the characteristic roots are $\lambda_1 = 2i, \lambda_2 = -2i$. A general solution of the associated homogeneous equation is $\varphi(t) = c_1 \cos 2t + c_2 \sin 2t$. Suppose that the equation has a particular solution

$$\varphi_p(t) := t(a \sin 2t + b \cos 2t).$$

The initial values give $a = -\frac{1}{4}, b = 0$. So $\varphi_p(t) = -\frac{t}{4} \cos 2t, x(t) = c_1 \cos 2t + c_2 \sin 2t - \frac{t}{4} \cos 2t$ is a general solution. $\square$

Problem 3.4.2. $x'' + 4x' + 3x = e^{-3t}$.

Solution. The characteristic equation is $\lambda^2 + 4\lambda + 3 = 0$, which gives the characteristic roots $\lambda_1 = -1, \lambda_2 = -3$. A general solution of the associated homogeneous equation is $\varphi(t) = c_1 e^{-t} + c_2 e^{-3t}$. Suppose that the equation has a particular solution $\varphi_p(t) := ate^{-3t}$. The initial values give that $a = -\frac{1}{2}$. So $\varphi_p(t) = -\frac{t}{2}e^{-3t}, x(t) = c_1 e^{-t} + c_2 e^{-3t} - \frac{t}{2}e^{-3t}$ is the general solution. $\square$

Problem 3.4.3. $x'' + x = \tan t$.

Solution. The characteristic equation is $\lambda^2 + 1 = 0$. Thus, the characteristic roots are $\lambda_1 = -i, \lambda_2 = i$. A general solution of the associated homogeneous equation is $\varphi(t) = c_1 \cos t + c_2 \sin t$. Suppose that the equation has a particular solution

$$\varphi_p(t) := a(t) \cos t + b(t) \sin t$$

Then we know that $a(t)$ and $b(t)$ satisfy the following linear system

$$\begin{cases} a'(t) \cos t + b'(t) \sin t = 0, \\ -a'(t) \sin t + b'(t) \cos t = \tan t, \end{cases}$$

Then,

$$\begin{cases} a'(t) &= -\sin t, \\ b'(t) &= \sin t, \\ a(t) &= -\ln |\sec t + \tan t| + \sin t, \\ b(t) &= -\cos t. \end{cases}$$

So $\varphi_p(t) = -\ln |\sec t + \tan t| \cos t$, $x(t) = c_1 \cos t + c_2 \sin t - \ln |\sec t + \tan t| \cos t$ is a general solution. $\square$

Problem 3.4.4. Suppose that $x_1(t)$ and $x_2(t)$ are two functions such that

$$L[x_1(t)] = f_1(t) \quad \text{and} \quad L[x_2(t)] = f_2(t).$$

Show that their sum $x = x_1(t) + x_2(t)$ satisfies the nonhomogeneous equation

$$L[x(t)] = f_1(t) + f_2(t).$$

Proof. Obviously. $\square$

Problem 3.4.5. Let the functions

$$3e^t + te^t, 7e^t + te^t, 5e^t + te^t + e^{-t^3}$$

be three solutions of the second-order linear equation

$$x''(t) + p(t)x' + q(t)x = f(t),$$

where $p(t)$, $q(t)$ and $f(t)$ are continuous on the real line. Find a general solution and a particular solution satisfying the initial condition $x(0) = 0$, $x'(0) = 1$ of this equation.

Solution. Firstly, we get $x_1(t) = 4e^t$, $x_2(t) = 2e^t + e^{-t^3}$ are solutions of $x''(t) + p(t)x' + q(t)x = 0$ so the general solution of $x''(t) + p(t)x' + q(t)x = f(t)$ can be written by

$$x(t) = c_1 e^t + c_2 e^{-t^3} + te^t$$

and then we get

$$x(0) = c_1 + c_2 = 0, \quad x'(0) = c_1 + 1 = 1$$

so we get $c_1 = 0$, $c_2 = 0$. The particular solution is $x_p(t) = te^t$. $\square$

Problem 3.4.6. Verify that $x_1(t) = t$ and $x_2(t) = \frac{1}{t}$ are two solutions of

$$t^3 x'' + t^2 x' - tx = 0.$$

Use this information and the method of variation of parameters to find the general solution of

$$t^3 x'' + t^2 x' - tx = \frac{t}{1+t}.$$

Solution. Verify that $x_1(t) = t$ and $x_2(t) = \frac{1}{t}$ satisfy the equation $t^3 x'' + t^2 x' - tx = 0$. Calculate the Wronskian $W(t) = -\frac{2}{t}$, convert the non-homogeneous equation to the standard form $x'' + \frac{1}{t}x' - \frac{1}{t^2}x = \frac{1}{t^2(1+t)}$. Apply the variation of parameters method: $x_p(t) = -t \int \frac{\frac{1}{t} \cdot \frac{1}{t^2(1+t)}}{-\frac{2}{t}} dt + \frac{1}{t} \int \frac{t \cdot \frac{1}{t^2(1+t)}}{-\frac{2}{t}} dt = \frac{t}{2} \int \frac{1}{t^2(1+t)} dt - \frac{1}{2t} \int \frac{1}{1+t} dt$, using the partial fraction $\frac{1}{t^2(1+t)} = -\frac{1}{t} + \frac{1}{t^2} + \frac{1}{1+t}$, we obtain the particular solution $x_p(t) = \frac{1}{2}(t - \frac{1}{t}) \ln|1+t| - \frac{t}{2} \ln|t| - \frac{1}{2}$, thus the general solution is $x(t) = c_1 t + \frac{c_2}{t} + \frac{1}{2}(t - \frac{1}{t}) \ln|1+t| - \frac{t}{2} \ln|t| - \frac{1}{2}$. $\square$

Problem 3.4.7. Prove that for any solution $\varphi(t)$ of the differential equation

$$x'' + 3x' + 2x = f(t),$$

we have

$$\lim_{t \rightarrow +\infty} \varphi(t) = 0,$$

where $f(t)$ is continuous on the interval $[a, +\infty)$ and $\lim_{t \rightarrow +\infty} f(t) = 0$.

Remark 3.3. 这个问题无需借助求解微分方程即可证明. 求解微分方程的步骤相当繁琐. 相反, 我们采用洛必达法则的上下限形式来给出一个简洁的证明.

Proof. Using the L'Hôpital's rule with upper and lower limits, we have

$$\begin{aligned} \overline{\lim}_{t \rightarrow +\infty} \varphi(t) &= \overline{\lim}_{t \rightarrow +\infty} \frac{e^{2t} \varphi(t)}{e^{2t}} \leq \overline{\lim}_{t \rightarrow +\infty} \frac{e^{2t} [\varphi'(t) + 2\varphi(t)]}{e^{2t}} = \frac{1}{2} \overline{\lim}_{t \rightarrow +\infty} [\varphi'(t) + 2\varphi(t)] \\ &= \frac{1}{2} \overline{\lim}_{t \rightarrow +\infty} \frac{e^t [\varphi'(t) + 2\varphi(t)]}{e^t} \\ &\leq \frac{1}{2} \overline{\lim}_{t \rightarrow +\infty} \frac{e^t [\varphi''(t) + 3\varphi'(t) + 2\varphi(t)]}{e^t} = \frac{1}{2} \overline{\lim}_{t \rightarrow +\infty} [\varphi''(t) + 3\varphi'(t) + 2\varphi(t)] \\ &= \frac{1}{2} \overline{\lim}_{t \rightarrow +\infty} f(t) = 0, \end{aligned}$$

and

$$\begin{aligned}\lim_{t \rightarrow +\infty} \varphi(t) &= \lim_{t \rightarrow +\infty} \frac{e^{2t}\varphi(t)}{e^{2t}} \geq \lim_{t \rightarrow +\infty} \frac{e^{2t}[\varphi'(t) + 2\varphi(t)]}{2e^{2t}} = \frac{1}{2} \lim_{t \rightarrow +\infty} [\varphi'(t) + 2\varphi(t)] \\ &= \frac{1}{2} \lim_{t \rightarrow +\infty} \frac{e^t[\varphi'(t) + 2\varphi(t)]}{e^t} \geq \frac{1}{2} \lim_{t \rightarrow +\infty} \frac{e^t[\varphi''(t) + 3\varphi'(t) + 2\varphi(t)]}{e^t} \\ &= \frac{1}{2} \lim_{t \rightarrow +\infty} [\varphi''(t) + 3\varphi'(t) + 2\varphi(t)] = \frac{1}{2} \lim_{t \rightarrow +\infty} f(t) = 0.\end{aligned}$$

And that completes the proof. $\square$

Exercise 3.5

Find the general solutions of the differential equations in Problems 1 through 10.

Problem 3.5.1. $x'' - 4x' = 0$.

Solution. Solving the characteristic equation $\lambda^2 - 4\lambda = 0$ gives $\lambda_1 = 0$ and $\lambda_2 = 4$. Since we have two distinct real roots, the general solution is $x(t) = c_1 + c_2e^{4t} = c_1 + c_2e^{4t}$, where c_1 and c_2 are arbitrary constants. $\square$

Problem 3.5.2. $x'' - 4x = 0$.

Solution. Solving the characteristic equation $\lambda^2 - 4 = 0$ gives $\lambda_1 = 2, \lambda_2 = -2$. Therefore, the general solution is $x(t) = c_1e^{2t} + c_2e^{-2t} = c_1e^{2t} + c_2e^{-2t}$, where c_1 and c_2 are arbitrary constants. $\square$

Problem 3.5.3. $x'' + 6x' + 9x = 0$.

Solution. Solving the characteristic equation $\lambda^2 + 6\lambda + 9 = 0$ gives $\lambda_1 = \lambda_2 = -3$. Therefore, the general solution is $x(t) = (c_1 + c_2t)e^{-3t}$, where c_1 and c_2 are arbitrary constants. $\square$

Problem 3.5.4. $x^{(4)} + 3x^{(3)} = 0$.

Solution. $x(t) = c_4t^2 + c_3t - \frac{1}{27}c_1e^{-3t} + c_2$. $\square$

Problem 3.5.5. $x^{(4)} - 8x'' + 16x = 0$.

Solution. $x(t) = c_1e^{-2t} + c_2e^{-2t}t + c_3e^{2t} + c_4e^{2t}t$. $\square$

Problem 3.5.6. $x^{(4)} - 16x = 0$.

Solution. $x(t) = c_1e^{2t} + c_3e^{-2t} + c_2\cos(2t) + c_4\sin(2t)$. $\square$

Problem 3.5.7. $x^{(7)} - 2x^{(5)} + x^{(3)} = 0$.

Solution. $x(t) = c_7t^2 + c_6t + e^{-t}(-c_2(t+3) + e^{2t}(c_4(t-3) + c_3) - c_1) + c_5$. $\square$

Problem 3.5.8. $x''' - 5x'' + 8x' - 4x = 0$.

Solution. $x(t) = c_1e^t + c_2e^{2t} + c_3e^{2t}t$. $\square$

Problem 3.5.9. $x''' + x'' - x' - x = 0$.

Solution. $x(t) = c_1 e^{-t} + c_2 e^{-t} t + c_3 e^t$. □

Problem 3.5.10. $x'' + 8x' + 25x = 0$.

Solution. $x(t) = c_2 e^{-4t} \cos(3t) + c_1 e^{-4t} \sin(3t)$. □

Solve the initial value problem given in Problems 11 through 15.

Problem 3.5.11. $x'' - 4x' + 3x = 0; x(0) = 7, x'(0) = 11$.

Solution. Its characteristic equation is $\lambda^2 - 4\lambda + 3 = 0$, which gives $\lambda_1 = 3, \lambda_2 = 1$. Therefore, its general solution is $x(t) = c_1 e^{3t} + c_2 e^t$. With the initial values, we get $c_1 = -3, c_2 = 10$. Then the solution is $x(t) = -3e^{3t} + 10e^t$. □

Problem 3.5.12. $x'' - 6x' + 25x = 0; x(0) = 3, x'(0) = 1$.

Solution. Its characteristic equation is $\lambda^2 - 6\lambda + 25 = 0$, which gives $\lambda_1 = 3 - 4i, \lambda_2 = 3 + 4i$. Therefore, its general solution is $x(t) = c_2 e^{3t} \cos(4t) + c_1 e^{3t} \sin(4t)$. With the initial values, we get $c_1 = -2, c_2 = 3$. Then the solution is $x(t) = e^{3t}(3 \cos(4t) - 2 \sin(4t))$. □

Problem 3.5.13. $x'' + x' = 0; x(0) = 2, x'(0) = 5$.

Solution. Its characteristic equation is $\lambda^2 + \lambda = 0$, which gives $\lambda_1 = 0, \lambda_2 = -1$. Therefore, its general solution is $x(t) = c_2 - c_1 e^{-t}$. With the initial values, we get $c_1 = -1, c_2 = 4$. Then the solution is $x(t) = e^{-t}(e^t - 1)$. □

Problem 3.5.14. $x'' + 4x' + 4x = 0; x(2) = 4, x'(2) = 0$.

Solution. $x(t) = 4e^{4-2t}(2t - 3)$. □

Problem 3.5.15. $2x''' - 3x'' - 2x' = 0; x(0) = 1, x'(0) = -1, x''(0) = 3$.

Solution. $x(t) = \frac{1}{2} \left(8e^{-t/2} + e^{2t} - 7 \right)$. □

In Problems 16 through 18, find a linear homogeneous constant coefficient equation with the given general solution.

Problem 3.5.16. $x(t) = (c_1 + c_2 t + c_3 t^2 + c_4 t^3) e^{2t}$.

Solution. Through the discussion of repeated real roots, we can easily see that $\lambda = 2$ is a repeated real root of multiplicity 4. Then a linear homogeneous constant coefficient equation is

$$x^{(4)} - 8x''' + 24x'' - 32x' + 16x = 0.$$

□

Problem 3.5.17. $x(t) = c_1 e^{-2t} + c_2 \cos t + c_3 \sin t$.

Solution. Through the discussion of repeated real roots, we can easily see that $\lambda_1 = -2, \lambda_2 = i, \lambda_3 = -i$. Then a linear homogeneous constant coefficient equation is

$$x''' + 2x'' + x' + 2x = 0.$$

□

Problem 3.5.18. $x(t) = (c_1 + c_2t + c_3t^2) \cos 2t + (c_4 + c_5t + c_6t^2) \sin 2t$.

Solution. The general solution can be rewrite into

$$x(t) = (c_1 \cos 2t + c_4 \sin 2t) + t(c_2 \cos 2t + c_5 \sin 2t) + t^2(c_3 \cos 2t + c_6 \sin 2t).$$

So a linear homogeneous constant coefficient equation is

$$x^{(6)} + 12x^{(4)} + 48x'' + 64x = 0.$$

□

Problem 3.5.19. Prove Theorem 3.5.2, i.e.,

Theorem 3.5.1. Suppose that

$$P(\lambda) = (\lambda - \lambda_1)^{k_1} (\lambda - \lambda_2)^{k_2} \cdots (\lambda - \lambda_m)^{k_m} = 0,$$

that is, $\lambda_i (i = 1, 2, \dots, m)$ is a repeated real root of multiplicity k_i , $k_1 + k_2 + \cdots + k_m = n$. Then the following functions

$$\begin{aligned} & e^{\lambda_1 t}, te^{\lambda_1 t}, \dots, t^{k_1-1} e^{\lambda_1 t}; \\ & e^{\lambda_2 t}, te^{\lambda_2 t}, \dots, t^{k_2-1} e^{\lambda_2 t}; \\ & \dots \\ & e^{\lambda_m t}, te^{\lambda_m t}, \dots, t^{k_m-1} e^{\lambda_m t}; \end{aligned}$$

are n linearly independent solutions of the equation

$$x^{(n)} + a_1 x^{(n-1)} + a_{n-1} x' + a_n x = 0. \quad (3.7)$$

Proof. It is noted that the following fact holds: If η is a k -th root of $p(\lambda) = 0$, then

$$p(\eta) = p'(\eta) = \cdots = p^{(k-1)}(\eta) = 0, \quad p^{(k)}(\eta) \neq 0.$$

Due to

$$L_n(e^{\lambda x}) = p(\lambda)e^{\lambda x},$$

Taking the derivative of λ with respect to k on both sides k times, we obtain

$$L_n(x^k e^{\lambda x}) = L_n\left(\frac{\partial^k}{\partial \lambda^k} e^{\lambda x}\right) = \frac{\partial^k}{\partial \lambda^k} L_n(e^{\lambda x}) = \frac{\partial^k}{\partial \lambda^k} (p(\lambda)e^{\lambda x}).$$

So

$$L_n(x^k e^{\lambda x}) = \left(p^{(k)}(\lambda) + C_k^1 p^{(k-1)}(\lambda)x + \cdots + C_k^k p(\lambda)x^k \right) e^{\lambda x}.$$

Therefore, it can be concluded that if λ_j is an n_j -fold root of $p(\lambda) = 0$, then

$$L_n(x^{k_j} e^{\lambda_j x}) = 0, \quad k_j = 0, 1, \cdots, n_j - 1.$$

So

$$x^{k_j} e^{\lambda_j x} \quad (k_j = 0, 1, \cdots, n_j - 1; j = 1, 2, \cdots, s)$$

These are the n solutions of equation (3.7).

Let's start with the analysis. Rewrite the above equation as

$$\sum_{j=1}^s P_j(x) e^{\lambda_j x} \equiv 0,$$

Here, $P_j(x) (j = 1, 2, \cdots, s)$ are $(n_j - 1)$ -degree polynomials of x , with coefficients being $c_{j0}, \cdots, c_{j, n_j-1}$. Dividing both sides of this identity by $e^{\lambda_1 x}$, we obtain

$$P_1(x) + \sum_{j=2}^s P_j(x) e^{(\lambda_j - \lambda_1)x} \equiv 0.$$

Taking the n_1 th derivative on both sides of the above equation, we obtain

$$\sum_{j=2}^s Q_j(x) e^{(\lambda_j - \lambda_1)x} \equiv 0, \tag{3.8}$$

Here, $Q_j(x)$ is an $(n_j - 1)$ -degree polynomial of x .

Among them, $Q_j(x) (j = 2, 3, \cdots, s)$ are polynomials of the same degree as $P_j(x)$, that is, the degree of $Q_2(x)$ is $n_2 - 1$, the degree of $Q_3(x)$ is $n_3 - 1$, and so on, and the coefficient of the highest degree term of $Q_j(x)$ is $(\lambda_j - \lambda_1)^{n_1} c_{j, n_j-1}$. After dividing both sides of (3.8) by $e^{(\lambda_2 - \lambda_1)x}$, we obtain

$$Q_2(x) + \sum_{j=3}^s Q_j(x) e^{(\lambda_j - \lambda_2)x} \equiv 0.$$

Repeat the same process as before, and eventually you will obtain

$$R_s(x) e^{(\lambda_s - \lambda_{s-1})x} \equiv 0,$$

Here, $R_s(x)$ is an $(n_s - 1)$ -degree polynomial, and the coefficient of its highest degree term is

$$c_{s, n_s-1} (\lambda_2 - \lambda_1)^{n_1} \cdots (\lambda_s - \lambda_{s-1})^{n_{s-1}}.$$

From the above identity, we can conclude that $R_s(x) \equiv 0$, thus

$$c_{s, n_s-1} = 0.$$

With the analysis from the previous paragraph, let's now prove it.

$$x^{k_j} e^{\lambda_j x} \quad (k_j = 0, 1, \dots, n_j - 1; j = 1, 2, \dots, s)$$

The linear independence. Suppose there exist n non-zero constants c_{jk_j} (where $j = 1, 2, \dots, s; k_j = 0, 1, \dots, n_j - 1$), such that

$$\sum_{j=1}^s \sum_{k_j=0}^{n_j-1} c_{jk_j} x^{k_j} e^{\lambda_j x} \equiv 0.$$

Let the non-zero coefficient with the largest subscript (in lexicographical order) in the above equation be $c_{\sigma k_\sigma}$, then

$$\sum_{j=1}^{\sigma} P_j(x) e^{\lambda_j x} \equiv 0,$$

And the highest degree term of $P_\sigma(x)$ is $c_{\sigma k_\sigma} x^{k_\sigma}$. By repeating the analysis and reasoning process of the previous paragraph, we obtain

$$R_\sigma(x) e^{(\lambda_\sigma - \lambda_{\sigma-1})x} \equiv 0,$$

And the coefficient of the highest degree term of $R_\sigma(x)$ is $c_{\sigma k_\sigma} (\lambda_2 - \lambda_1)^{n_1} \dots (\lambda_\sigma - \lambda_{\sigma-1})^{n_{\sigma-1}} \neq 0$. This contradicts the condition that $R_\sigma(x) \equiv 0$. $\square$

Problem 3.5.20. Show that any solution $x(t)$ of the differential equation $x'' + ax' + bx = 0$, one has $\lim_{t \rightarrow +\infty} x(t) = 0$, where $a > 0, b > 0$.

Proof. We can get $p(\lambda) = \lambda^2 + a\lambda + b = 0$.

- If $a^2 - 4b = 0$, then $\lambda_1 = \lambda_2 = -\frac{a}{2}$. So $x(t) = c_1 e^{-\frac{a}{2}t} + c_2 t e^{-\frac{a}{2}t}$, we can get $\lim_{t \rightarrow +\infty} x(t) = 0$.
- If $a^2 - 4b < 0$, then $p(\lambda)$ have two complex solutions

$$\lambda_1 = -\frac{a}{2} + \frac{\sqrt{4b - a^2}}{2}i, \lambda_2 = -\frac{a}{2} - \frac{\sqrt{4b - a^2}}{2}i.$$

So $x(t) = c_1 \cos \frac{\sqrt{4b - a^2}}{2} e^{-\frac{a}{2}t} + c_2 \sin \frac{\sqrt{4b - a^2}}{2} e^{-\frac{a}{2}t}$. We can get $\lim_{t \rightarrow +\infty} x(t) = 0$.

- If $a^2 - 4b > 0$, then $\lambda_1 = -\frac{a}{2} + \frac{\sqrt{a^2 - 4b}}{2}$, $\lambda_2 = -\frac{a}{2} - \frac{\sqrt{a^2 - 4b}}{2}$, so $x(t) = c_1 e^{\frac{a}{2} + \frac{\sqrt{a^2 - 4b}}{2}t} + c_2 e^{\frac{a}{2} - \frac{\sqrt{a^2 - 4b}}{2}t}$, we can get $\lim_{t \rightarrow +\infty} x(t) = 0$.

In all, $\lim_{t \rightarrow +\infty} x(t) = 0$. $\square$

Problem 3.5.21. Show that if the real parts of all solutions of $P(\lambda) = \lambda^n + a_1 \lambda^{n-1} + \dots + a_{n-1} \lambda + a_n = 0$ are negative, then

$$\lim_{t \rightarrow +\infty} x(t) = 0$$

for every solution of (3.7).

Proof. If all the roots $\lambda_i = \alpha_i + i\beta_i$ of the characteristic equation $P(\lambda) = 0$ satisfy $\alpha_i < 0$, then any solution of the differential equation can be expressed as $x(t) = \sum c_{ij} t^j e^{\lambda_i t}$. For each term, we have $|t^j e^{\lambda_i t}| = t^j e^{\alpha_i t}$, since the exponential decay $e^{\alpha_i t}$ (with $\alpha_i < 0$) decays to zero faster than any polynomial growth t^j . Therefore, $\lim_{t \rightarrow +\infty} t^j e^{\alpha_i t} = 0$. As a result, all the terms tend to zero, and thus $\lim_{t \rightarrow +\infty} x(t) = 0$. $\square$

Problem 3.5.22. Show that the substitution $x = e^t$ ($x > 0$) can transform the second-order Euler's equation

$$ax^2y'' + bxy' + cy = 0$$

to a constant coefficient homogeneous linear equation. Further, show similarly that this same substitution transforms the third-order Euler's equation

$$ax^3y''' + bx^2y'' + cxy' + dy = 0$$

into the constant coefficient equation

$$a \frac{d^3y}{dt^3} + (b - 3a) \frac{d^2y}{dt^2} + (c - b + 2a) \frac{dy}{dt} + dy = 0,$$

where a, b, c, d are constants.

Proof. By substituting $x = e^t$ (where $x > 0$), using the chain rule, we obtain $\frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{dt}{dx} = e^{-t} \frac{dy}{dt}$. For the second derivative:

$$\frac{d^2y}{dx^2} = \frac{d}{dx} \left(e^{-t} \frac{dy}{dt} \right) = e^{-t} \frac{d}{dt} \left(e^{-t} \frac{dy}{dt} \right) = e^{-2t} \left(\frac{d^2y}{dt^2} - \frac{dy}{dt} \right).$$

Substitute the second-order Euler equation:

$$ax^2y'' + bxy' + cy = ae^{2t} \cdot e^{-2t} \left(\frac{d^2y}{dt^2} - \frac{dy}{dt} \right) + be^t \cdot e^{-t} \frac{dy}{dt} + cy = a \frac{d^2y}{dt^2} + (b - a) \frac{dy}{dt} + cy = 0.$$

For the third-order derivative:

$$\frac{d^3y}{dx^3} = e^{-t} \frac{d}{dt} \left[e^{-2t} \left(\frac{d^2y}{dt^2} - \frac{dy}{dt} \right) \right] = e^{-3t} \left(\frac{d^3y}{dt^3} - 3 \frac{d^2y}{dt^2} + 2 \frac{dy}{dt} \right)$$

Substitute the third-order Euler equation:

$$\begin{aligned} & ax^3y''' + bx^2y'' + cxy' + dy \\ &= ae^{3t} \cdot e^{-3t} \left(\frac{d^3y}{dt^3} - 3 \frac{d^2y}{dt^2} + 2 \frac{dy}{dt} \right) + be^{2t} \cdot e^{-2t} \left(\frac{d^2y}{dt^2} - \frac{dy}{dt} \right) + ce^t \cdot e^{-t} \frac{dy}{dt} + dy \\ &= a \frac{d^3y}{dt^3} + (b - 3a) \frac{d^2y}{dt^2} + (c - b + 2a) \frac{dy}{dt} + dy = 0. \end{aligned}$$

This thus proves that replacing $x = e^t$ can transform the Euler equation into a linear differential equation with constant coefficients. $\square$

Find the general solutions (for $x > 0$) of the Euler's equations in Problems 23 through 26.

Problem 3.5.23. $x^2 y'' + xy' + 9y = 0$.

Solution. The substitution $x = e^t$ ($x > 0$) can get $y'' + 9y = 0$. The characteristic equation is

$$\lambda^2 + 9 = 0.$$

It has the roots $\lambda = \pm 3i$, we substitute $t = \ln x$. Thus a general solution is given by

$$y = (c_1 \cos 3 \ln x + c_2 \sin 3 \ln x). \quad \square$$

Problem 3.5.24. $x^3 y''' + 6x^2 y'' + 4xy' = 0$.

Solution. The substitution $x = e^t$ ($x > 0$) can get $y''' + 3y'' = 0$. The characteristic equation is

$$\lambda^3 + 3\lambda^2 = \lambda^2(\lambda + 3) = 0.$$

It has the double root $\lambda = 0$ and the simple root $\lambda = -3$. The part of a general solution corresponding to $\lambda = 0$ is of the form

$$c_1 + c_2 t = c_1 + c_2 t,$$

While the simple root $\lambda = -3$ contributes $c_3 e^{-3t}$ to the solution. we substitute $t = \ln x$. Thus a general solution is given by

$$y = c_1 + c_2 \ln x - c_3 / 3x^{-3}. \quad \square$$

Problem 3.5.25. $x^2 y'' + 7xy' + 25y = 0$.

Solution. $y = \frac{c_2 \cos(4 \ln(x))}{x^3} + \frac{c_1 \sin(4 \ln(x))}{x^3}.$ $\square$

Problem 3.5.26. $x^3 y''' - 3x^2 y'' + xy' = 0$.

Solution. $y = \frac{c_2 x^{3+\sqrt{3}}}{3+\sqrt{3}} + \frac{c_1 x^{3-\sqrt{3}}}{3-\sqrt{3}} + c_3.$ $\square$

Exercise 3.6

In Problems 1 through 10, find a particular solution x_p of the given equation.

Problem 3.6.1. $x'' + 16x = 2e^{3t}$.

Solution. For the equation $x'' + 16x = 2e^{3t}$, the roots of the characteristic equation $\lambda^2 + 16 = 0$ are purely imaginary and do not coincide with e^{3t} . Let the particular solution be $x_p = Ae^{3t}$. Substituting this in, we get $9Ae^{3t} + 16Ae^{3t} = 25Ae^{3t} = 2e^{3t}$. Solving for A , we obtain $A = \frac{2}{25}$, thus the particular solution is $x_p = \frac{2}{25}e^{3t}$. $\square$

Problem 3.6.2. $x'' - x' - 2x = t^2$.

Solution. For the equation $x'' - x' - 2x = t^2$, the characteristic equation $\lambda^2 - \lambda - 2 = 0$ has roots $\lambda = 2$ and -1 , which are different from the polynomial form. Let the particular solution be $x_p = At^2 + Bt + C$. Substituting and comparing coefficients gives $-2A = 1$, $-2A - 2B = 0$, and $2A - B - 2C = 0$. Solving these equations yields $A = -\frac{1}{2}$, $B = \frac{1}{2}$, and $C = -\frac{3}{4}$. Therefore, the particular solution is $x_p = -\frac{1}{2}t^2 + \frac{1}{2}t - \frac{3}{4}$. $\square$

Problem 3.6.3. $x'' + x' + x = \sin^2 t$.

Solution. For the equation $x'' + x' + x = \sin^2 t$, using $\sin^2 t = \frac{1 - \cos 2t}{2}$, the characteristic equation $\lambda^2 + \lambda + 1 = 0$ has complex roots which are different from the form of the right-hand term. Let the particular solution be $x_p = A + B \cos 2t + C \sin 2t$. Substituting and comparing coefficients, we obtain $A = \frac{1}{2}$, $B = \frac{3}{26}$, and $C = -\frac{1}{13}$. Therefore, the particular solution is $x_p = \frac{1}{2} + \frac{3}{26} \cos 2t - \frac{1}{13} \sin 2t$. $\square$

Problem 3.6.4. $2x'' + 4x' + 7x = 2t^2$.

Solution. For the equation $2x'' + 4x' + 7x = 2t^2$, the characteristic equation $2\lambda^2 + 4\lambda + 7 = 0$ has complex roots, which is different from the polynomial form. Let the particular solution be $x_p = At^2 + Bt + C$. Substituting and comparing coefficients gives $7A = 2$, $8A + 7B = 0$, and $4A + 4B + 7C = 0$. Solving these equations yields $A = \frac{2}{7}$, $B = -\frac{16}{49}$, and $C = \frac{8}{343}$. Therefore, the particular solution is $x_p = \frac{2}{7}t^2 - \frac{16}{49}t + \frac{8}{343}$. $\square$

Problem 3.6.5. $x'' + 2x' + 5x = 2e^t \sin t$.

Solution. For the equation $x'' + 2x' + 5x = 2e^t \sin t$, the characteristic equation $\lambda^2 + 2\lambda + 5 = 0$ has complex roots, which are different from the form of e^t . Let the particular solution be $x_p = ((At + B) \cos t + (Ct + D) \sin t) e^t$. Substituting this in, we get

$$\begin{cases} 4A + 7B + 2C + 4D = 0, \\ 7A + 4C = 0, \\ -4A + 7C = 0, \\ -2A - 4B + 4C + 7D = 1, \end{cases}$$

which gives $A = C = 0$, $B = -\frac{8}{65}$, $D = \frac{14}{65}$. Therefore, the particular solution is $x_p = \left(-\frac{8}{65} \cos t + \frac{14}{65} \sin t\right) e^t$. $\square$

Problem 3.6.6. $x^{(4)} - 2x'' + x = te^t$.

Solution. $x(t) = \frac{1}{48}e^t(2t^3 - 6t^2 + 9t - 6)$. $\square$

Problem 3.6.7. $x'' + x = \sin t + 5 \cos 2t$.

Solution. For the equation $x'' + x = \sin t + 5 \cos 2t$, the characteristic equation $\lambda^2 + 1 = 0$ has roots $\lambda = \pm i$ which are the same as the form of $\sin t$. Let the particular solution be $x_p = At \cos t + B \sin 2t + C \cos 2t$. Substituting this into the equation and comparing coefficients, we obtain $A = -\frac{1}{2}$, $B = 0$, and $C = -\frac{5}{3}$. Therefore, the particular solution is $x_p = -\frac{1}{2}t \cos t - \frac{5}{3} \cos 2t$. $\square$

Problem 3.6.8. $x'' - 2x' + 10x = t \sin 2t$.

Solution. $x_p = \frac{1}{338}((39t + 29) \sin(2t) + (26t + 2) \cos(2t))$. $\square$

Problem 3.6.9. $x'' - x = 2e^t + 1$.

Solution. $x_0 = \frac{1}{8}e^{-t}(8e^{2t}t - 8e^t - 4e^{2t} - 1)$. $\square$

Problem 3.6.10. $x^{(4)} - 5x'' + 4x = e^t + te^{2t}$.

Solution. $x_p = \frac{e^t(72e^t t^2 - 228e^t t - 288t + 265e^t - 48)}{1728}$. $\square$

In Problems 11 through 18, set up the appropriate form of a particular solution x_p , but do not determine the values of the coefficients.

Problem 3.6.11. $x'' - 2x' + 2x = e^t \sin t + t^2$.

Solution. The root of the characteristic equation

$$\lambda^2 - 2\lambda + 2 = 0$$

is given by

$$\lambda_1 = 1 + i, \quad \lambda_2 = 1 - i,$$

and neither of them is 0 nor 1. So the appropriate form of the particular solution x_p is given by

$$x_p = e^t(c_1 \cos t + c_2 \sin t) + c_3 t^2 + c_4 t + c_5,$$

where c_1, c_2, c_3, c_4, c_5 are undetermined coefficients. $\square$

Remark 3.4. In fact, its general solution is

$$x(t) = \frac{1}{4}(2(t+1)^2 - 2e^t(t - 2c_2) \cos(t) + (1 + 4c_1)e^t \sin(t)).$$

Problem 3.6.12. $x^{(5)} - x''' = 2e^t + t^2 + 4$.

Solution. Homogeneous equation $x^{(5)} - x''' = 0$, its characteristic equation is:

$$\lambda^5 - \lambda^3 = 0.$$

We can get

$$\lambda_1 = \lambda_2 = \lambda_3 = 0, \lambda_4 = 1, \lambda_5 = -1.$$

$f_1(t) = 2e^t, f_2(t) = (t^2 + 4)$. So, the appropriate form of the particular solution is given by:

$$\varphi_p^1 = tc_1 e^t, \quad \varphi_p^2 = t^3(c_2 t^2 + c_3 t + c_4).$$

Hence,

$$\varphi_p = tc_1e^t + t^3(c_2t^2 + c_3t + c_4) \quad (c_i \text{ are constants}). \quad \square$$

Remark 3.5. In fact, its general solution is

$$x(t) = -\frac{t^5}{60} - t^3 + c_5t^2 + c_4t + e^t \left(t - \frac{7}{2} + c_1 \right) - c_2e^{-t} + c_3.$$

Problem 3.6.13. $x'' + 4x = t^2 \cos 2t + 3 \sin t + 4$.

Solution. Homogeneous equation $x'' + 4x = 0$, its characteristic equation is: $\lambda^2 + 4 = 0$. We can get

$$\lambda_1 = 2i, \lambda_2 = -2i,$$

$f_1(t) = e^{(0t)} \cos 2t \cdot t^2, \alpha_1 = 0, \beta_1 = 2$. We can know that $2i$ is a single eigenvalue. So, φ_p^1 is given by:

$$\varphi_p^1 = t(c_1t^2 + c_2t + c_3)(c_4 \cos 2t + c_5 \sin 2t).$$

Note that $f_2(t) = 3 \sin t, \alpha_2 = 0, \beta_2 = 1$, i is not an eigenvalue,

$$\varphi_p^2 = c_6 \sin t + c_7 \cos t$$

$f_3(t) = 4, \alpha_3 = 0, \beta_3 = 0$, 0 is not an eigenvalue, $\varphi_p^3 = c_8$. So,

$$\varphi_p = t(c_1t^2 + c_2t + c_3)(c_4 \cos 2t + c_5 \sin 2t) + c_6 \sin t + c_7 \cos t + c_8 \quad (c_i \text{ are constants}). \quad \square$$

Remark 3.6. In fact, its general solution is

$$x(t) = \frac{1}{6}t^3 \sin(t) \cos(t) + \left(\frac{t^2}{16} - \frac{1}{128} + c_1 \right) \cos(2t) - \frac{1}{32}t \sin(2t) + \sin(t) + c_2 \sin(2t) + 1.$$

Problem 3.6.14. $x''' - x'' - 12x' = t^2 - 2te^{3t}$.

$$\text{Solution. } x(t) = -\frac{t^3}{36} + \frac{t^2}{144} - \frac{13t}{864} + \frac{1}{864}e^{3t}(96t + 48) - \frac{1}{3}c_1e^{-3t} + \frac{1}{4}c_2e^{4t} + c_3. \quad \square$$

Problem 3.6.15. $x'' + 3x' + 2x = t^2e^t - te^{3t}$.

$$\text{Solution. } x(t) = \frac{e^t(-1800t^2 + 540e^{2t}t + 3000t - 243e^{2t} - 1900)}{10800} + c_1e^{-2t} + c_2e^{-t}. \quad \square$$

Problem 3.6.16. $x^{(4)} - 2x'' + x = \cos t + e^t \sin t + t^2$.

$$\text{Solution. } x(t) = \frac{1}{16}(2e^tt^2 + 16t^2 - 4e^tt + 3e^t + 4 \cos(t) + 64) + c_1e^{-t} + c_2e^{-t}t + c_3e^t + c_4e^tt. \quad \square$$

Problem 3.6.17. $x^{(4)} + 9x'' = (t^2 + 1) \sin 3t + \cos t$.

Solution.

$$x(t) = \frac{6(6t^3 + t - 9 - 108c_1) \cos(3t) - (90t^2 + 41 + 648c_2) \sin(3t) - 729 \cos(t)}{5832} + c_4t + c_3. \quad \square$$

Problem 3.6.18. $x^{(4)} + 5x''' + 4x = 3 \sin t + \cos 2t$.

Solution. $x(t) = -\frac{\sin(t)}{12} + \left(-\frac{11}{72} + c_1\right) \cos(2t) + c_4 \sin(t) + c_2 \sin(2t) - \frac{1}{6} \cos(t)(3t + t \sin(t) - 6c_3).$ $\square$

Solve the initial value problems in Problems 19 through 23.

Problem 3.6.19. $x'' + 9x = \sin 2t; x(0) = 1, x'(0) = 0.$

Solution. The general solution is given by

$$x^*(t) = \frac{1}{5} \sin(2t) + c_1 \cos(3t) + c_2 \sin(3t).$$

With the initial values, we get $c_1 = 1, c_2 = -2/15$. Finally, we obtain that

$$x(t) = \frac{1}{5} \sin(2t) - \frac{2}{15} \sin(3t) + \cos(3t).$$

as the required solution. $\square$

Problem 3.6.20. $x'' + 2x' + 2x = \sin 3t; x(0) = 2, x'(0) = 0.$

Solution. The general solution is given by

$$x^*(t) = -\frac{7}{85} \sin(3t) - \frac{6}{85} \cos(3t) + c_2 e^{-t} \cos(t) + c_1 e^{-t} \sin(t).$$

With the initial values, we get $c_1 = 176/85, c_2 = 197/85$. Finally, we obtain that

$$x(t) = \frac{1}{85} e^{-t} (197 \sin(t) - 7e^t \sin(3t) + 176 \cos(t) - 6e^t \cos(3t)).$$

as the required solution. $\square$

Problem 3.6.21. $x'' + x = \cos t; x(0) = 1, x'(0) = -1.$

Solution. $x(t) = \frac{1}{2}(t-2) \sin(t) + \cos(t).$ $\square$

Problem 3.6.22. $x''' + x'' = t + e^{-t}; x(0) = 1, x'(0) = 0, x''(0) = 1.$

Solution. $x(t) = \frac{1}{6}(t^3 - 3t^2 + 18t - 18) + e^{-t}(t + 4).$ $\square$

Problem 3.6.23. $x'' + 4x = 2t; x(0) = 1, x'(0) = 2.$

Solution. $x(t) = \cos(2t) + \frac{1}{2}(t + 3 \sin(t) \cos(t)).$ $\square$

Use trigonometric identities to find general solutions of the equations in Problems 24 through 27.

Problem 3.6.24. $x'' + 4x = 2 \cos^3 t.$

Solution. We have $2 \cos^3 t = \cos t(1 + \cos 2t) + \cos t + \cos t \cos 2t = \frac{3}{2} \cos t + \frac{1}{2} \cos 3t$. Therefore, the general solution is $x(t) = \frac{\cos(t)}{2} - \frac{1}{10} \cos(3t) + c_1 \cos(2t) + c_2 \sin(2t).$ $\square$

Problem 3.6.25. $x'' + x' + x = \cos t \sin 2t.$

Solution. We have $\cos t \sin 2t = 2 \sin t \cos^2 t = \sin t(1 + \cos 2t) = \frac{1}{2} \sin t + \frac{1}{2} \sin 3t$. Therefore, the general solution is $x(t) = -\frac{4}{73} \sin(3t) - \frac{\cos(t)}{2} - \frac{3}{146} \cos(3t) + c_2 e^{-t/2} \cos\left(\frac{\sqrt{3}t}{2}\right) + c_1 e^{-t/2} \sin\left(\frac{\sqrt{3}t}{2}\right)$. $\square$

Problem 3.6.26. $x'' + 9x = \cos^4 t$.

Solution. We have $\cos^4 t = \left(\frac{1 + \cos 2t}{2}\right)^2 = \frac{1 + 2 \cos 2t + \cos^2 2t}{4} = \frac{3}{8} + \frac{1}{2} \cos 2t + \frac{1}{8} \cos 4t$. Therefore, the general solution is $x(t) = \frac{1}{10} \cos(2t) - \frac{1}{56} \cos(4t) + c_1 \cos(3t) + c_2 \sin(3t) + \frac{1}{24}$. $\square$

Problem 3.6.27. $x'' + x = t \sin^3 t$.

Solution. We have $t \sin^3 t = t \sin t \cdot \frac{1 - \cos 2t}{2} = \frac{3}{4} t \sin t - \frac{t}{4} \sin 3t$. Therefore, the general solution is $x(t) = \frac{1}{128} (-4(6t^2 - 3 - 32c_1) \cos(t) + 3 \cos(3t) + 4 \sin(t)(7t + 2t \cos(2t) + 32c_2))$. $\square$

Use the method of variation of parameters to find a particular solution of the given differential equation in Problems 28 through 30.

Problem 3.6.28. $x'' + 9x = 3 \sec 3t$.

Solution. The general solution of the homogeneous equation $x'' + 9x = 0$ is $x_h = c_1 \cos 3t + c_2 \sin 3t$.

Using the parameter variation method, let the particular solution be $x_p = u_1(t) \cos 3t + u_2(t) \sin 3t$. The Wronskian $W = \begin{vmatrix} \cos 3t & \sin 3t \\ -3 \sin 3t & 3 \cos 3t \end{vmatrix} = 3$. Calculation gives

$$u_1' = -\frac{\sin 3t \cdot 3 \sec 3t}{3} = -\tan 3t, \quad u_2' = \frac{\cos 3t \cdot 3 \sec 3t}{3} = 1.$$

Integrating both sides yields $u_1 = \frac{1}{3} \ln |\cos 3t|$, $u_2 = t$. Therefore, the particular solution is: $x_p = \frac{1}{3} \cos 3t \ln |\cos 3t| + t \sin 3t$. $\square$

Problem 3.6.29. $x'' + x = \csc^2 t$.

Solution. The general solution of the homogeneous equation $x'' + x = 0$ is $x_h = c_1 \cos t + c_2 \sin t$.

Using the parameter variation method, let the particular solution be $x_p = u_1(t) \cos t + u_2(t) \sin t$. The Wronskian $W = \begin{vmatrix} \cos t & \sin t \\ -\sin t & \cos t \end{vmatrix} = 1$. Calculation gives

$$u_1' = -\sin t \cdot \csc^2 t = -\csc t, \quad u_2' = \cos t \cdot \csc^2 t = \cot t \csc t,$$

Integrating both sides yields

$$u_1 = \ln |\csc t + \cot t|, \quad u_2 = -\csc t.$$

Therefore, the particular solution is $x_p = \cos t \ln |\csc t + \cot t| - 1$. □

Problem 3.6.30. $x'' + x = \tan t$.

Solution. The general solution of the homogeneous equation $x'' + x = 0$ is $x_h = c_1 \cos t + c_2 \sin t$.

Using the parameter variation method, let the particular solution be $x_p = u_1(t) \cos t + u_2(t) \sin t$. The Wronskian $W = \begin{vmatrix} \cos t & \sin t \\ -\sin t & \cos t \end{vmatrix} = 1$. Calculation gives

$$u_1' = -\sin t \cdot \tan t = -\frac{\sin^2 t}{\cos t}, \quad u_2' = \cos t \cdot \tan t = \sin t,$$

Integrating both sides yields

$$u_1 = -\ln |\sec t + \tan t| + \sin t, \quad u_2 = -\cos t.$$

Therefore, the particular solution is: $x_p = -\cos t \ln |\sec t + \tan t|$. □

Chapter 4

General Theory

Exercise 4.1

不在考试范围内, 不写.

Exercise 4.2

教材上给出的存在唯一性定理只是基于二元的, 事实上可以推广到高维. 我们直接给出高维的定理叙述, 方便引用.

Theorem 4.2.1 (Picard 定理). 设初值问题

$$(E): \quad \frac{d\mathbf{y}}{dx} = \mathbf{f}(x, \mathbf{y}), \quad \mathbf{y}(x_0) = \mathbf{y}_0,$$

其中 $f(x, y)$ 在矩形区域

$$R: \quad |x - x_0| \leq a, \quad \|\mathbf{y} - \mathbf{y}_0\| \leq b$$

内连续, 而且对 y 满足 Lipschitz 条件, 则初值问题 (E) 在区间 $I = [x_0 - h, x_0 + h]$ 上有且只有一个解 $\mathbf{y} = \varphi(x)$, 其中 $h = \min \left\{ a, \frac{b}{M} \right\}$, $M = \max_{(x, y) \in R} \|\mathbf{f}(x, \mathbf{y})\|$. 这时也说方程组 $\frac{d\mathbf{y}}{dx} = \mathbf{f}(x, \mathbf{y})$ 过点 (x_0, y_0) 有并且只有一个解.

Problem 4.2.1. Prove the uniqueness of solution in Theorem 4.2.1 by Gronwall's inequality.

Proof. Let $x = \varphi(t), x = \psi(t)$ be two different solutions to the integral equation

$$x(t) = x_0 + \int_{t_0}^t f(s, x(s)) \, ds, \quad t \in I.$$

By the Lipschitz condition, we get

$$|\varphi(t) - \psi(t)| \leq L \left| \int_{t_0}^t |\varphi(s) - \psi(s)| \, ds \right|, \quad t \in I.$$

Then by Gronwall's inequality, we have

$$0 \leq |\varphi(t) - \psi(t)| \leq 0, \quad t \in I,$$

which implies $\varphi(t) \equiv \psi(t)$. □

Problem 4.2.2. Show that the initial value problem

$$\begin{cases} x'' = f(t, x), \\ x(t_0) = x_0, x'(t_0) = x_1, \end{cases}$$

where $f(t, x)$ is continuous in a domain $R \supset (t_0, x_0)$, is equivalent to the integral equation

$$x(t) = x_0 + (t - t_0)x_1 + \int_{t_0}^t (t - s)f(s, x(s)) \, ds.$$

Proof. This problem is exactly the same as Problem 2.3.2 in §2.3. □

Problem 4.2.3. By computing appropriate Lipschitz constants, show that the following functions satisfy the Lipschitz condition in the given domains.

- (1) $x \sin y + y \cos x$, $R : |x| \leq a, |y| \leq b$;
- (2) $t^2 e^{t+x}$, $R : |t| \leq a, |x| \leq b$;
- (3) $p(t)x + q(t)$, $R : |t| \leq 1, |x| < \infty$, where $p(t)$ and $q(t)$ are continuous in the interval $|t| \leq 1$.

Proof. (1) Let $f_1(x, y) := x \sin y + y \cos x$, then $\forall |x| \leq a, |y| \leq b$, we have

$$\begin{aligned} |f_1(x, y_1) - f_1(x, y_2)| &= |x(\sin y_1 - \sin y_2) + (y_1 - y_2) \cos x| \\ &\leq |x| |\sin y_1 - \sin y_2| + |\cos x| |y_1 - y_2| \leq M |y_1 - y_2|, \end{aligned}$$

where $M_1 = \max\{a, 1\}$. Similarly, we have

$$|f_1(x_1, y) - f_1(x_2, y)| \leq M_2 |x_1 - x_2|,$$

where $M_2 = \max\{b, 1\}$. Therefore, f_1 satisfy the Lipschitz condition.

- (2) Let $f_2(t, x) := t^2 e^{t+x}$, then $\forall x_1, x_2 (|x_1| \leq a, |x_2| \leq b)$, we have

$$\begin{aligned} |f_2(t, x_1) - f_2(t, x_2)| &= |t^2 e^{t+x_1} - t^2 e^{t+x_2}| = |t^2 e^t| |e^{x_1} - e^{x_2}| \\ &= |t^2 e^t| |e^\xi| |x_1 - x_2| \leq \max_{|t| \leq a} \{t^2 e^t\} e^b |x_1 - x_2|, \end{aligned}$$

where ξ is between x_1, x_2 . Therefore, f_2 satisfy the Lipschitz condition.

- (3) Note that $p \in C[-1, 1]$, then p have maximum values on $[-1, 1]$, denoted as M . Hence, briefly note $f_3(t, x) := p(t)x + q(t)$, we have

$$|f_3(t, x_1) - f_3(t, x_2)| = |p(t)x_1 + q(t) - p(t)x_2 - q(t)| \leq M |x_1 - x_2|,$$

which implies that f_3 satisfy the Lipschitz condition. □

Problem 4.2.4. Let $\varphi(t)$ be a nonnegative continuous function in the interval $|t - t_0| \leq b$, and $k \geq 0$ be a given constant, and

$$\varphi(t) \leq \left| \int_{t_0}^t k\varphi^\alpha(s) ds \right|, 0 < \alpha < 1.$$

Show that for all t in $|t - t_0| \leq b$,

$$\varphi(t) \leq [k(1 - \alpha)|t - t_0|]^{\frac{1}{1-\alpha}}.$$

Proof. Let $F(t) := \int_{t_0}^t \varphi^\alpha(s) ds$, and define

$$g(t) := \frac{1}{1 - \alpha} F^{1-\alpha}(t) - k^\alpha(t - t_0).$$

Note that

$$g'(t) = \frac{F'(t)}{F^\alpha(t)} - k^\alpha = \frac{\varphi^\alpha(t) - k^\alpha \left(\int_{t_0}^t \varphi^\alpha(s) ds \right)^\alpha}{F^\alpha(t)} \leq 0.$$

Therefore, $g(t) \leq g(t_0) = 0$ for all $t \geq t_0$, and thus $\varphi(t) \leq (k(1 - \alpha)(t - t_0))^{1/(1-\alpha)}$. The proof can be similarly established when $t < t_0$. Thus, we have completed the proof. $\square$

Problem 4.2.5. Compute the first three Picard iterates for the following initial value problems.

$$(1) \quad \frac{dx}{dt} = t - x^2, x(0) = 0 \text{ with } x_0(t) \equiv 0;$$

$$(2) \quad \frac{dx}{dt} = x^2 + t^2, x(0) = 1 \text{ with } x_0(t) \equiv 1;$$

$$(3) \quad \frac{dx}{dt} = t^2 - x^2 - 1, x(0) = 0 \text{ with } x_0(t) \equiv t;$$

$$(4) \quad \frac{dx}{dt} = t^2 + x^2, x(0) = 0 \text{ with } x_0(t) \equiv t.$$

Solution. (1) Here $f(t, x) = t - x^2$. We have

$$x_1(t) = 0 + \int_0^t (s - x_0^2(s)) ds = \int_0^t s ds = \frac{1}{2}t^2,$$

$$x_2(t) = \int_0^t (s - x_1^2(s)) ds = \frac{1}{2}t^2 - \frac{1}{20}t^5.$$

(2) Here $f(t, x) = t^2 + x^2$. We have

$$x_1(t) = 1 + \int_0^t (s^2 + x_0^2(s)) ds = 1 + \int_0^t (s^2 + 1) ds = \frac{1}{3}t^3 + t + 1,$$

$$x_2(t) = 1 + \int_0^t (s^2 + x_1^2(s)) ds = \frac{1}{63}t^7 + \frac{2}{15}t^5 + \frac{1}{6}t^4 + \frac{2}{3}t^3 + t^2 + t + 1.$$

(3) Here $f(t, x) = t^2 - x^2 - 1$. We have

$$\begin{aligned}x_1(t) &= 0 + \int_0^t (s^2 - x_0^2(s) - 1) \, ds = -t, \\x_2(t) &= 0 + \int_0^t (s^2 - x_1^2(s) - 1) \, ds = -t.\end{aligned}$$

(4) Here $f(t, x) = t^2 + x^2$. We have

$$\begin{aligned}x_1(t) &= 0 + \int_0^t (s^2 + x_0^2(s)) \, ds = \int_0^t (2s^2) \, ds = \frac{2}{3}t^3, \\x_2(t) &= \int_0^t (s^2 + x_1^2(s)) \, ds = \frac{4}{63}t^7 + \frac{1}{3}t^3.\end{aligned}$$

□

Problem 4.2.6. Discuss the existence and uniqueness of solutions of the following initial value problems.

(1) $x' = 1 + x^{\frac{2}{3}}, x(0) = 0;$

(2) $\frac{dx}{dt} = \frac{3}{2}x^{\frac{1}{3}}, x(0) = 0;$

(3) $\frac{dx}{dt} = \sqrt{|x|}, x(0) = 0;$

(4) $\frac{dx}{dt} = (t + x)t^2x^2, x(0) = 1.$

Solution. (1) $1 + x^{2/3}$ 在 0 附近不满足 Lipschitz 条件, 但是解是存在且唯一的, 具体可以由下面定理直接导出:

Theorem 4.2.2. 考虑微分方程的初值问题, 即

$$\frac{dy}{dx} = f(x, y), \quad y(x_0) = y_0. \quad (4.1)$$

设 $f(x, y)$ 在平面区域 $D \subset \mathbb{R}^2$ 内连续, (x_0, y_0) 为区域 D 内的任意一点, 且 $f(x_0, y_0) \neq 0$, 则初值问题 (4.1) 存在局部唯一的解当且仅当初值问题

$$\frac{dx}{dy} = \frac{1}{f(x, y)}, \quad x(y_0) = x_0. \quad (4.2)$$

有局部唯一的解. 这里 (4.1) 有局部唯一解的含义是: 存在 $\alpha > 0$, 使得方程的初值问题 (4.1) 在 $[x_0 - \alpha, x_0 + \alpha]$ 上有解 $y = \phi(x)$, 且 (4.1) 的任何解 $y = \eta(x)$ ($x \in J$) 都满足 $\phi(x) = \eta(x), \forall x \in [x_0 - \alpha, x_0 + \alpha] \cap J$.

(2) It is easy to check that for any $c \geq 0$, $x_c(t) = \begin{cases} 0 & t \leq c \\ \frac{(t-c)^2}{4} & t > c \end{cases}$ is a solution to the original equation.

- (3) Existence and Uniqueness: The function $f(t, x) = (t + x)t^2x^2$ is a polynomial, so it's continuous everywhere. Its partial derivative with respect to x is:

$$\frac{\partial f}{\partial x} = t^2x^2 + 2(t + x)t^2x = 3t^2x^2 + 2t^3x,$$

which is continuous everywhere, which means f is Lipschitz with respect to x in any bounded region, including a neighborhood of $(0, 1)$. By the Picard's theorem, a unique solution exists in some interval around $t = 0$. $\square$

Problem 4.2.7. Show that the initial value problem

$$\begin{cases} \frac{dx}{dt} = f(t, x), \\ x(t_0) = x_0, \end{cases} \quad (4.3)$$

where $f(t, x)$ is continuous in $R^+ : t_0 \leq t \leq t_0 + a, |x - x_0| \leq b$ and nonincreasing in x for each fixed t in $t_0 \leq t \leq t_0 + a$ exists at most one solution in $t_0 \leq t \leq t_0 + a$.

Proof. By using the method of contradiction, assume that the initial value problem has two solutions $x = x_1(t)$ and $x = x_2(t)$, and there exists $t_1 > t_0$ such that $x_1(t_1) \neq x_2(t_1)$. Without loss of generality, let $x_1(t_1) > x_2(t_1)$. Let's define

$$\bar{t} := \sup\{t \in [t_0, t_1] : x_1(t) = x_2(t)\},$$

Then it is obvious that $t_0 \leq \bar{t} < t_1$, and

$$r(t) := x_1(t) - x_2(t) > 0, \quad \text{if } \bar{t} < t \leq t_1,$$

and $r(\bar{t}) = 0$. However,

$$r'(t) = x_1'(t) - x_2'(t) = f(t, x_1(t)) - f(t, x_2(t)) \leq 0, \quad \bar{t} < t \leq t_1.$$

From the mean value theorem, we can further conclude that $r(t) \leq 0$, where $\bar{t} < t \leq t_1$. This is a contradiction. $\square$

Remark 4.1. 左侧解不一定唯一. 例如, $y' = -\frac{3}{2}y^{\frac{1}{3}}$ 有特解 $y = 0$ 及通解

$$-y^{\frac{4}{3}} = x + C, \quad x + C \leq 0.$$

因而过 $y = 0$ 上每点的解是不唯一的.

Problem 4.2.8. Show that the initial value problem

$$\begin{cases} \frac{dx}{dt} = (t^2 - x^2) \sin x + x^2 \cos x, \\ x(0) = 0. \end{cases}$$

has a unique solution $x(t) \equiv 0$ in a closed domain R :

$$|t| \leq a, |x| \leq b.$$

Proof. For the initial value problem $\frac{dx}{dt} = (t^2 - x^2) \sin x + x^2 \cos x, x(0) = 0$ within the closed region $R: |t| \leq a, |x| \leq b$, direct substitution and verification show that $x(t) \equiv 0$ is a solution; the function $f(t, x)$ is continuous, and its partial derivative $\frac{\partial f}{\partial x} = t^2 \cos x + x \cos x(2 - x) - x \sin x(2 + x)$ is continuous and bounded on R , thus f satisfies the Lipschitz condition with respect to x . By the Picard's theorem, the solution is unique, so $x(t) \equiv 0$ is the unique solution. $\square$

Problem 4.2.9. Let $f(t, x)$ be continuous in $\mathbb{R}$ and for all $(t, x_1), (t, x_2) \in \mathbb{R}$ it satisfies

$$|f(t, x_1) - f(t, x_2)| \leq \omega(|x_1 - x_2|),$$

where $\omega(s)$ is a continuous function and $\omega(0) = 0, \omega(s) > 0$ for $s > 0$ and the improper integral

$$\int_0^r \frac{ds}{\omega(s)} = \infty^1 \quad (r > 0).$$

Then the initial value problem has at most one solution in $|t - t_0| \leq a$.

Proof. Suppose there exists $(t_0, x_0) \in D$ such that the initial value problem (4.1) has two solutions.

$$x = \varphi_1(t), \quad x = \varphi_2(t),$$

And there exists $t_1 \neq t_0$ such that $\varphi_1(t_1) > \varphi_2(t_1)$. Let's assume $t_1 > t_0$. Define

$$\bar{t} := \sup_{t \in [t_0, t_1]} \{x: \varphi_1(t) = \varphi_2(t)\},$$

Then $\bar{t} \in [t_0, t_1)$, and

$$\varphi_1(\bar{t}) = \varphi_2(\bar{t}), \quad \varphi_1(t) > \varphi_2(t), \quad t \in (\bar{t}, t_1].$$

Define

$$r(t) := \varphi_1(t) - \varphi_2(t),$$

Then, when t is in the interval $(\bar{t}, t_1)$, $r(t) > 0$, and

$$r'(t) = f(t, \varphi_1(t)) - f(t, \varphi_2(t)) \leq \omega(|\varphi_1(t) - \varphi_2(t)|) = \omega(r(t)).$$

So

$$\frac{1}{\omega(s)} ds \leq dt.$$

Therefore

$$\int_0^{r_1} \frac{1}{\omega(s)} ds \leq t_1 - \bar{t},$$

¹这里应该理解成正无穷.

Here, $r_1 = r(t_1) > 0$. This contradicts the Osgood condition. Therefore, the initial value problem has at most one solution in the interval $|t - t_0| \leq a$. $\square$

Remark 4.2. 这个题目上的条件实际上叫作 **Osgood 条件**. 实际上, 由 *Peano* 存在定理, 保证了解的存在性, *Osgood* 条件保证了解不会超过两个, 因此 *Osgood* 条件保证了解的唯一性. 容易验证 *Osgood* 条件比 *Lipschitz* 条件弱.

Problem 4.2.10. Let $f(t, x)$ be continuous and satisfy the generalized Lipschitz condition

$$|f(t, x_1) - f(t, x_2)| \leq L(t)|x_1 - x_2|$$

for all $(t, x_1), (t, x_2) \in \mathbb{R}$, where $L(t)$ is such that the integral $\int_{t_0-a}^{t_0+a} L(s) ds$ exists. Show that the initial value problem has at most one solution in the interval $|t - t_0| \leq a$.

Proof. Assume $x_1(t)$ and $x_2(t)$ are two distinct solutions to the IVP in the interval $|t - t_0| \leq a$. Define $u(t) := |x_1(t) - x_2(t)|$, then $u(t_0) = 0$. For any t in the interval $|t - t_0| \leq a$, we have:

$$\begin{aligned} u(t) &= |x_1(t) - x_2(t)| = \left| \int_{t_0}^t (x_1'(s) - x_2'(s)) ds \right| \\ &= \left| \int_{t_0}^t (f(s, x_1(s)) - f(s, x_2(s))) ds \right| \\ &\leq \int_{t_0}^t |f(s, x_1(s)) - f(s, x_2(s))| ds \\ &\leq \int_{t_0}^t L(s)|x_1(s) - x_2(s)| ds \\ &= \int_{t_0}^t L(s)u(s) ds. \end{aligned}$$

So we have:

$$u(t) \leq \int_{t_0}^t L(s)u(s) ds \quad \text{for } t \geq t_0.$$

Similarly, for $t < t_0$, we can write:

$$u(t) = \left| \int_t^{t_0} (x_2'(s) - x_1'(s)) ds \right| \leq \int_t^{t_0} L(s)u(s) ds.$$

If $t \geq t_0$, applying Gronwall's inequality to $u(t) \leq \int_{t_0}^t L(s)u(s) ds$ with $\alpha = 0$, we get:

$$u(t) \leq 0 \cdot \exp \left(\int_{t_0}^t L(s) ds \right) = 0.$$

Therefore, $u(t) = 0$ for all $t \geq t_0$ in the interval.

If $t < t_0$, define $w(t) = \int_t^{t_0} L(s)u(s) ds$. Then $w(t_0) = 0$ and:

$$\frac{dw}{dt} = -L(t)u(t) \geq -L(t)w(t).$$

Multiplying by the integrating factor $e^{\int_t^{t_0} L(s) ds}$:

$$\frac{d}{dt} \left(w(t) e^{\int_t^{t_0} L(s) ds} \right) \geq 0.$$

This means $w(t) e^{\int_t^{t_0} L(s) ds}$ is non-decreasing. Since $w(t_0) = 0$, for $t < t_0$:

$$w(t) e^{\int_t^{t_0} L(s) ds} \leq 0.$$

But $w(t) \geq 0$ and the exponential term is positive, so $w(t) = 0$ for all $t \leq t_0$, which implies $u(t) = 0$.

Since $u(t) = 0$ for all t in the interval $|t - t_0| \leq a$, we have $x_1(t) = x_2(t)$ for all such t . Therefore, the initial value problem has at most one solution in this interval. $\square$

Problem 4.2.11. Apply the iterative formula in

$$x_{n+1}(t) = x_0 + \int_{t_0}^t f(s, x_n(s)) ds, \quad t \in I, \quad x_0(t) \equiv x_0, \quad n = 0, 1, 2, \dots$$

to compute the first three successive approximations to the solution of the initial value problem

$$\begin{cases} \frac{dx}{dt} = 2x - y, & x(0) = 1, \\ \frac{dy}{dt} = 3x - 2y, & y(0) = -1. \end{cases}$$

Solution. The coefficient matrix is $\mathbf{A} = \begin{pmatrix} 2 & -1 \\ 3 & -2 \end{pmatrix}$, we transform the system to vector form.

$$\frac{d\mathbf{x}}{dt} = \mathbf{A}\mathbf{x}$$

Where $\mathbf{x}_0 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$, so we have

$$\mathbf{x}_1(t) = \mathbf{x}_0 + \int_{t_0}^t \mathbf{A} \cdot \mathbf{x}_0 dt = \begin{pmatrix} 3t + 1 \\ 5t - 1 \end{pmatrix},$$

$$\mathbf{x}_2(t) = \mathbf{x}_0 + \int_{t_0}^t \mathbf{A} \cdot \mathbf{x}_1(t) dt = \begin{pmatrix} \frac{1}{2}t^2 + 3t + 1 \\ -\frac{1}{2}t^2 + 5t - 1 \end{pmatrix},$$

$$\mathbf{x}_3(t) = \mathbf{x}_0 + \int_{t_0}^t \mathbf{A} \cdot \mathbf{x}_2(t) dt = \begin{pmatrix} \frac{1}{2}t^3 + \frac{1}{2}t^2 + 3t + 1 \\ \frac{5}{6}t^3 - \frac{1}{2}t^2 + 5t - 1 \end{pmatrix}. \quad \square$$